FISHER–RAO AND WASSERSTEIN–OTTO GEOMETRIES OF PARAMETRIC PROBABILITY FAMILIES: A COMPARATIVE ATLAS

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Abstract. We compare the Fisher–Rao and Wasserstein–Otto metrics by pulling them back onto the same parametric families of probability densities. After recalling the measure-theoretic and differential-geometric framework, we derive the two ambient constructions and make their relation explicit on the real line. The comparative atlas covers location–dispersion models, multivariate Gaussians, linear and nonlinear Gibbs families, and push-forward families. For each model we specify its parameter and tangent spaces, compute the density differential and both induced metrics, and interpret the resulting geometry through densities, metric indicatrices, and geodesics. Exact paths are distinguished from numerical approximations and from paths constrained to a subfamily. The figures and their Python scripts are available at https://github.com/PierreGuilbertF/fisher-rao-wasserstein-atlas.

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1. Introduction

Geometry gives form to mathematical objects and allows us to reason about their deformations. Once a smooth manifold is equipped with a Riemannian metric, each tangent space carries an inner product, and one can define lengths, angles, distances, geodesics and, in finite dimensions, the associated notions of area and volume. These geometric quantities are not only descriptive: they provide tools for moving through and optimizing over nonlinear spaces. This point of view is now standard in many areas of applied mathematics and engineering, including constrained and Riemannian optimization, continuum mechanics and transport, control and robotics, signal and image processing, statistics, and machine learning.

In this manuscript, we apply this geometric viewpoint to the space of strictly positive probability densities. Let E be a finite-dimensional vector space, equipped when necessary with an inner product, and let

$$\mathcal{P}_+(E) = \left\{ p > 0 : \int_E p(x)\,dx = 1 \right\}.$$

We shall equip this space successively with two classical and complementary geometries: the Fisher–Rao geometry, whose Riemannian interpretation in statistics goes back to Rao [4], and the Wasserstein–Otto geometry, which gives the quadratic Wasserstein distance its formal Riemannian structure on spaces of densities [5]. We use these metrics as complementary geometric descriptions of the same probability families.

If $t \mapsto p_t \in \mathcal{P}_+(E)$ is a differentiable curve with $p_0 = p$, its velocity

$$u = \partial_t p_t|_{t=0}$$

must preserve the normalization of the density. Hence the admissible tangent directions satisfy

$$T_p\mathcal{P}_+(E) = \left\{ u : \int_E u(x)\,dx = 0 \right\},$$

with the additional regularity and integrability assumptions required by the metric under consideration. The same tangent variation may also be described through a velocity field on the underlying space. Indeed, if the density evolves according to the continuity equation

$$\partial_t p_t + \operatorname{div}(p_t v_t) = 0,$$

then at $t = 0$,

$$u = -\operatorname{div}(pv).$$

This second representation will be fundamental for the Wasserstein–Otto geometry, where tangent vectors are interpreted through the motion of probability mass.

The Fisher–Rao metric acts directly on density variations:

$$g_p^{\mathrm{FR}}(u, w) = \int_E \frac{u(x)w(x)}{p(x)}\,dx.$$

It measures the relative local variation of the density and does not require a notion of distance between points of E. By contrast, the Wasserstein–Otto metric is defined through the kinetic cost of transporting the density. Its squared norm can be written formally as [5]

$$\|u\|^2_{p,\mathrm{WO}} = \inf_{\substack{v \\ u=-\operatorname{div}(pv)}} \int_E \|v(x)\|^2 p(x)\,dx.$$

Under the usual regularity assumptions, the minimizing velocity field is a gradient, $v = \nabla\phi_u$ [5], where

$$-\operatorname{div}(p\nabla\phi_u) = u.$$

The corresponding bilinear form is therefore

$$g_p^{\mathrm{WO}}(u, w) = \int_E p(x)\,\langle \nabla\phi_u(x), \nabla\phi_w(x)\rangle\,dx.$$

Fisher–Rao measures local relative density change, whereas Wasserstein–Otto measures mass displacement through E. Figure 1 illustrates this contrast on Gaussian densities. Pulling back these ambient metrics can therefore give the same parametric family different lengths, distances, curvature and geodesics.

More precisely, for a smooth parametric family

$$\theta \longmapsto p_\theta \in \mathcal{P}_+(E),$$

and tangent directions $d\theta_1, d\theta_2 \in T_\theta\Theta$, each ambient metric induces a finite-dimensional metric through

$$g_\theta^{\mathrm{FR}}(d\theta_1, d\theta_2) = g_{p_\theta}^{\mathrm{FR}}\left(dp_\theta(d\theta_1), dp_\theta(d\theta_2)\right),$$

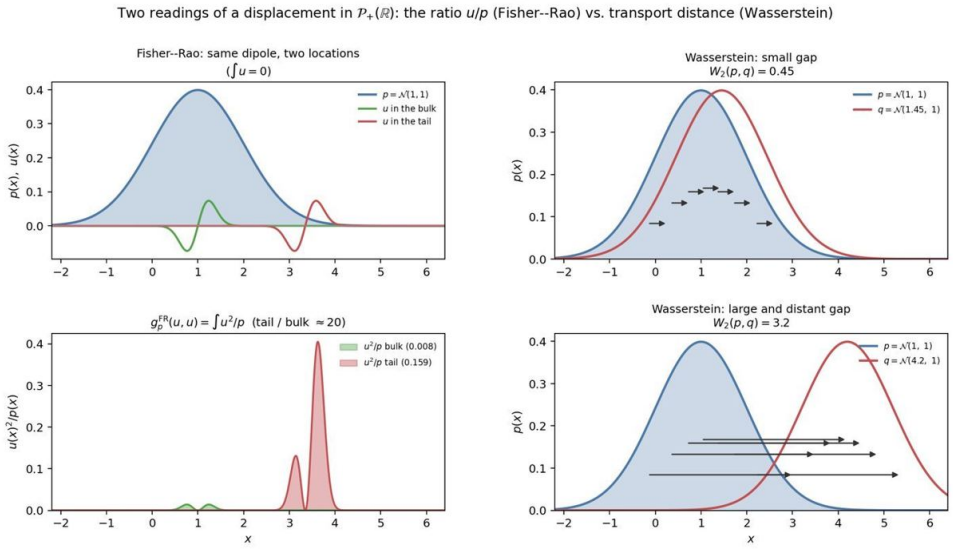


FIGURE 1. Two readings of a displacement in $\mathcal{P}_+(\mathbb{R})$. Left: the same mass-preserving perturbation costs more in a density's tail under Fisher–Rao, whose integrand is u^2/p. Right: Wasserstein distance records the displacement between translated Gaussian densities.

and

$$g_\theta^{\mathrm{WO}}(d\theta_1, d\theta_2) = g_{p_\theta}^{\mathrm{WO}}\left(dp_\theta(d\theta_1), dp_\theta(d\theta_2)\right).$$

The statistical model is therefore unchanged, while its geometry depends on the metric through which it is observed.

This expository Atlas gathers classical Fisher–Rao and Wasserstein–Otto constructions in a common pullback framework. Explicit calculations and visualizations compare their induced geometries on one-dimensional location–dispersion models, multivariate Gaussians, linear and nonlinear Gibbs families, and push-forward families. We emphasize the form and interpretation of each metric and, whenever tractable, its distances, geodesics, curvature and gradient dynamics.

2. Overview of the Atlas and Comparative Study

Table 1 summarizes the families studied in this Atlas and their Fisher–Rao and Wasserstein–Otto line elements on parameter space.

For the multivariate Gaussian family, the inverse Lyapunov operator is

$$\mathcal{L}_\Sigma(A) = X \qquad \Longleftrightarrow \qquad \Sigma X + X\Sigma = A, \qquad \Sigma \in S_n^{++}(\mathbb{R}).$$

Table 1 uses $ds^2 = g_\theta(d\theta, d\theta)$. The regularity, integrability, immersion, and Poisson solvability assumptions are those stated in the corresponding sections.

Location–dispersion families (Section 5.3)	
Family	$p_{\mu,\sigma}(x) = \sigma^{-1}p((x-\mu)/\sigma)$, $\Theta = \mathbb{R}\times(0,\infty)$; p has mean 0 and variance 1.
Fisher–Rao	$ds^2_{\mathrm{FR}} = \dfrac{A_p\, d\mu^2 + 2B_p\, d\mu\, d\sigma + C_p\, d\sigma^2}{\sigma^2}$.
Wasserstein–Otto	$ds^2_{\mathrm{WO}} = d\mu^2 + d\sigma^2$.
Comparison	Shape-dependent constants A_p, B_p, C_p are defined in Section 5.3. FR has curvature $-1/(C_p - B_p^2/A_p)$; WO is Euclidean, with straight location–scale geodesics.
Multivariate Gaussian family (Section 6)	
Family	$p_{\mu,\Sigma} = \mathcal{N}(\mu,\Sigma)$, $\Theta = \mathbb{R}^n \times S_n^{++}(\mathbb{R})$.
Fisher–Rao	$ds^2_{\mathrm{FR}} = d\mu^\top \Sigma^{-1} d\mu + \frac{1}{2}\operatorname{tr}(\Sigma^{-1}d\Sigma\,\Sigma^{-1}d\Sigma)$.
Wasserstein–Otto	$ds^2_{\mathrm{WO}} = \lVert d\mu\rVert^2 + \frac{1}{2}\operatorname{tr}(\mathcal{L}_\Sigma(d\Sigma)\,d\Sigma)$.
Comparison	FR couples mean and covariance along geodesics. WO splits into Euclidean mean and Bures–Wasserstein covariance geometry. Principal-scale costs are relative for FR and absolute for WO.
Gibbs families: nonlinear and linear (Section 7)	
Family	$p_\theta = Z_\theta^{-1}e^{-U_\theta}$, $\Theta \subset \mathbb{R}^k$ open; $\mathbb{E}_\theta$ is expectation under p_θ.
Notation	$h_\theta(d\theta) = d_\theta U_\theta(d\theta) - \mathbb{E}_\theta[d_\theta U_\theta(d\theta)]$; $L_\theta(f) = -p_\theta^{-1}\operatorname{div}(p_\theta \nabla f)$, with inverse on centered functions.
Fisher–Rao	$ds^2_{\mathrm{FR}} = \mathbb{E}_\theta[h_\theta(d\theta)^2]$.
Wasserstein–Otto	$ds^2_{\mathrm{WO}} = \mathbb{E}_\theta\big[h_\theta(d\theta)\, L_\theta^{-1}\big(h_\theta(d\theta)\big)\big]$.
Linear case	For $U_\theta = \phi^T\theta$, $G^{\mathrm{FR}} = \operatorname{Cov}_\theta(\phi) = \operatorname{Hess}_\theta \log Z_\theta$; $G^{\mathrm{WO}}_{ij} = \mathbb{E}_\theta\big[(\phi_i - \mathbb{E}_\theta\phi_i)L_\theta^{-1}(\phi_j - \mathbb{E}_\theta\phi_j)\big]$.
Comparison	FR measures energy fluctuations directly; WO weights them through an inverse spatial elliptic operator. Linear potentials retain the Poisson solve.
Push-forward families (Section 8)	
Family	$p_\theta = (\phi_\theta)_\sharp q$, $\Theta \subset \mathbb{R}^k$ open; q is fixed and ϕ_θ is a diffeomorphism.
Velocity	$v_{\theta,d\theta} = d_\theta\phi_\theta(d\theta)\circ\phi_\theta^{-1}$; $\operatorname{div}(p_\theta\nabla\psi_{\theta,d\theta}) = \operatorname{div}(p_\theta v_{\theta,d\theta})$.
Fisher–Rao	$ds^2_{\mathrm{FR}} = \displaystyle\int_E \frac{[\operatorname{div}(p_\theta v_{\theta,d\theta})]^2}{p_\theta}\,dy$.
Wasserstein–Otto	$ds^2_{\mathrm{WO}} = \displaystyle\int_E p_\theta \lVert\nabla\psi_{\theta,d\theta}\rVert^2\,dy$.
Comparison	FR measures relative density deformation; WO measures the kinetic energy of the gradient projection. The raw velocity gives the WO cost only when it is already an admissible gradient.

TABLE 1. Pullback metrics in the comparative Atlas. Each block gives the family, the two squared tangent norms, and their main geometric contrast. Gibbs families include the linear specialization.

For each family, the analysis follows the same general scheme: identification of the tangent directions, computation of the two pullback metrics, and comparison of the resulting geodesics, distances, curvature, and gradient dynamics whenever these quantities admit a tractable expression.

3. Framework and notation

The notation follows the conventions used in Guilbert [1]. Readers looking for a more detailed and pedagogical introduction to the objects used below may refer to that work.

3.1. Probability densities

Throughout the paper, E is a finite-dimensional real vector space equipped with an inner product, most often $E = \mathbb{R}^n$. We denote its Borel σ-algebra by $\mathcal{B}(E)$ and the associated Lebesgue measure by λ. Unless another measure is specified, all integrals on E are taken with respect to λ, and we write dx for $d\lambda(x)$.

Every probability measure P considered below is defined on $\mathcal{B}(E)$ and is absolutely continuous with respect to λ. Its density therefore depends on this reference measure and is written

$$p = \frac{dP}{d\lambda}, \qquad P(A) = \int_A p(x)\, d\lambda(x), \qquad A \in \mathcal{B}(E).$$

We write

$$\mathcal{P}_+(E) = \left\{ p \in L^1(E) : p > 0 \ \lambda\text{-a.e.}, \ \int_E p\, d\lambda = 1 \right\}.$$

The set $\mathcal{P}_+(E)$ is not open in $L^1(E)$, and no Banach manifold structure on it will be needed. We shall nevertheless consider differentiable curves $t \mapsto p_t \in \mathcal{P}_+(E)$. Their velocity

$$u = \partial_t p_t \big|_{t=0}$$

satisfies

$$\int_E u(x)\, dx = 0,$$

by differentiation of the normalization. We use $T_p\mathcal{P}_+(E)$ for the class of such admissible variations at p, with the additional integrability or regularity required by the metric under consideration.

A finite-dimensional parametric family is a map

$$p \colon \Theta \longrightarrow \mathcal{P}_+(E), \qquad \theta \longmapsto p_\theta,$$

where Θ is a finite-dimensional smooth manifold, usually an open subset of $\mathbb{R}^k$. A direction $d\theta \in T_\theta\Theta$ induces the density variation

$$dp_\theta(d\theta) \in T_{p_\theta}\mathcal{P}_+(E).$$

3.2. **Differentials, Jacobians, and gradients**

Let $f\colon E \to F$ be a map between normed vector spaces. Its Fréchet differential at $a \in E$ is the linear map $df_a \in \mathcal{L}(E, F)$ characterized by

$$f(a + h) = f(a) + df_a(h) + o(\|h\|).$$

This linear map is the intrinsic object used throughout the paper. If (e_i) is a basis of E, then

$$df_a(e_i) = \frac{\partial f}{\partial x_i}(a).$$

When E and F are finite-dimensional and equipped with bases B_E and B_F, the matrix representing the same linear map is the Jacobian

$$J_f(a) = \mathrm{Mat}_{B_E,B_F}(df_a), \qquad (J_f(a))_{ij} = \partial_j f_i(a).$$

If $F = \mathbb{R}$ and E carries an inner product, the same differential is represented by the gradient:

$$df_a(h) = \langle \nabla f(a),\, h \rangle .$$

We use df_a, $J_f(a)$, or $\nabla f(a)$ according to whether the intrinsic linear map, its representation in chosen bases, or its representation through an inner product is more convenient.

3.3. **Riemannian manifolds**

Manifolds are assumed smooth, connected, and finite-dimensional. We denote a manifold by $\mathcal{M}$ and its tangent space at $x \in \mathcal{M}$ by $T_x\mathcal{M}$. If $\mathcal{M}$ is immersed in a vector space F through a local parameterization $\phi\colon U \subset \mathbb{R}^k \to \mathcal{M} \subset F$, then

$$T_{\phi(a)}\mathcal{M} = \mathrm{Im}(d\phi_a),$$

and the vectors $\partial_{a_i}\phi(a)$ give the associated coordinate basis whenever ϕ is an immersion.

A Riemannian manifold is denoted by $(\mathcal{M}, g)$. The metric is a smooth field of inner products

$$g_x\colon T_x\mathcal{M} \times T_x\mathcal{M} \to \mathbb{R}.$$

Given a basis $B = (e_1, \dots, e_k)$ of $T_x\mathcal{M}$, the matrix of g_x is denoted by

$$G(x) = (g_{ij}(x))_{i,j} \in S^k_{++}(\mathbb{R}), \qquad g_{ij}(x) = g_x(e_i, e_j).$$

If $U, V \in \mathbb{R}^k$ are the coordinates of $u, v \in T_x\mathcal{M}$ in B, the intrinsic metric and its matrix representation are related by

$$g_x(u, v) = U^\top G(x) V.$$

The same metric is also written locally as the line element

$$ds^2 = \sum_{i,j} g_{ij}(x)\, dx_i\, dx_j,$$

where $(dx_1, \dots, dx_k)$ is the dual coordinate basis. This notation records the squared infinitesimal length: if $u = \sum_i U_i e_i$, then

$$g_x(u, u) = U^\top G(x) U = \sum_{i,j} g_{ij}(x) U_i U_j.$$

Thus ds^2 is the coordinate expression of g, not an additional metric.

For a smooth function $f\colon \mathcal{M} \to \mathbb{R}$, the Riemannian gradient $\nabla_g f(x) \in T_x\mathcal{M}$ is the representation of df_x through g_x:

$$df_x(h) = g_x(\nabla_g f(x), h), \qquad h \in T_x\mathcal{M}.$$

3.4. Induced metrics and pullbacks

Let $F\colon \mathcal{M} \to \mathcal{N}$ be a smooth map and let h be a Riemannian metric on $\mathcal{N}$. The pullback of h by F is the symmetric form

$$(F^*h)_x(u, v) = h_{F(x)}(dF_x(u), dF_x(v)), \qquad u, v \in T_x\mathcal{M}.$$

Figure 2 recalls the construction: tangent vectors on $\mathcal{M}$ are measured after push-forward by dF, using the metric of the target. If F is an immersion, F^*h is positive definite and therefore defines a Riemannian metric on $\mathcal{M}$. This is the mechanism used throughout the paper. For a parametric family $\theta \mapsto p_\theta$ and a metric h on admissible variations of densities, the induced metric on Θ is

$$g_\theta(d\theta_1, d\theta_2) = h_{p_\theta}(dp_\theta(d\theta_1), dp_\theta(d\theta_2)).$$

The same family may therefore inherit different geometries according to the metric chosen on $\mathcal{P}_+(E)$.

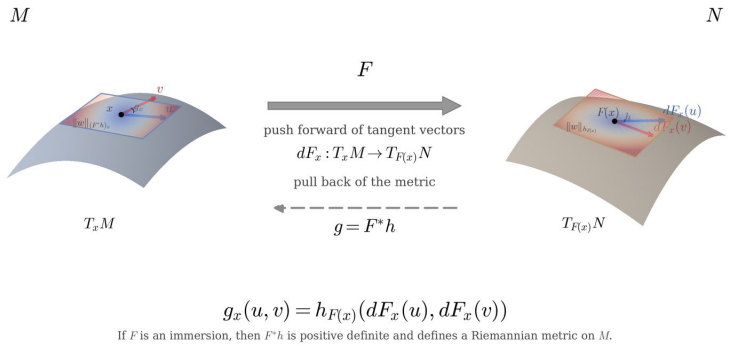


FIGURE 2. Illustration of a pullback metric. Tangent vectors $u, v \in T_x\mathcal{M}$ are pushed forward by dF_x to $T_{F(x)}\mathcal{N}$, where they are measured with h; the resulting inner product defines the pullback metric $g = F^*h$ on $\mathcal{M}$.

3.5. Length and distance

For a piecewise $\mathcal{C}^1$ curve $\gamma\colon [a, b] \to \mathcal{M}$, we use

$$L(\gamma) = \int_a^b \sqrt{g_{\gamma(t)}(\gamma'(t), \gamma'(t))}\, dt, \qquad d_g(x, y) = \inf_{\substack{\gamma(a)=x \\ \gamma(b)=y}} L(\gamma).$$

A geodesic is a constant-speed curve which is locally length-minimizing. Figure 3 illustrates this on a bump surface: between the same endpoints, several curves may be locally minimizing, while only the shortest among them realizes the Riemannian distance. The distance d_g induces the original topology of $\mathcal{M}$.

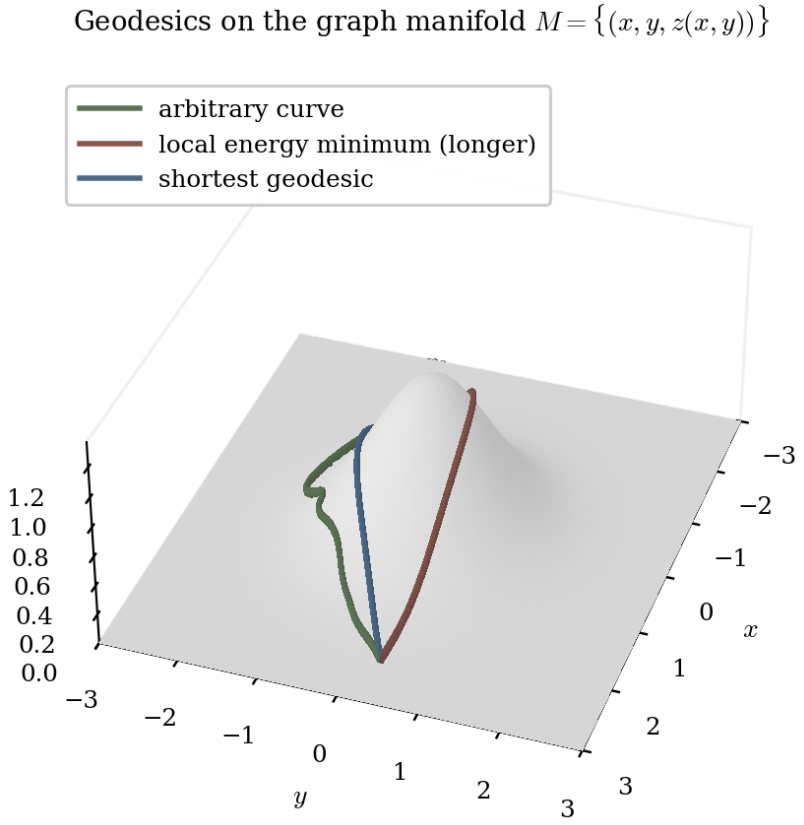


FIGURE 3. Numerical geodesics on a smooth bump surface between x_0 and x_1. The green curve is an arbitrary path. Energy minimization from opposite sides of the bump gives a longer local minimum (red) and the shortest path found (blue), illustrating distinct local minimizers with the same endpoints.

4. Construction of the metrics

4.1. Fisher-Rao

The Fisher–Rao metric is obtained by pulling back the L^2 metric on $\mathcal{P}_+(E)$ through the map $p \mapsto 2\sqrt{p}$. For a more detailed account of this construction, we refer the reader to [1]. For the square-root and spherical representation on smooth density spaces, see Khesin, Lenells, Misiołek, and Preston [13, Sections 3.1–3.3], in the compact-manifold setting.

4.1.1. Fisher–Rao as the pullback of the L^2 inner product

Positive densities live naturally in $L^1(E)$, which carries no preferred inner product. The square-root map

$$\psi(p) = 2\sqrt{p}$$

sends them into $L^2(E)$, with

$$\langle f, g \rangle_{L^2} = \int_E f\, g\, dx.$$

In the finite discrete case, ψ is a smooth immersion of the open simplex onto the positive orthant of the sphere of radius 2, and the pullback of the round metric is exact. Figure 4 makes this identification visible: the Fisher–Rao geometry on the simplex is the round geometry of the sphere, read back through ψ. In the continuous case we keep the same formulas on the directions for which they make sense. Differentiating gives

$$d\psi_p(u) = \frac{u}{\sqrt{p}},$$

defined whenever $u/\sqrt{p} \in L^2(E)$, and therefore

$$g_p^{\mathrm{FR}}(u, w) = \langle d\psi_p(u),\, d\psi_p(w) \rangle_{L^2} = \int_E \frac{u(x)\, w(x)}{p(x)}\, dx.$$

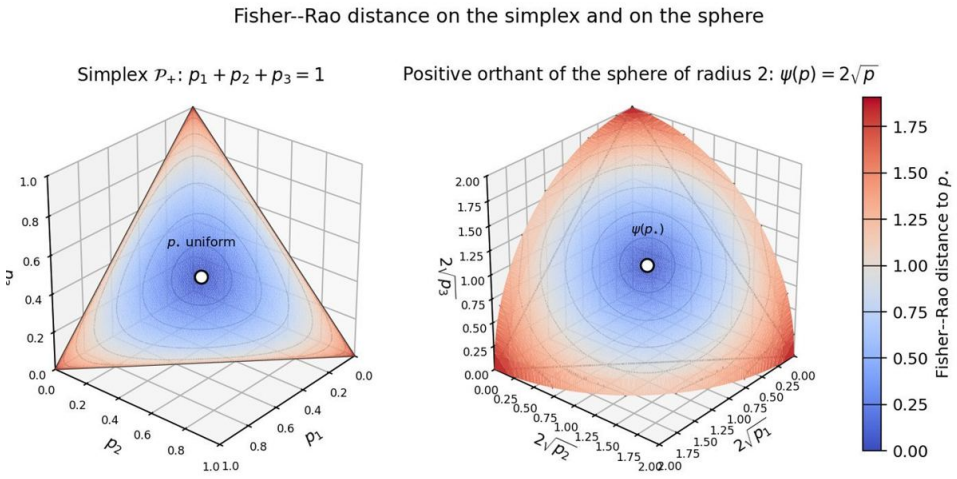


FIGURE 4. The probability simplex $p_1 + p_2 + p_3 = 1$ and its image under $p \mapsto 2\sqrt{p}$ on the positive octant of the sphere of radius 2. Colors and contours show Fisher–Rao distance from $p_\star = (1/3, 1/3, 1/3)$, illustrating the spherical geometry induced by the square-root map.

The admissible variations at p are those u with

$$\int_E u\, dx = 0 \qquad \text{and} \qquad \int_E \frac{u^2}{p}\, dx < \infty.$$

The first condition is conservation of mass; the second is finiteness of the Fisher–Rao norm. Equipped with g_p^{FR}, this space is a Hilbert space. We write it $T_p\mathcal{P}_+(E)$ by analogy with a tangent space, without claiming a Banach manifold structure on $\mathcal{P}_+(E)$.

On a parametric family $\theta \mapsto p_\theta \in \mathcal{P}_+(E)$, the same construction pulls back to Θ:

$$g^{\mathrm{FR}}_\theta(d\theta_1, d\theta_2) = g^{\mathrm{FR}}_{p_\theta}(dp_\theta(d\theta_1), dp_\theta(d\theta_2)).$$

In coordinates,

$$G^{\mathrm{FR}}_{ij}(\theta) = \int_E \frac{\partial_{\theta_i} p(\theta, x)\, \partial_{\theta_j} p(\theta, x)}{p(\theta, x)}\, dx = \mathbb{E}_{p_\theta}\Big[\partial_{\theta_i} \log p_\theta\, \partial_{\theta_j} \log p_\theta\Big],$$

which is the classical Fisher information matrix of Fisher [2] and Rao [4]. The matrix $G^{\mathrm{FR}}(\theta)$ defines a Riemannian metric on Θ as soon as it is smooth and positive definite, that is, as soon as dp_θ is injective.

4.1.2. **Analysis of the metric**

For $u \in T_p\mathcal{P}_+(E)$,

$$\|u\|^2_{p,\mathrm{FR}} = g^{\mathrm{FR}}_p(u, u) = \int_E \frac{u(x)^2}{p(x)}\, dx = \mathbb{E}_{x\sim p}\left[\left(\frac{u(x)}{p(x)}\right)^2\right].$$

What enters the cost is the relative variation u/p, not the absolute size of u. The same density increment is expensive where p is small and cheap where p is concentrated. The left column of Figure 1 makes this visible on a Gaussian: an identical mass-preserving dipole costs far more in the tail than near the mode.

The metric is also invariant under reparametrization of the observation space. If $\phi : E \to E'$ is a diffeomorphism and densities are transported by ϕ, the inner products g^{FR} are unchanged. Only the way densities differ from one another matters, not the coordinates chosen on E. This invariance characterizes the Fisher–Rao metric in several infinite-dimensional settings; see Bauer, Bruveris, and Michor [3].

On the parameter space, the same geometry is intrinsic in the usual Riemannian sense: under a change of coordinates $\theta \mapsto \widetilde{\theta}$, the matrix G^{FR} transforms as a covariant 2-tensor. Distances and geodesics on $(\Theta, g^{\mathrm{FR}})$ therefore do not depend on the chosen parametrization of the family.

4.2. **Wasserstein-Otto**

4.2.1. **Kantorovich formulation of quadratic optimal transport**

We begin directly from the Kantorovich formulation of optimal transport [7]. The general theory can be developed under substantially weaker assumptions, for instance for Borel probability measures on suitable metric spaces and for fairly general cost functions. Here we deliberately remain in the setting already used throughout this paper: E is a finite-dimensional real vector space equipped with an inner product, and the probability measures under consideration admit strictly positive densities with respect to Lebesgue measure.

Because the quadratic transport cost involves the distance travelled by the mass, we restrict, when necessary, to densities with finite second moment and write

$$\mathcal{P}_{2,+}(E) = \left\{p \in \mathcal{P}_+(E) : \int_E \|x\|^2 p(x)\, dx < \infty\right\}.$$

Let $p, q \in \mathcal{P}_{2,+}(E)$, and denote the corresponding probability measures by

$$d\mu(x) = p(x)\,dx, \qquad d\nu(y) = q(y)\,dy.$$

A transport plan, or coupling, between p and q is a probability measure π on $E \times E$ whose first and second marginals are μ and ν. Equivalently, if

$$\mathrm{pr}_1(x, y) = x, \qquad \mathrm{pr}_2(x, y) = y,$$

then

$$(\mathrm{pr}_1)_{\#}\pi = \mu, \qquad (\mathrm{pr}_2)_{\#}\pi = \nu.$$

We denote the set of such couplings by

$$\Pi(p, q) = \{\pi \in \mathcal{P}(E \times E) : (\mathrm{pr}_1)_{\#}\pi = p(x)\,dx,\ (\mathrm{pr}_2)_{\#}\pi = q(y)\,dy\}\,.$$

The quadratic Wasserstein distance is then defined by the Kantorovich minimization problem

$$\boxed{W_2^2(p, q) = \inf_{\pi \in \Pi(p,q)} \int_{E\times E} \|x - y\|^2\, d\pi(x, y).}$$

The interpretation is direct. The quantity $d\pi(x, y)$ describes how much probability mass is coupled between a position x in the first density and a position y in the second one, while $\|x - y\|^2$ measures the cost of moving that mass through the geometry of E. The marginal constraints ensure that all the mass initially distributed according to p is used and that the final distribution is exactly q. Figure 5 illustrates this structure on $\mathbb{R}$: the joint density of a coupling occupies the product space, its marginals recover p and q, and the quadratic cost vanishes on the diagonal $y = x$.

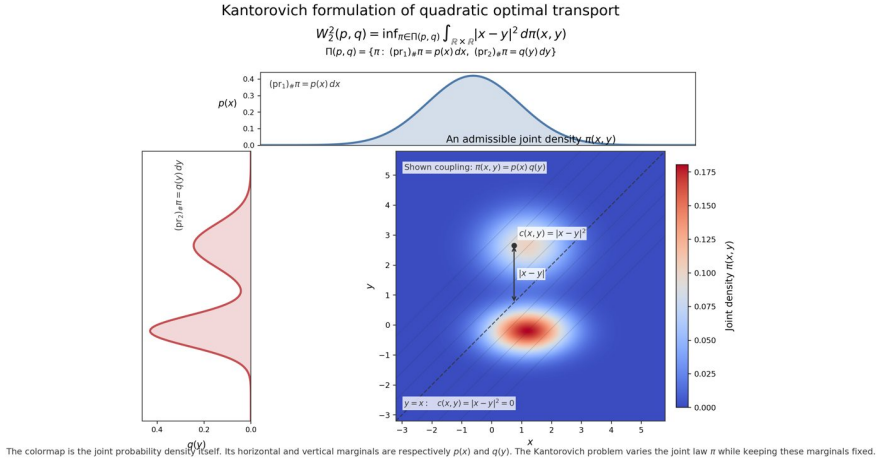


FIGURE 5. A transport coupling and its cost. The independent coupling $\pi(x, y) = p(x)q(y)$ has marginals p and q. The quadratic cost $|x-y|^2$ vanishes on the diagonal and increases with displacement; minimizing its integral over all admissible couplings gives $W_2^2(p, q)$.

In the present finite-dimensional setting, and for p, q with finite second moments, this gives a finite distance and the infimum is attained under the

standard hypotheses. We shall not need here the more general measure-theoretic formulation of the Kantorovich problem. What matters for the comparison with Fisher–Rao is that the Wasserstein cost explicitly depends on the geometry of the underlying space through the factor $\|x-y\|^2$. In particular, changing the spatial separation between two pieces of mass changes the transport cost even if the corresponding variations of probability density have the same local amplitude.

The static formula above compares two complete probability distributions. To obtain the infinitesimal Riemannian structure used by Otto, we next describe a differentiable path of densities $t \mapsto p_t$ through the velocity field that transports its mass. This leads to the continuity equation and, after minimizing the kinetic energy over all velocity fields producing the same tangent variation, to the Wasserstein–Otto metric.

4.2.2. **Construction of the Wasserstein–Otto metric**

The Kantorovich formulation is static: it assigns a transport cost between two complete probability distributions. To obtain a Riemannian metric, we now examine the same geometry at the infinitesimal scale. If a density p changes with instantaneous variation u, we ask how much spatial motion of the probability mass is required to produce that variation.

On the moving density and conservation of mass.

Let $t \mapsto p_t$ be a differentiable path of strictly positive probability densities, with $p_0 = p$. We describe the motion of the mass by a time-dependent velocity field v_t and its flow

$$\frac{d}{dt}\Phi_t(x) = v_t(\Phi_t(x)), \qquad \Phi_0(x) = x.$$

For a measurable region $\Omega_0 \subset E$, let

$$\Omega_t = \Phi_t(\Omega_0).$$

If the density is transported by the flow, then the amount of probability mass contained in the moving region is conserved:

$$\int_{\Omega_t} p_t(y)\, dy = \int_{\Omega_0} p_0(x)\, dx.$$

This relation is required for every admissible initial region Ω_0. In particular, it is the local form of the push-forward relation

$$p_t(y)\, dy = (\Phi_t)_{\#}(p_0(x)\, dx).$$

Figure 6 illustrates the same identity at finite time: a material region and its particles are transported by a diffeomorphism, the density is updated by push-forward, and the probability mass of the region is preserved.

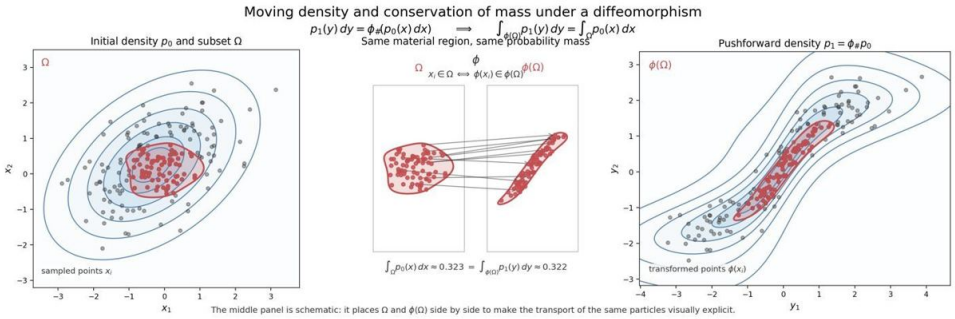


FIGURE 6. Conservation of probability under transport. The map ϕ carries the reference region Ω and its particles to $\phi(\Omega)$, while $p_1 = \phi_\# p_0$. The density changes with local expansion or compression, but the transported region retains its mass: $\int_{\phi(\Omega)} p_1\,dx = \int_\Omega p_0\,dx$.

Applying the Reynolds transport theorem to the moving domain Ω_t gives [6]

$$\frac{d}{dt}\int_{\Omega_t} p_t(y)\,dy = \int_{\Omega_t} \left(\partial_t p_t(y) + \operatorname{div}(p_t(y)v_t(y))\right) dy.$$

The left-hand side vanishes because the mass of every material region is constant. Therefore

$$\int_{\Omega_t} \left(\partial_t p_t + \operatorname{div}(p_t v_t)\right) dy = 0$$

for every transported region Ω_t. Since the flow is locally a diffeomorphism, such regions range over arbitrary local subsets of E. Hence

$$\boxed{\partial_t p_t + \operatorname{div}(p_t v_t) = 0.}$$

This is the continuity equation.

At $t = 0$, the tangent variation

$$u := \partial_t p_t|_{t=0}$$

is therefore represented by the instantaneous velocity field $v = v_0$ through

$$\boxed{u = -\operatorname{div}(pv).}$$

Integrating over E, under the usual decay or boundary assumptions, gives

$$\int_E u\,dx = 0,$$

which is exactly the first-order conservation of total probability mass.

The continuity equation is also the constraint appearing in the dynamic formulation of quadratic optimal transport. More precisely, the Benamou–Brenier formulation [8] writes

$$W_2^2(p_0, p_1) = \inf_{(p_t, v_t)} \int_0^1 \int_E p_t(x)\|v_t(x)\|^2\,dx\,dt,$$

where the infimum is taken over sufficiently regular paths satisfying

$$\partial_t p_t + \operatorname{div}(p_t v_t) = 0, \qquad p_{t=0} = p_0, \qquad p_{t=1} = p_1.$$

The kinetic quantity below is thus the infinitesimal energy density underlying the quadratic Wasserstein distance.

On the infinitesimal transport cost.

The quadratic Kantorovich cost measures squared spatial displacement. Along the flow,

$$\Phi_\varepsilon(x) = x + \varepsilon v(x) + o(\varepsilon),$$

so

$$\|\Phi_\varepsilon(x) - x\|^2 = \varepsilon^2 \|v(x)\|^2 + o(\varepsilon^2).$$

Averaging with respect to the density p gives the local quadratic cost

$$\varepsilon^2 \int_E \|v(x)\|^2 p(x)\, dx + o(\varepsilon^2).$$

This suggests the weighted kinetic norm

$$\|v\|_p^2 = \int_E \|v(x)\|^2 p(x)\, dx.$$

The tangent vector, however, is the density variation u, not the velocity field v. Moreover, the relation

$$u = -\operatorname{div}(pv)$$

does not determine v uniquely. Different particle motions may produce the same first-order variation of the density.

The Wasserstein–Otto norm is therefore defined by the least kinetic energy among all velocity fields that realize the same tangent variation:

$$\boxed{\|u\|_{p,\mathrm{WO}}^2 := \inf_{\substack{v \\ -\operatorname{div}(pv)=u}} \int_E \|v(x)\|^2 p(x)\, dx.}$$

Figure 7 illustrates this cost for a mass-preserving variation u: the minimizing velocity field $v = \nabla\phi$ moves the density, and the integrand $\|v\|^2 p$ is the local kinetic contribution.

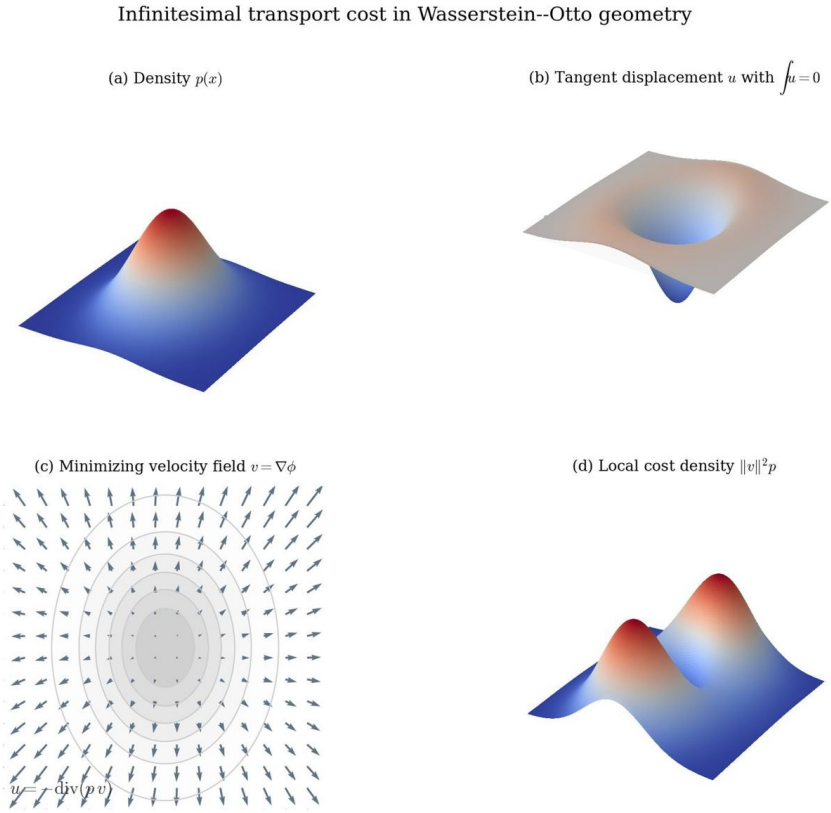


FIGURE 7. Infinitesimal Wasserstein transport cost. A tangent variation u with $\int u = 0$ is represented by the minimizing velocity field $v = \nabla\phi$ through $u = -\operatorname{div}(pv)$; the local kinetic cost is $\|v\|^2 p$, whose integral gives the squared Wasserstein–Otto norm.

Formally, the admissible tangent variations are thus

$$T_p^{\mathrm{WO}}\mathcal{P}_+(E) = \left\{ u : \int_E u\,dx = 0, \quad u = -\operatorname{div}(pv) \text{ for some } v \in L^2(E, p\,dx; E) \right\}.$$

At this stage, the geometry is already visible. A tangent variation u records only the net first-order redistribution of probability mass. It does not remember all particle trajectories that may realize it. The Wasserstein–Otto norm retains, among all such realizations, the one with the smallest kinetic energy.

The constraint does not determine the velocity field uniquely. The next subsection identifies geometrically the part of a velocity field that is relevant to the density variation and shows that the minimizer is necessarily a gradient field.

4.2.3. **Weighted Helmholtz decomposition and the minimal velocity field**

The continuity equation does not determine a unique velocity field. Indeed, if v realizes the tangent variation u,

$$u = -\operatorname{div}(pv),$$

then $v + w$ realizes the same variation whenever

$$\operatorname{div}(pw) = 0.$$

Such a field w may move points of the underlying space while remaining invisible at the level of the density, at least to first order. Figure 8 shows a concrete example: for the isotropic Gaussian $p = \mathcal{N}(0, I_2)$ and the rotational field $v(y) = (-y_2, y_1)$, one has $\operatorname{div} v = 0$ and $\langle \nabla p, v \rangle = 0$, hence $\operatorname{div}(pv) = 0$. The points circulate along the level sets of p, yet the density itself does not change. The Wasserstein–Otto construction selects, among all velocity fields producing the same density variation, the one with minimal kinetic energy.

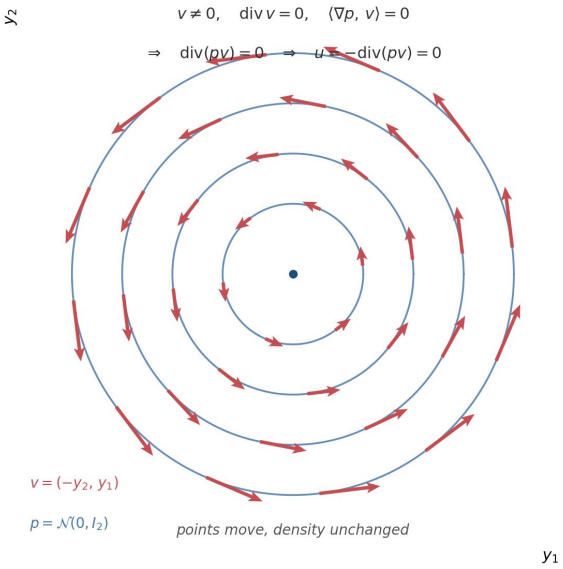


FIGURE 8. Rotation of the isotropic Gaussian $p = \mathcal{N}(0, I_2)$. The field $v(y) = (-y_2, y_1)$ is tangent to density levels and satisfies $\operatorname{div}(pv) = 0$. It belongs to K_p: particles move while the density remains unchanged.

To make this geometric, fix a strictly positive density p and consider the weighted Hilbert space

$$\mathcal{H}_p = L^2(E, p(x)\,dx; E),$$

with inner product

$$\langle v_1, v_2 \rangle_p = \int_E p(x)\, \langle v_1(x), v_2(x) \rangle\, dx.$$

We assume throughout this discussion sufficient regularity and either sufficient decay at infinity or boundary conditions eliminating boundary terms.

Let

$$\mathcal{G}_p = \overline{\{\nabla \varphi : \varphi \in C_c^\infty(E)\}}^{L^2(p)}$$

denote the closed subspace generated by gradient fields. Its orthogonal complement has a simple interpretation. A vector field $w \in \mathcal{H}_p$ belongs to $\mathcal{G}_p^\perp$

precisely when

$$\int_E p \, \langle w, \nabla \psi \rangle \, dx = 0 \qquad \text{for every admissible } \psi.$$

By integration by parts,

$$\int_E p \, \langle w, \nabla \psi \rangle \, dx = - \int_E \psi \, \operatorname{div}(pw) \, dx.$$

Hence, in the weak sense,

$$\mathcal{G}_p^{\perp} = \mathcal{K}_p, \qquad \mathcal{K}_p := \{ w \in \mathcal{H}_p : \operatorname{div}(pw) = 0 \}.$$

We therefore obtain the weighted Helmholtz decomposition [5]

$$\boxed{\mathcal{H}_p = \mathcal{G}_p \oplus^{\perp} \mathcal{K}_p.}$$

Under the usual regularity assumptions, the projection onto $\mathcal{G}_p$ is represented by a gradient. Thus any admissible velocity field may be written

$$\boxed{v = \nabla \phi + w, \qquad \operatorname{div}(pw) = 0, \qquad \langle \nabla \phi, w \rangle_p = 0.}$$

The two components have very different meanings. The gradient part changes the density, while the weighted divergence-free part is invisible to the continuity equation:

$$- \operatorname{div}(pv) = - \operatorname{div}(p \nabla \phi).$$

Consequently, if v realizes a prescribed tangent vector u, then $\nabla \phi$ realizes the same u. Orthogonality gives the Pythagorean identity

$$\|v\|_p^2 = \|\nabla \phi\|_p^2 + \|w\|_p^2.$$

The additional component w can therefore only increase the kinetic energy without changing the induced density variation. The minimizing representative is obtained by setting $w = 0$:

$$v_{\min} = \nabla \phi_u,$$

where ϕ_u solves the weighted Poisson equation

$$\boxed{- \operatorname{div}(p \nabla \phi_u) = u.}$$

It follows that

$$\boxed{\|u\|_{p,\mathrm{WO}}^2 = \inf_{v : \, - \operatorname{div}(pv) = u} \int_E p \|v\|^2 \, dx = \int_E p \|\nabla \phi_u\|^2 \, dx.}$$

Figure 9 shows the same decomposition in coordinates: a general velocity field splits into a gradient part that changes the density and a weighted divergence-free part that does not.

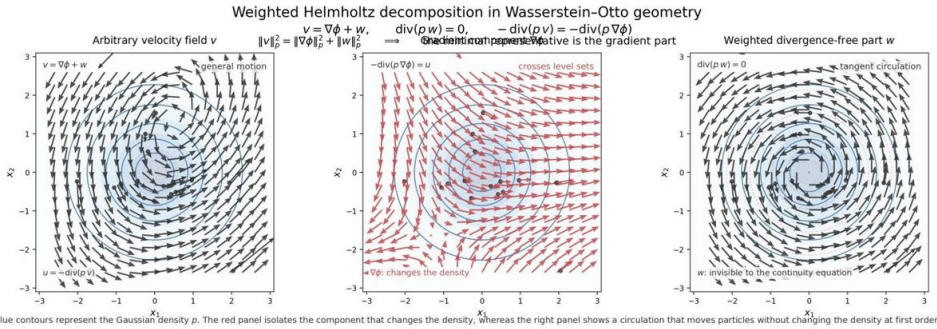


FIGURE 9. Weighted Helmholtz decomposition $v = \nabla\phi + w$: prescribed velocity (left), gradient component (middle), and weighted divergence-free component (right), with $\operatorname{div}(pw) = 0$. The components are orthogonal in $L^2(p)$, so $\int p\|v\|^2 = \int p\|\nabla\phi\|^2 + \int p\|w\|^2$. Only the gradient component contributes to the Wasserstein–Otto norm of $-\operatorname{div}(pv)$.

This also gives a quotient interpretation. At a fixed density p, two velocity fields represent the same tangent variation whenever they differ by an element of $\mathcal{K}_p$. Formally,

$$T_p\mathcal{P}_+(E) \simeq \mathcal{H}_p/\mathcal{K}_p,$$

and the gradient field $\nabla\phi_u$ is the canonical representative of minimal $L^2(p)$-norm in the corresponding equivalence class. In this sense, the Wasserstein tangent geometry keeps the part of a velocity field that actually transports the density and discards weighted divergence-free circulations.

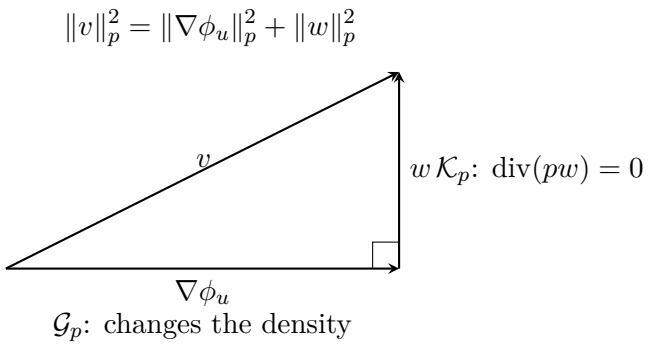


FIGURE 10. Weighted Helmholtz decomposition in $L^2(E, p\,dx; E)$. The gradient component $\nabla\phi_u$ produces the density variation u, whereas the orthogonal component w satisfies $\operatorname{div}(pw) = 0$ and is invisible to the continuity equation. Removing w leaves the same tangent variation and strictly decreases the kinetic energy unless $w = 0$.

A variational reading. The same conclusion follows directly from the constrained minimization problem. Introduce a Lagrange multiplier ϕ and consider

$$\mathcal{L}(v,\phi) = \frac{1}{2}\int_E p\|v\|^2\,dx - \int_E \phi\,(-\operatorname{div}(pv) - u)\,dx.$$

After integration by parts,

$$\mathcal{L}(v,\phi) = \frac{1}{2}\int_E p\|v\|^2\,dx - \int_E p\,\langle\nabla\phi, v\rangle\,dx + \int_E \phi u\,dx.$$

Stationarity with respect to v gives

$$v = \nabla\phi,$$

and the constraint then yields

$$-\operatorname{div}(p\nabla\phi) = u.$$

Thus the Euler–Lagrange condition and the weighted Helmholtz decomposition express the same geometry: the Wasserstein–Otto norm is obtained by orthogonally projecting an arbitrary admissible velocity field onto the gradient subspace.

The Wasserstein–Otto metric. We can now complete the construction. For tangent variations u, w, let ϕ_u and ϕ_w be the potentials of their minimal velocity fields, that is,

$$-\operatorname{div}(p\nabla\phi_u) = u, \qquad -\operatorname{div}(p\nabla\phi_w) = w.$$

The Wasserstein–Otto inner product is defined by the weighted $L^2(p)$ inner product of these minimal representatives:

$$\boxed{g_p^{\mathrm{WO}}(u,w) = \int_E p(x)\,\langle\nabla\phi_u(x), \nabla\phi_w(x)\rangle\,dx.}$$

In particular,

$$g_p^{\mathrm{WO}}(u,u) = \|u\|_{p,\mathrm{WO}}^2 = \int_E p(x)\|\nabla\phi_u(x)\|^2\,dx.$$

This is the formal Riemannian metric associated with the quadratic Wasserstein distance [5, 9].

Pullback to a parametric family. Let

$$p : \Theta \longrightarrow \mathcal{P}_+(E), \qquad \theta \longmapsto p_\theta,$$

be a smooth finite-dimensional family. A parameter displacement $d\theta \in T_\theta\Theta$ induces the density variation

$$u_{d\theta} = dp_\theta(d\theta).$$

Let $\phi_{d\theta}$ be defined, up to an additive constant, by

$$-\operatorname{div}(p_\theta\nabla\phi_{d\theta}) = dp_\theta(d\theta).$$

Pulling back the Wasserstein–Otto metric to the parameter space gives

$$\boxed{g_\theta^{\mathrm{WO}}(d\theta_1, d\theta_2) = \int_E p_\theta(x)\,\langle\nabla\phi_{d\theta_1}(x), \nabla\phi_{d\theta_2}(x)\rangle\,dx.}$$

In coordinates

$$\theta = (\theta^1, \dots, \theta^k),$$

let ϕ_i solve

$$-\operatorname{div}(p_\theta \nabla \phi_i) = \partial_{\theta^i} p_\theta.$$

Then the matrix of the pullback metric is

$$\boxed{G_{ij}^{\mathrm{WO}}(\theta) = \int_E p_\theta(x)\, \langle \nabla \phi_i(x), \nabla \phi_j(x) \rangle \, dx.}$$

This is the expression that will be used throughout the Atlas. The construction is therefore complete: a parameter variation first produces a density variation, this variation determines a minimal transport field through a weighted elliptic equation, and the metric measures the weighted kinetic pairing of the resulting fields. This parametric Wasserstein construction is developed by Chen and Li [12, Section 3.1].

4.2.4. **Analysis of the metric**

The construction above already reveals the main geometric feature of the Wasserstein–Otto metric. Fisher–Rao measures a density variation directly, whereas Wasserstein–Otto measures the spatial motion required to realize that variation. The geometry of the observation space E therefore enters the metric explicitly.

A nonlocal metric. For Fisher–Rao, the cost of a tangent variation is read pointwise from u/p. For Wasserstein–Otto, the variation u first has to be converted into its minimal velocity field by solving

$$-\operatorname{div}(p \nabla \phi_u) = u.$$

The value of $\nabla \phi_u$ at one point generally depends on the variation of the density over the whole domain. The metric is therefore nonlocal: a local change in u can modify the transport field away from the point where the change was introduced.

Equivalently, introducing formally the weighted elliptic operator

$$\mathcal{L}_p \phi = -\operatorname{div}(p \nabla \phi),$$

one may write, on zero-mass variations and modulo additive constants,

$$\phi_u = \mathcal{L}_p^{-1} u.$$

This gives the formal identity

$$g_p^{\mathrm{WO}}(u, w) = \int_E \phi_u(x) w(x)\, dx = \int_E u(x) \phi_w(x)\, dx,$$

after integration by parts. In this sense, the Wasserstein–Otto norm has the character of a weighted negative-order Sobolev norm [5, 9].

Dependence on the geometry of E. The metric depends on the inner product of the observation space through the kinetic quantity

$$\int_E p(x) \|\nabla \phi_u(x)\|^2 \, dx.$$

This makes Wasserstein geometry sensitive to spatial scale and spatial distance.

For example, if

$$T(x) = Qx + b,$$

with Q orthogonal, then T is an isometry of E. Transporting both densities by T leaves the quadratic transport cost unchanged, and the same invariance is inherited infinitesimally by the Wasserstein–Otto metric.

A general diffeomorphism of the observation space does not have this property if the original metric on E is kept fixed. A dilation already shows the difference. If

$$S_a(x) = ax, \qquad a \neq 0,$$

then

$$W_2\big((S_a)_{\#}p, (S_a)_{\#}q\big) = |a|\, W_2(p, q),$$

so the squared infinitesimal cost scales by a^2. The physical scale of the observation space is therefore part of the Wasserstein geometry.

This contrasts sharply with Fisher–Rao, whose metric is invariant under smooth reparametrizations of the observation space. The distinction is not a failure of coordinate invariance on the parameter manifold Θ: once either metric has been pulled back to Θ, its matrix transforms as a covariant 2-tensor under a change of parameter coordinates. The difference lies in what the ambient metric regards as geometrically meaningful before the pullback. Translation as the elementary transport direction. The simplest example is a pure translation. Let $a \in E$ and consider

$$p_t(x) = p(x - ta).$$

Then

$$u(x) = \partial_t p_t(x)|_{t=0} = -\langle a, \nabla p(x)\rangle = -\operatorname{div}(p(x)a).$$

The constant field

$$v(x) = a$$

realizes this variation. Since

$$a = \nabla\langle a, x\rangle,$$

it already belongs to the gradient subspace and is therefore the minimal representative. Hence

$$\|u\|^2_{p,\mathrm{WO}} = \int_E p(x)\|a\|^2\, dx = \|a\|^2.$$

The infinitesimal cost of translating a probability distribution is thus exactly the squared Euclidean speed of the translation, independently of the shape of the density.

This elementary case already anticipates a recurrent feature of the Atlas. For location parameters, Wasserstein–Otto often retains the Euclidean geometry of the underlying space, while Fisher–Rao weights the same parameter displacement according to the relative change it induces in the density.

4.3. **Recapitulation**

We now summarize the two ambient geometries before applying them to the parametric families of the Atlas. Let $p \in \mathcal{P}_+(E)$ and let u be an admissible tangent variation, so that

$$\int_E u(x)\, dx = 0.$$

For Fisher–Rao, the squared norm is

$$\boxed{\|u\|_{p,\mathrm{FR}}^2 = \int_E \frac{u(x)^2}{p(x)}\,dx.}$$

It acts directly on the density variation. Equivalently,

$$\|u\|_{p,\mathrm{FR}}^2 = \mathbb{E}_{x\sim p}\left[\left(\frac{u(x)}{p(x)}\right)^2\right],$$

so Fisher–Rao measures the size of the *relative* statistical variation u/p.

For Wasserstein–Otto, the same density variation is represented through a velocity field. Among all v satisfying

$$u = -\operatorname{div}(pv),$$

the metric selects the field of minimal kinetic energy. The weighted Helmholtz decomposition shows that this minimizer is a gradient field,

$$\boxed{v_u = \nabla\phi_u,}$$

where the potential ϕ_u, defined up to an additive constant, solves

$$\boxed{-\operatorname{div}(p\nabla\phi_u) = u.}$$

Hence

$$\boxed{\|u\|_{p,\mathrm{WO}}^2 = \int_E p(x)\|\nabla\phi_u(x)\|^2\,dx.}$$

Equivalently,

$$\|u\|_{p,\mathrm{WO}}^2 = \inf_{\substack{v\\ -\operatorname{div}(pv)=u}} \int_E p(x)\|v(x)\|^2\,dx.$$

This is the infinitesimal kinetic structure associated with the quadratic Wasserstein distance [5, 9].

The two metrics therefore assign a very different meaning to the same tangent variation. Figure 11 shows this on a one-dimensional example: the same mass-preserving u is read either through the relative integrand u^2/p or through the kinetic density $\|v\|^2 p$ of the minimal transport field.

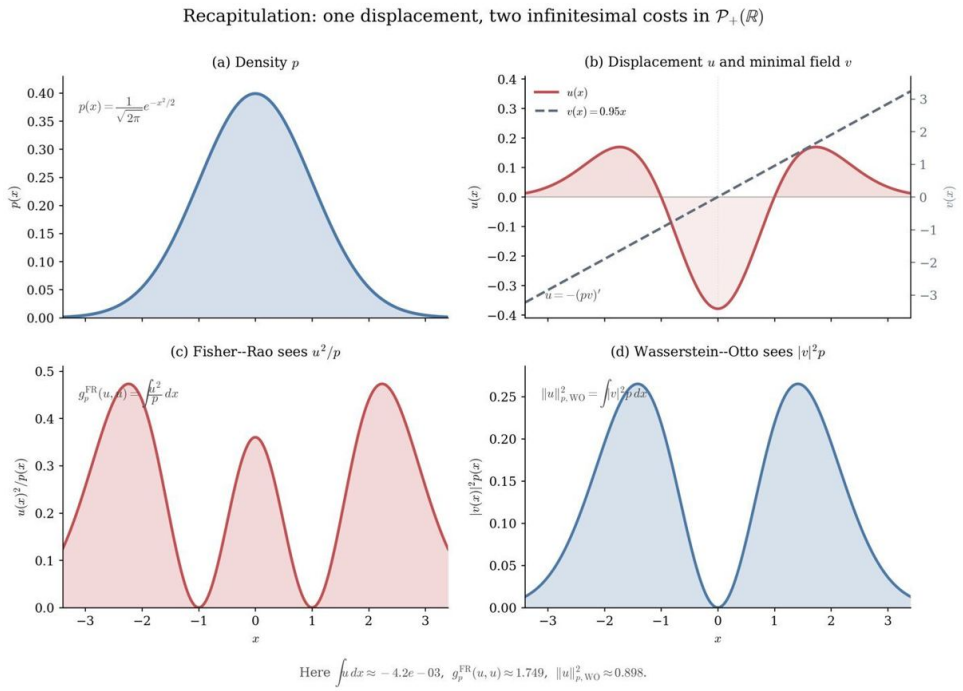


FIGURE 11. One tangent displacement u viewed through the two ambient geometries. Fisher–Rao measures the local relative variation u^2/p, whereas Wasserstein–Otto represents u through the minimal field v satisfying $u = -(pv)'$ and measures the kinetic density $\|v\|^2 p$.

Fisher–Rao: statistical variation without spatial geometry. Fisher–Rao compares how probability density changes relative to its current value. Its cost is pointwise: the same absolute variation is more expensive where p is small and less expensive where p is large. More fundamentally, Fisher–Rao does not require a notion of distance between points of the observation space. If the observation space is smoothly reparameterized and the densities are pushed forward accordingly, the metric is unchanged. In this sense, Fisher–Rao is agnostic to the particular coordinate system used on E; its geometry is attached to the statistical variation itself rather than to distances in the sample space. This diffeomorphism invariance is a characteristic property of the Fisher–Rao metric in broad classes of smooth density manifolds [3].
Wasserstein–Otto: transport through the geometry of E. Wasserstein–Otto instead asks how probability mass must move in order to produce the variation u. The norm depends explicitly on the inner product of E through the kinetic energy

$$\int_E p\|\nabla\phi_u\|^2\,dx,$$

and the potential ϕ_u is obtained by solving a weighted elliptic equation on E. The metric is therefore sensitive to spatial distance, scale, and direction. Euclidean isometries preserve the transport geometry, whereas a general

deformation of the observation space changes the metric unless the geometry of E is transformed with it [5, 9].

The contrast can be summarized schematically as

Fisher–Rao	Wasserstein–Otto
$\frac{u}{p}$	$u = -\operatorname{div}(p\nabla\phi_u)$
relative density change	minimal motion of probability mass
local	nonlocal
independent of the geometry of E	depends on the geometry of E
invariant under smooth reparameterizations of E	preserved by isometries of E

After pullback to a parametric family

$$\theta \longmapsto p_\theta,$$

both constructions define ordinary Riemannian metrics on the same parameter space. The statistical family is unchanged; what changes is the geometric notion of an infinitesimal displacement. The examples that follow make this difference explicit: the same parameter direction can be short, long, flat, curved, or even geometrically decoupled depending on whether it is measured through relative statistical variation or through transport in E.

5. The one-dimensional bridge

On the real line, the continuity equation can be integrated explicitly. This gives a direct expression for the Wasserstein–Otto metric in terms of primitives of density variations, and then in terms of quantiles. Recall that

$$\mathcal{P}_{2,+}(E) = \left\{ p \in \mathcal{P}_+(E) : \int_E \|x\|^2 p(x)\,dx < \infty \right\}$$

is the space of strictly positive probability densities with finite second moment. In this section, $E = \mathbb{R}$, so the moment condition is $\int_{\mathbb{R}} x^2 p(x)\,dx < \infty$.

5.1. From a density variation to a flux

Let $p \in \mathcal{P}_{2,+}(\mathbb{R})$, and let u, w be integrable density variations with zero total mass. Define their primitives by

$$U(x) = \int_{-\infty}^{x} u(s)\,ds, \qquad W(x) = \int_{-\infty}^{x} w(s)\,ds.$$

The one-dimensional continuity equation for u is

$$u(x) = -\frac{d}{dx}(p(x)v_u(x)).$$

Imposing vanishing flux at infinity and integrating from $-\infty$ to x gives

$$U(x) = -p(x)v_u(x), \qquad v_u(x) = -\frac{U(x)}{p(x)}.$$

Likewise, $v_w = -W/p$. Substitution into the kinetic inner product yields

$$\boxed{g_p^{\mathrm{WO}}(u,w) = \int_{\mathbb{R}} p(x) v_u(x) v_w(x)\,dx = \int_{\mathbb{R}} \frac{U(x)W(x)}{p(x)}\,dx.} \tag{1}$$

This formula requires $U/\sqrt{p}, W/\sqrt{p} \in L^2(\mathbb{R})$. Its parameter-space version is given in [12, Section 3.3, Proposition 5]. The zero-mass condition ensures $U(-\infty) = U(+\infty) = 0$, and likewise for W, but does not by itself imply finite kinetic energy.

The finite-energy velocity representing u is unique. Indeed, two such fields differ by c/p, since the derivative of their flux difference vanishes. But

$$\int_{\mathbb{R}} p(x) \left|\frac{c}{p(x)}\right|^2 dx = c^2 \int_{\mathbb{R}} \frac{dx}{p(x)}, \qquad \int_{\mathbb{R}} \frac{dx}{p(x)} = \infty.$$

For the last assertion, Cauchy–Schwarz on $[-R, R]$ gives

$$(2R)^2 \le \left(\int_{-R}^{R} p(x)\,dx\right)\left(\int_{-R}^{R} \frac{dx}{p(x)}\right) \le \int_{-R}^{R} \frac{dx}{p(x)}.$$

Thus finite energy forces $c = 0$. The field v_u is consequently the minimal representative used by the Wasserstein–Otto metric.

The last integral in (1) has the same algebraic form as the Fisher–Rao inner product, applied to the primitives rather than to the density variations. However, U and W need not have zero integral, or even belong to $L^1(\mathbb{R})$, so they need not be Fisher–Rao tangent vectors. We therefore do not identify this expression literally with $g_p^{\mathrm{FR}}(U, W)$ on its tangent space. Wasserstein–Otto measures the weighted size of an *integrated* density change: positive and negative variations far apart leave a nonzero flux between them.

5.2. Distribution functions and quantiles

Consider a family $p : \Theta \to \mathcal{P}_{2,+}(\mathbb{R})$, with $\Theta \subset \mathbb{R}^k$ open and $d\theta \in T_\theta\Theta \simeq \mathbb{R}^k$. Assume sufficient regularity to differentiate under the integrals below, that p_θ is continuous and strictly positive, and that the quantile map is differentiable into $L^2(0,1)$. Define

$$F_\theta(x) = \int_{-\infty}^{x} p_\theta(s)\,ds, \qquad Q_\theta(r) = F_\theta^{-1}(r), \qquad 0 < r < 1.$$

Strict positivity makes F_θ strictly increasing from $\mathbb{R}$ onto $(0,1)$. We distinguish differentiation with respect to the parameter θ from differentiation with respect to the spatial argument x. At fixed x,

$$d_\theta F_\theta(d\theta)(x) = \frac{d}{dt} F_{\theta + t\,d\theta}(x)\Big|_{t=0} = \int_{-\infty}^{x} d_\theta p_\theta(d\theta)(s)\,ds,$$

whereas the spatial differential, applied to $h \in \mathbb{R}$, is

$$d_x F_\theta(x)(h) = p_\theta(x) h.$$

Similarly, at fixed quantile level r, we write explicitly

$$d_\theta Q_\theta(d\theta)(r) = \frac{d}{dt} Q_{\theta + t\,d\theta}(r)\Big|_{t=0}.$$

Figure 12 gathers these objects for a Gaussian location variation: the density increment $u = d_\theta p_\theta(d\theta)$, its primitive $U = d_\theta F_\theta(d\theta)$, and the inverse relation between F_θ and Q_θ.

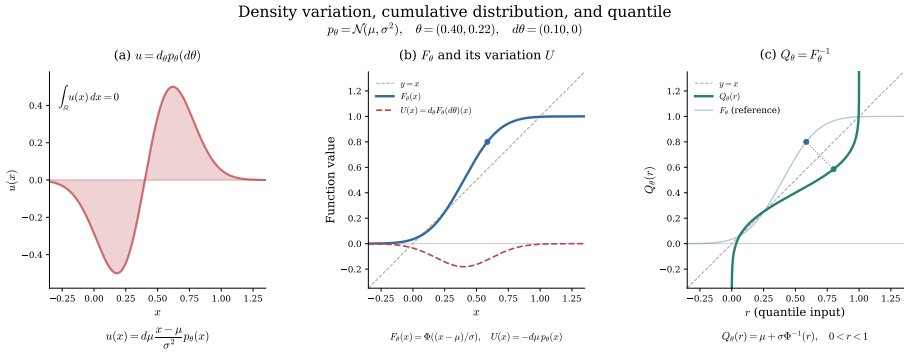


FIGURE 12. Density, cumulative distribution function, and quantile for a Gaussian with $\theta = (0.40, 0.22)$ and $d\theta = (0.10, 0)$. Left: density variation $u = d_\theta p_\theta(d\theta)$. Middle: F_θ and its variation $U = d_\theta F_\theta(d\theta)$. Right: $Q_\theta = F_\theta^{-1}$; reflection across $y = x$ displays the inverse relation between F_θ and Q_θ.

Differentiating the inverse relation. Fix a quantile level r. For every small t, we have

$$F_{\theta+td\theta}(Q_{\theta+td\theta}(r)) = r.$$

The chain rule has two terms: one from the parameter dependence of F_θ at fixed spatial argument, and one from the motion of that argument. Thus

$$\begin{aligned} 0 &= \frac{d}{dt} F_{\theta+td\theta}(Q_{\theta+td\theta}(r))\Big|_{t=0} \\ &= d_\theta F_\theta(d\theta)(Q_\theta(r)) + d_x F_\theta(Q_\theta(r))\big(d_\theta Q_\theta(d\theta)(r)\big) \\ &= d_\theta F_\theta(d\theta)(Q_\theta(r)) + p_\theta(Q_\theta(r))\, d_\theta Q_\theta(d\theta)(r). \end{aligned}$$

Here $d_\theta F_\theta$ acts on the parameter direction $d\theta$, while $d_x F_\theta$ acts on the scalar displacement of the quantile. Since $p_\theta(Q_\theta(r)) > 0$, solving for the latter gives

$$\boxed{d_\theta Q_\theta(d\theta)(r) = -\frac{d_\theta F_\theta(d\theta)(Q_\theta(r))}{p_\theta(Q_\theta(r))}.}$$

In particular, if $u = d_\theta p_\theta(d\theta)$, its primitive is $U = d_\theta F_\theta(d\theta)$, and this identity reads $d_\theta Q_\theta(d\theta)(r) = v_u(Q_\theta(r))$. The derivative of the quantile is the velocity of the probability mass labelled by r.

From the flux formula to the pullback metric. For $i = 1, 2$, put

$$u_i = d_\theta p_\theta(d\theta_i), \qquad U_i(x) = \int_{-\infty}^{x} u_i(s)\, ds = d_\theta F_\theta(d\theta_i)(x).$$

By the definition of the pullback and (1),

$$g_\theta^{\mathrm{WO}}(d\theta_1, d\theta_2) = g_{p_\theta}^{\mathrm{WO}}(u_1, u_2) = \int_{\mathbb{R}} \frac{U_1(x) U_2(x)}{p_\theta(x)}\, dx.$$

Now hold θ fixed and change the integration variable to $r = F_\theta(x)$. This is a spatial change of variable, so

$$dr = d_x F_\theta(x) = p_\theta(x)\,dx, \qquad x = Q_\theta(r), \qquad dx = \frac{dr}{p_\theta(Q_\theta(r))},$$

where $d_x F_\theta(x)$ is written as the one-form $p_\theta(x)\,dx$. Consequently,

$$\begin{aligned} g_\theta^{\mathrm{WO}}(d\theta_1, d\theta_2) &= \int_0^1 \frac{U_1(Q_\theta(r))U_2(Q_\theta(r))}{p_\theta(Q_\theta(r))^2}\,dr \\ &= \int_0^1 \left(-\frac{U_1(Q_\theta(r))}{p_\theta(Q_\theta(r))}\right)\left(-\frac{U_2(Q_\theta(r))}{p_\theta(Q_\theta(r))}\right)\,dr. \end{aligned}$$

Applying the quantile differential identity to each factor gives

$$\boxed{g_\theta^{\mathrm{WO}}(d\theta_1, d\theta_2) = \int_0^1 d_\theta Q_\theta(d\theta_1)(r)\, d_\theta Q_\theta(d\theta_2)(r)\,dr.} \tag{2}$$

The spaces and the maps involved. Equation (2) expresses the metric on the parameter space Θ. To keep track of the objects used to obtain it, consider the three maps

$$\theta \longmapsto p_\theta, \qquad \theta \longmapsto F_\theta, \qquad \theta \longmapsto Q_\theta \in L^2(0,1).$$

Here $p_\theta \in \mathcal{P}_{2,+}(\mathbb{R})$ is a density, F_θ is its cumulative distribution function on $\mathbb{R}$, and Q_θ is its inverse function on $(0,1)$. The finite second moment ensures that

$$\int_0^1 Q_\theta(r)^2\,dr = \int_{\mathbb{R}} x^2 p_\theta(x)\,dx < \infty.$$

For a parameter direction $d\theta \in T_\theta\Theta$, the corresponding differentials are, successively,

$$\begin{aligned} u = d_\theta p_\theta(d\theta) &\in T_{p_\theta}\mathcal{P}_{2,+}(\mathbb{R}), \\ U(x) = d_\theta F_\theta(d\theta)(x) &= \int_{-\infty}^x u(s)\,ds, \\ d_\theta Q_\theta(d\theta)(r) &= -\frac{U(Q_\theta(r))}{p_\theta(Q_\theta(r))}. \end{aligned}$$

The first line is a density variation of zero total mass, subject to the finite-energy assumptions above. The second is a variation of a cumulative distribution function; it is not generally a density tangent vector. The third is a variation of a quantile function, belonging to $L^2(0,1)$. We invert F_θ to obtain Q_θ; we do not invert the tangent variation U. The relation between their differentials follows from the chain rule already established.

The image of the parametric family in quantile space is

$$\{Q_\theta : \theta \in \Theta\} \subset L^2(0,1).$$

Assuming that the family is immersed, its admissible quantile velocities at Q_θ form the finite-dimensional subspace $\{d_\theta Q_\theta(d\theta) : d\theta \in T_\theta\Theta\}$ of $L^2(0,1)$. Thus equation (2) means precisely

$$g_\theta^{\mathrm{WO}}(d\theta_1, d\theta_2) = \langle d_\theta Q_\theta(d\theta_1), d_\theta Q_\theta(d\theta_2)\rangle_{L^2(0,1)}.$$

Figure 13 checks this equality on a Gaussian example: the Wasserstein–Otto pullback norm of $d\theta$ coincides with the $L^2(0,1)$ norm of $d_\theta Q_\theta(d\theta)$. The

pullback measures parameter tangents through their images under $d_\theta Q_\theta$. The quantile image $\{Q_\theta : \theta \in \Theta\}$ need not be affine.

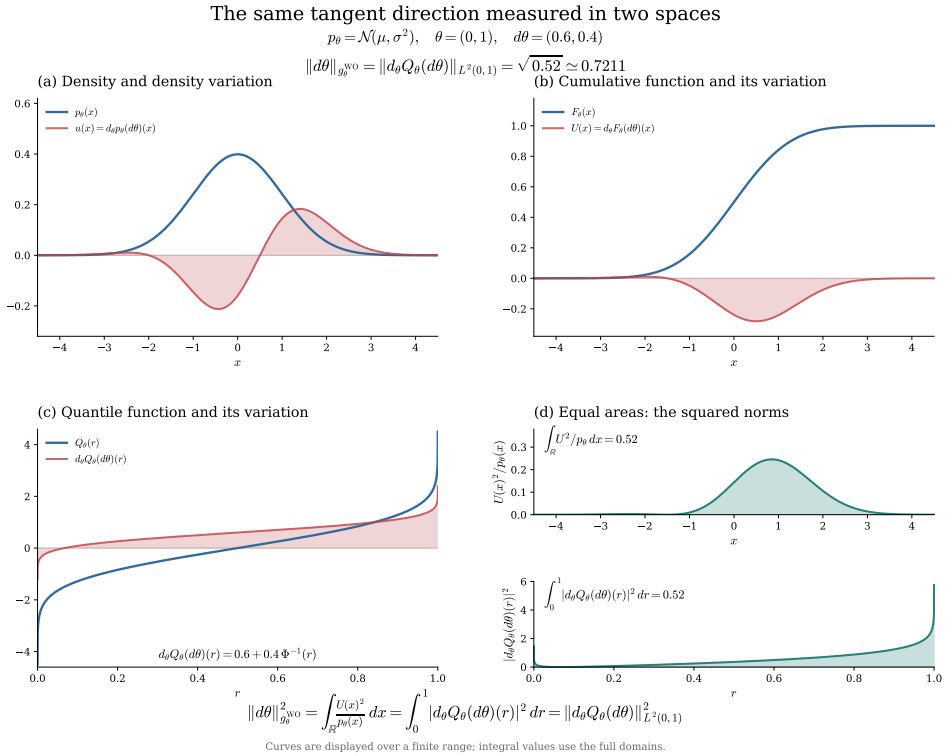


FIGURE 13. The quantile differential evaluates the Wasserstein–Otto pullback norm. For $p_\theta = \mathcal{N}(\mu, \sigma^2)$ at $\theta = (0, 1)$ and $d\theta = (0.6, 0.4)$, panels (a–c) show p_θ, F_θ, and Q_θ with their parameter variations u, U, and $d_\theta Q_\theta(d\theta)(r) = 0.6 + 0.4\Phi^{-1}(r)$, where Φ is the standard normal CDF. Panel (d) compares $\int_{\mathbb{R}} U^2/p_\theta\,dx$ and $\int_0^1 |d_\theta Q_\theta(d\theta)|^2\,dr$. Both full-domain integrals equal 0.52, giving $\|d\theta\|_{g_\theta^{\mathrm{WO}}} = \|d_\theta Q_\theta(d\theta)\|_{L^2(0,1)} = \sqrt{0.52}$.

Lengths and geodesics within the parametric family. Let $t \mapsto \theta(t)$ be a smooth parameter path, with $\theta(0) = \theta_0$ and $\theta(1) = \theta_1$. Its associated density, cumulative distribution function, and quantile function are explicitly $p_{\theta(t)}$, $F_{\theta(t)}$, and $Q_{\theta(t)}$. Their velocities satisfy

$$\partial_t p_{\theta(t)} = d_\theta p_{\theta(t)}(\dot\theta(t)), \qquad \partial_t Q_{\theta(t)} = d_\theta Q_{\theta(t)}(\dot\theta(t)).$$

By definition, the length of the parameter path in the pullback metric is

$$\begin{aligned} L_{g^{\mathrm{WO}}}(\theta) &= \int_0^1 \sqrt{g^{\mathrm{WO}}_{\theta(t)}(\dot\theta(t), \dot\theta(t))}\,dt \\ &= \int_0^1 \left(\int_0^1 \left| d_\theta Q_{\theta(t)}(\dot\theta(t))(r) \right|^2 dr \right)^{1/2} dt. \end{aligned}$$

The second line uses (2), with path parameter t and quantile label r. The intrinsic distance is

$$d_{g^{\mathrm{WO}}}(\theta_0, \theta_1) = \inf_{\substack{\theta(0)=\theta_0,\ \theta(1)=\theta_1 \\ \theta(t)\in\Theta}} L_{g^{\mathrm{WO}}}(\theta).$$

An intrinsic minimizing geodesic, when it exists, minimizes length among paths in the family. Its quantile image is constrained to $\{Q_\theta : \theta \in \Theta\}$ and need not be straight; the induced geometry may be curved.

Ambient distance and the straight quantile segment. We now distinguish this constrained problem from transport in the full space of probability distributions on $\mathbb{R}$ with finite second moment. For the same endpoint densities, the monotone optimal coupling gives [7]

$$W_2(p_{\theta_0}, p_{\theta_1}) = \left(\int_0^1 |Q_{\theta_1}(r) - Q_{\theta_0}(r)|^2 \, dr \right)^{1/2}.$$

Figure 14 shows how this integral is read from the cumulative distribution functions: equal ranks are matched, and the squared horizontal displacements accumulate into W_2^2. The straight segment between these two functions is

$$(1-t)Q_{\theta_0} + tQ_{\theta_1} \quad \text{in } L^2(0,1), \qquad 0 \le t \le 1.$$

It remains an increasing, square-integrable function and therefore represents a probability distribution with finite second moment at each t. It is the ambient Wasserstein geodesic [19], before imposing membership in the parametric family.

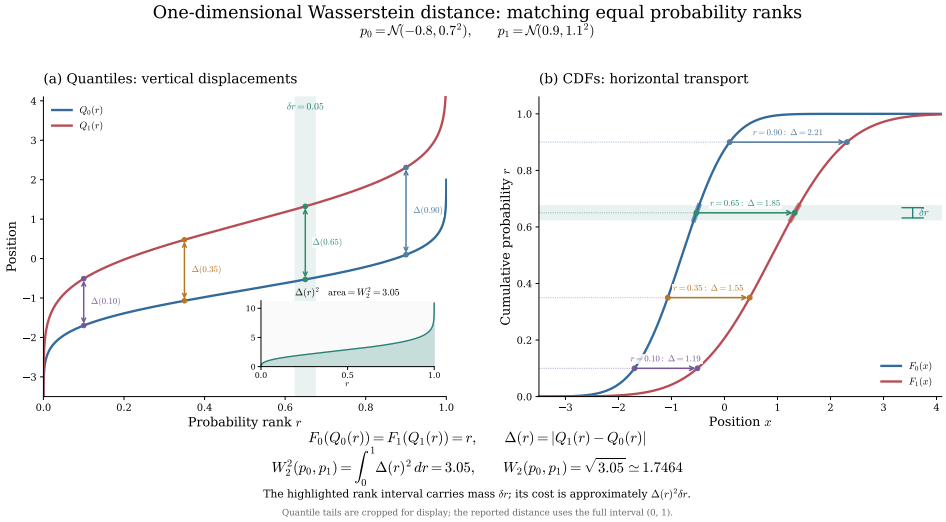


FIGURE 14. One-dimensional optimal transport matches equal probability ranks. Left: quantile gaps $Q_1(r) - Q_0(r)$. Right: the same displacements appear as horizontal arrows between F_0 and F_1. The highlighted rank interval $[0.625, 0.675]$ carries mass 0.05 in each density. Integrating squared gaps over $(0, 1)$ gives $W_2^2 = 3.05$ and $W_2 \approx 1.7464$; quantile tails are cropped only for display.

Indeed, for any parameter path, the fundamental theorem of calculus in $L^2(0,1)$ and the triangle inequality give

$$\begin{aligned} \|Q_{\theta_1} - Q_{\theta_0}\|_{L^2(0,1)} &= \left\| \int_0^1 d_\theta Q_{\theta(t)}(\dot\theta(t))\, dt \right\|_{L^2(0,1)} \\ &\le \int_0^1 \left\| d_\theta Q_{\theta(t)}(\dot\theta(t)) \right\|_{L^2(0,1)} dt \\ &= L_{g^{\mathrm{WO}}}(\theta). \end{aligned}$$

In the ambient quantile space, the straight segment attains this lower bound: its velocity is the constant function $Q_{\theta_1} - Q_{\theta_0}$, where "constant" refers to time t. In the parametric family, it is admissible only if there exists a smooth parameter path satisfying

$$\boxed{Q_{\theta(t)} = (1-t)Q_{\theta_0} + tQ_{\theta_1} \quad \text{for every } t \in [0,1].}$$

When such a path exists, it attains the ambient lower bound and is a minimizing geodesic of the pullback metric. Even then, the parameter path $\theta(t)$ need not be linear in the chosen coordinates. In general,

$$d_{g^{\mathrm{WO}}}(\theta_0, \theta_1) \ge W_2(p_{\theta_0}, p_{\theta_1}).$$

If an intrinsic minimizing path exists but no parameter path realizes the straight quantile segment, this inequality is strict.

Under this section's regularity assumptions, spatially differentiating the inverse of an interpolating quantile recovers an ambient density, whose membership in the family must still be established. Density and density variation are distinguished along a parameter path by

$$p_{\theta(t)}(x) = \partial_x F_{\theta(t)}(x), \qquad u(t,x) = \partial_t p_{\theta(t)}(x) = d_\theta p_{\theta(t)}(\dot\theta(t))(x).$$

An ambient variation need not lie in $d_\theta p_\theta(T_\theta\Theta)$, even at a density in the family; the last equality requires a parameter path.

5.3. **Location–dispersion families on the line**

Let p be a fixed, strictly positive probability density on $\mathbb{R}$ with finite second moment. Its location–dispersion family is

$$p_{\mu,\sigma}(x) = \frac{1}{\sigma}\, p\!\left(\frac{x-\mu}{\sigma}\right), \qquad (\mu,\sigma) \in \mathbb{R}\times(0,\infty).$$

The parameter μ translates the reference distribution and σ dilates it. Normalization follows by the substitution $x = \mu + \sigma s$:

$$\int_{\mathbb{R}} p_{\mu,\sigma}(x)\,dx = \int_{\mathbb{R}} p(s)\,ds = 1.$$

Choosing p to be a Gaussian or a Laplace density gives the corresponding Gaussian or Laplace location–dispersion family. More generally, the shape of p is fixed throughout the family; only its position and scale change. Unless p is centered with unit variance, μ and σ are not respectively the mean and standard deviation. Indeed,

$$\int_{\mathbb{R}} x p_{\mu,\sigma}(x)\,dx = \mu + \sigma\int_{\mathbb{R}} s p(s)\,ds,$$
$$\operatorname{Var}_{p_{\mu,\sigma}}(X) = \sigma^2\left(\int_{\mathbb{R}} s^2 p(s)\,ds - \left(\int_{\mathbb{R}} s p(s)\,ds\right)^2\right).$$

This normalization can always be imposed without changing the family of distributions. Indeed, if the original reference density has mean m and standard deviation $a > 0$, its centered, unit-variance version is

$$\bar p(x) = a\,p(m + ax).$$

The corresponding location–dispersion densities satisfy

$$\bar p_{\mu+\sigma m,\,\sigma a}(x) = \frac{1}{\sigma a}\,\bar p\!\left(\frac{x-\mu-\sigma m}{\sigma a}\right) = \frac{1}{\sigma}\,p\!\left(\frac{x-\mu}{\sigma}\right) = p_{\mu,\sigma}(x).$$

The change of parameters $(\mu,\sigma) \mapsto (\mu+\sigma m, \sigma a)$ is a bijection of the upper half-plane. Thus standardizing the reference changes only the parameterization, not the family. From now on, we denote this standardized reference density by p and assume

$$\int_{\mathbb{R}} x p(x)\,dx = 0, \qquad \int_{\mathbb{R}} x^2 p(x)\,dx = 1.$$

Consequently, μ is the mean of $p_{\mu,\sigma}$, σ its standard deviation, and σ^2 its variance. Figure 15 shows this on an asymmetric reference density: after centering and unit-variance standardization, the four panels only change location and dispersion.

The parameter domain is the upper half-plane. A tangent direction $(d\mu, d\sigma)$ belongs to $\mathbb{R}^2$; in particular, $d\sigma$ may be negative even though $\sigma > 0$.

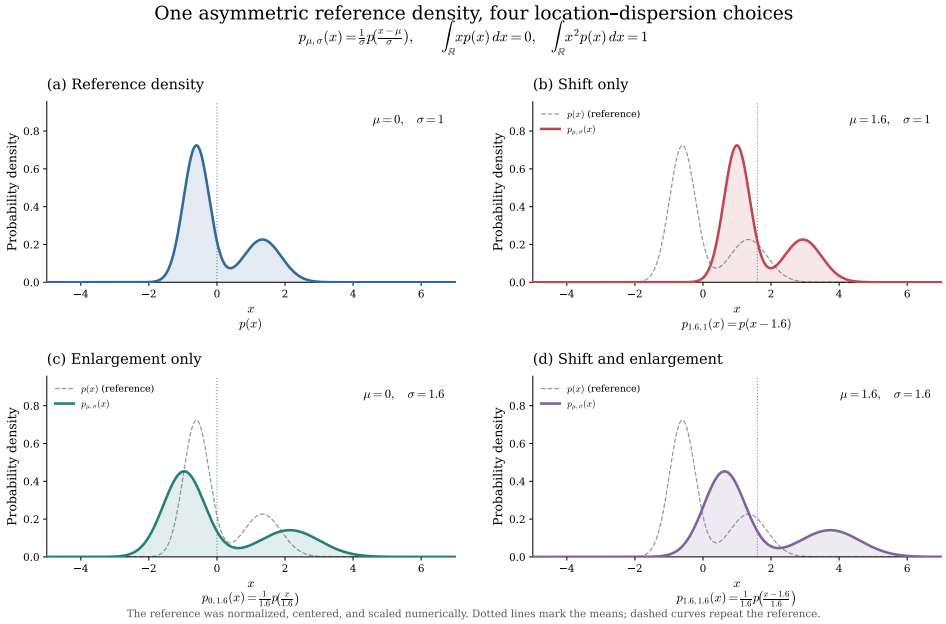


FIGURE 15. Location–dispersion transformations of an asymmetric density. The profile $q(x) = 0.8\exp(-(x + 0.8)^2/(2 \cdot 0.45^2)) + 0.25\exp(-(x - 1.5)^2/(2 \cdot 0.65^2))$ is numerically normalized and standardized to give p with mean zero and variance one. Panels show $p_{0,1}$, $p_{1.6,1}$, $p_{0,1.6}$, and $p_{1.6,1.6}$, where $p_{\mu,\sigma}(x) = \sigma^{-1}p((x - \mu)/\sigma)$. Translation and dilation preserve the reference shape and total mass; the resulting mean and standard deviation are μ and σ.

5.3.1. **Parameter derivatives of the density**

Assume that p is locally absolutely continuous, that the Fisher integrals below are finite, and that $p(s)$ and $sp(s)$ vanish as $s \to \pm\infty$. Derivatives involving p' are understood almost everywhere, which includes the Laplace density despite its corner at the origin. At fixed x, ordinary partial differentiation gives

$$\partial_\mu p_{\mu,\sigma}(x) = -\frac{1}{\sigma^2}\, p'\!\left(\frac{x-\mu}{\sigma}\right).$$

For the scale derivative, both the prefactor and the argument depend on σ, so

$$\begin{aligned}\partial_\sigma p_{\mu,\sigma}(x) &= -\frac{1}{\sigma^2}\, p\!\left(\frac{x-\mu}{\sigma}\right) - \frac{x-\mu}{\sigma^3}\, p'\!\left(\frac{x-\mu}{\sigma}\right) \\ &= -\frac{1}{\sigma^2}\left[p\!\left(\frac{x-\mu}{\sigma}\right) + \frac{x-\mu}{\sigma}\, p'\!\left(\frac{x-\mu}{\sigma}\right)\right].\end{aligned}$$

Thus the density variation associated with $(d\mu, d\sigma)$ is

$$\boxed{dp_{\mu,\sigma}(d\mu, d\sigma)(x) = -\frac{1}{\sigma^2}\left[p'\Big(\frac{x-\mu}{\sigma}\Big)\, d\mu + \Big(p\Big(\frac{x-\mu}{\sigma}\Big) + \frac{x-\mu}{\sigma}\, p'\Big(\frac{x-\mu}{\sigma}\Big)\Big)\, d\sigma\right].}$$

The decay assumptions ensure that both partial derivatives have zero integral: after substitution, the relevant integrals are $\int_{\mathbb{R}} p'(s)\, ds = 0$ and $\int_{\mathbb{R}} (p(s) + sp'(s))\, ds = [sp(s)]_{-\infty}^{+\infty} = 0$.

5.3.2. **Fisher–Rao pullback**

We first evaluate the squared norm of $(d\mu, d\sigma)$. Substituting the previous differential and the density into the Fisher–Rao formula gives

$$\begin{aligned}
&g^{\mathrm{FR}}_{\mu,\sigma}((d\mu, d\sigma), (d\mu, d\sigma)) \\
&\quad = \int_{\mathbb{R}} \frac{\big(dp_{\mu,\sigma}(d\mu, d\sigma)(x)\big)^2}{p_{\mu,\sigma}(x)}\, dx \\
&\quad = \frac{1}{\sigma^3} \int_{\mathbb{R}} \frac{\Big[p'\Big(\frac{x-\mu}{\sigma}\Big)\, d\mu + \Big(p\Big(\frac{x-\mu}{\sigma}\Big) + \frac{x-\mu}{\sigma} p'\Big(\frac{x-\mu}{\sigma}\Big)\Big)\, d\sigma\Big]^2}{p\Big(\frac{x-\mu}{\sigma}\Big)}\, dx \\
&\quad = \frac{1}{\sigma^2} \int_{\mathbb{R}} \frac{\big[p'(s) d\mu + (p(s) + sp'(s)) d\sigma\big]^2}{p(s)}\, ds.
\end{aligned}$$

The last line uses $x = \mu + \sigma s$ and $dx = \sigma\, ds$; s is only the integration variable for the reference density. Expanding the square leaves the three coefficients of the Fisher–Rao metric displayed in Table 1 of Section 2 of this manuscript (where the reference density was denoted by f):

$$\begin{aligned}
A_p &= \int_{\mathbb{R}} \frac{p'(s)^2}{p(s)}\, ds, \\
B_p &= \int_{\mathbb{R}} \frac{p'(s)(p(s) + sp'(s))}{p(s)}\, ds = \int_{\mathbb{R}} s\, \frac{p'(s)^2}{p(s)}\, ds, \\
C_p &= \int_{\mathbb{R}} \frac{(p(s) + sp'(s))^2}{p(s)}\, ds = \int_{\mathbb{R}} s^2 \frac{p'(s)^2}{p(s)}\, ds - 1.
\end{aligned}$$

The simplified expressions follow by expanding the numerators and using

$$\int_{\mathbb{R}} p'(s)\, ds = 0, \qquad \int_{\mathbb{R}} sp'(s)\, ds = [sp(s)]_{-\infty}^{+\infty} - \int_{\mathbb{R}} p(s)\, ds = -1.$$

These integration-by-parts identities use the normalization and decay assumptions. Centering and unit variance do not determine A_p, B_p, and C_p: they still depend on the reference shape. The Fisher–Rao pullback is therefore

$$\boxed{ds^2_{\mathrm{FR}} = \frac{A_p\, d\mu^2 + 2B_p\, d\mu\, d\sigma + C_p\, d\sigma^2}{\sigma^2}.}$$

Here, as throughout the paper, the line element records the metric; its evaluation on a tangent direction is the squared norm just computed. For two

directions, the same expression is read bilinearly:

$$g^{\mathrm{FR}}_{\mu,\sigma}\big((d\mu_1, d\sigma_1), (d\mu_2, d\sigma_2)\big) = \frac{A_p d\mu_1 d\mu_2 + B_p(d\mu_1 d\sigma_2 + d\sigma_1 d\mu_2) + C_p d\sigma_1 d\sigma_2}{\sigma^2}.$$

As for any pullback of a positive-definite metric, nondegeneracy follows when the parameterization $(\mu, \sigma) \mapsto p_{\mu,\sigma}$ is an immersion. Indeed, injectivity of its differential implies that a nonzero $(d\mu, d\sigma)$ produces a nonzero density variation, whose Fisher–Rao squared norm is strictly positive. Under this immersion assumption, the coefficient matrix is positive definite and hence

$$A_p > 0, \qquad A_p C_p - B_p^2 > 0.$$

If p is even, the integrand defining B_p is odd and $B_p = 0$: location and scale are then orthogonal for Fisher–Rao. Centering alone does not imply this symmetry or the vanishing of B_p.

5.3.3. **Wasserstein–Otto pullback**

Let F and Q denote the cumulative distribution function and quantile function of the reference density p. Direct integration gives

$$F_{\mu,\sigma}(x) = F\left(\frac{x-\mu}{\sigma}\right).$$

Differentiating with respect to the parameters at fixed x yields

$$\partial_\mu F_{\mu,\sigma}(x) = -p_{\mu,\sigma}(x), \qquad \partial_\sigma F_{\mu,\sigma}(x) = -\frac{x-\mu}{\sigma}\, p_{\mu,\sigma}(x).$$

Consequently, the primitive of the density variation is

$$\int_{-\infty}^{x} dp_{\mu,\sigma}(d\mu, d\sigma)(s)\, ds = -p_{\mu,\sigma}(x)\left(d\mu + \frac{x-\mu}{\sigma} d\sigma\right).$$

Using the flux formula (1), we obtain

$$\begin{aligned} g^{\mathrm{WO}}_{\mu,\sigma}\big((d\mu, d\sigma), (d\mu, d\sigma)\big) &= \int_{\mathbb{R}} p_{\mu,\sigma}(x)\left(d\mu + \frac{x-\mu}{\sigma} d\sigma\right)^2 dx \\ &= \int_{\mathbb{R}} p(s)(d\mu + s\, d\sigma)^2\, ds. \end{aligned}$$

Expanding the square and using the centered, unit-variance normalization, we obtain

$$\begin{aligned} \int_{\mathbb{R}} p(s)(d\mu + s\, d\sigma)^2\, ds &= d\mu^2 \int_{\mathbb{R}} p(s)\, ds + 2 d\mu\, d\sigma \int_{\mathbb{R}} s p(s)\, ds \\ &\quad + d\sigma^2 \int_{\mathbb{R}} s^2 p(s)\, ds \\ &= d\mu^2 + d\sigma^2. \end{aligned}$$

The Wasserstein–Otto pullback is therefore

$$\boxed{ds^2_{\mathrm{WO}} = d\mu^2 + d\sigma^2.}$$

Equivalently, its bilinear form is

$$g^{\mathrm{WO}}_{\mu,\sigma}\big((d\mu_1, d\sigma_1), (d\mu_2, d\sigma_2)\big) = d\mu_1 d\mu_2 + d\sigma_1 d\sigma_2.$$

The reference moments have removed the mixed term and set both diagonal coefficients to one. Every standardized reference density satisfying our assumptions therefore induces the same Wasserstein–Otto metric in these mean and standard-deviation coordinates.

The quantile calculation gives the same result and also identifies the geodesics. Solving $F_{\mu,\sigma}(x) = r$ for x gives

$$Q_{\mu,\sigma}(r) = \mu + \sigma Q(r), \qquad \partial_\mu Q_{\mu,\sigma}(r) = 1, \qquad \partial_\sigma Q_{\mu,\sigma}(r) = Q(r).$$

Thus the quantile variation associated with $(d\mu, d\sigma)$ is $d\mu + Q(r)d\sigma$. Its squared $L^2(0,1)$ norm equals $\int_0^1 (d\mu + Q(r)d\sigma)^2\, dr$; the substitution $r = F(s)$, $dr = p(s)\, ds$, recovers exactly the pullback norm above.

5.3.4. **Geometry and comparison**

Wasserstein–Otto: a constant metric and straight parameter paths. The coefficients of ds^2_{WO} do not depend on μ or σ. The parameter half-plane therefore has the Euclidean metric directly in (μ, σ) coordinates. Location and scale are orthogonal, and the Gaussian curvature is zero.

For endpoints (μ_0, σ_0) and (μ_1, σ_1), the path

$$\mu(t) = (1-t)\mu_0 + t\mu_1, \qquad \sigma(t) = (1-t)\sigma_0 + t\sigma_1, \qquad 0 \le t \le 1,$$

remains in the parameter domain because both endpoint scales are positive. Moreover,

$$\begin{aligned} Q_{\mu(t),\sigma(t)}(r) &= \mu(t) + \sigma(t)Q(r) \\ &= (1-t)Q_{\mu_0,\sigma_0}(r) + tQ_{\mu_1,\sigma_1}(r). \end{aligned}$$

This verifies, for this specific family, the condition established in the previous subsection: the ambient straight quantile segment is realized by a parameter path. The path is consequently a constant-speed minimizing geodesic for the pullback metric, and its intrinsic distance equals the ambient Wasserstein distance:

$$\begin{aligned} d_{g^{\mathrm{WO}}}\big((\mu_0,\sigma_0),(\mu_1,\sigma_1)\big)^2 &= W_2(p_{\mu_0,\sigma_0}, p_{\mu_1,\sigma_1})^2 \\ &= (\mu_1-\mu_0)^2 + (\sigma_1-\sigma_0)^2. \end{aligned}$$

The boundary $\sigma = 0$ is at finite transport distance. Indeed, at fixed μ, the length required to decrease the scale from σ_0 to zero is σ_0. The limiting distribution is the point mass at μ, outside our space of strictly positive densities. The open parameter half-plane is therefore not complete for this metric. Figure 16 shows how different the Fisher–Rao and Wasserstein–Otto geodesics already are between the same two Gaussian endpoints: the Wasserstein–Otto path translates at fixed variance, while the Fisher–Rao path broadens and then narrows again.

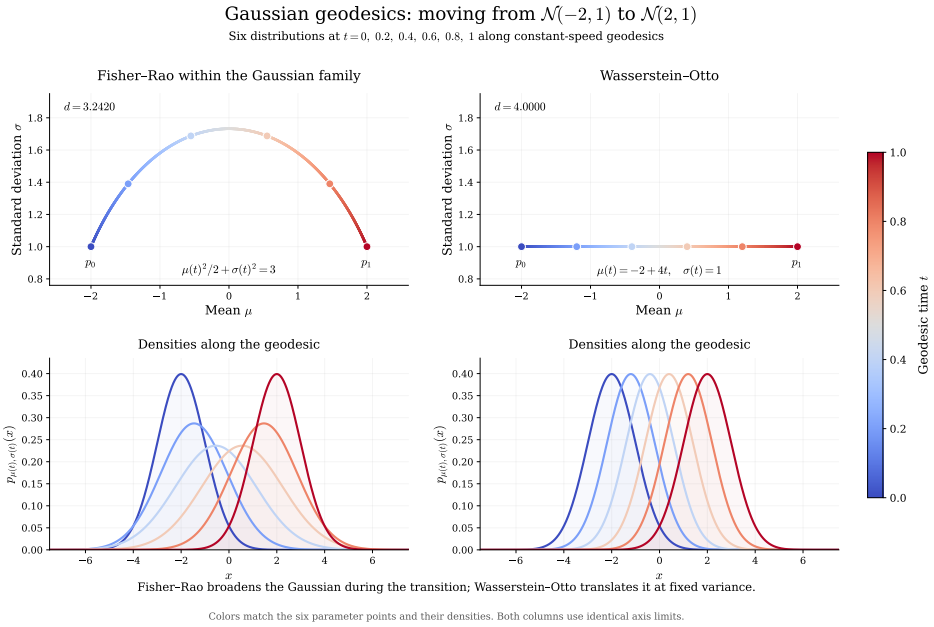


FIGURE 16. Gaussian geodesics from $\mathcal{N}(-2,1)$ to $\mathcal{N}(2,1)$. Top: parameter paths in (μ,σ); bottom: six densities at $t = 0, 0.2, \ldots, 1$. Left: the intrinsic Fisher–Rao geodesic follows $\mu^2/2 + \sigma^2 = 3$, broadening to $\sigma = \sqrt{3}$. Right: the Wasserstein–Otto geodesic translates at constant scale, $\mu = -2 + 4t$, $\sigma = 1$, and is also an ambient Wasserstein geodesic. Equal times refer to constant metric speed.

Fisher–Rao: a hyperbolic metric. The factor $1/\sigma^2$ changes the geometry. At a smaller scale, the same parameter displacement has a larger Fisher–Rao norm. Completing the square gives

$$ds^2_{\mathrm{FR}} = \frac{A_p}{\sigma^2}\left(d\mu + \frac{B_p}{A_p}d\sigma\right)^2 + \frac{C_p - B_p^2/A_p}{\sigma^2}\,d\sigma^2.$$

Under the linear change of half-plane coordinates

$$(\mu,\sigma) \longmapsto \left(\mu + \frac{B_p}{A_p}\sigma,\ \frac{\sqrt{A_pC_p - B_p^2}}{A_p}\sigma\right),$$

this is the standard hyperbolic half-plane metric multiplied by $C_p - B_p^2/A_p$. Its Gaussian curvature is therefore the negative constant

$$\boxed{K_{\mathrm{FR}} = -\frac{1}{C_p - B_p^2/A_p}, \qquad K_{\mathrm{WO}} = 0.}$$

In the displayed half-plane coordinates, Fisher–Rao geodesics are vertical lines or semicircles orthogonal to the boundary. They are geodesics of the pullback metric on this family; no claim is made that they are ambient Fisher–Rao geodesics in the full density space.

For an even reference density, $B_p = 0$, and their traces in the original parameter coordinates are vertical lines $\mu =$ constant or the upper halves of ellipses

$$\frac{A_p}{C_p}(\mu - \mu_c)^2 + \sigma^2 = R^2,$$

where $\mu_c \in \mathbb{R}$ and $R > 0$ specify the geodesic. Thus changing location can be less costly by passing through broader densities. Unlike the Wasserstein case, the boundary $\sigma = 0$ is at infinite intrinsic distance: for every path,

$$L_{g^{\mathrm{FR}}} \geq \sqrt{C_p - B_p^2/A_p} \int \frac{|\dot{\sigma}(t)|}{\sigma(t)}\, dt.$$

The hyperbolic half-plane description also shows that the Fisher–Rao pullback is complete. For the hyperbolic description with a symmetric reference density, see [14, Example 5].

Gaussian and Laplace examples. For the standard Gaussian reference density,

$$p(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \qquad p'(x) = -xp(x),$$

the Gaussian moments give $A_p = 1$, $B_p = 0$, and $C_p = 2$. Its mean is zero and its second moment is one, hence

$$\boxed{\begin{aligned} ds^2_{\mathrm{FR}} &= \frac{d\mu^2 + 2\, d\sigma^2}{\sigma^2}, & K_{\mathrm{FR}} &= -\tfrac{1}{2}, \\ ds^2_{\mathrm{WO}} &= d\mu^2 + d\sigma^2, & K_{\mathrm{WO}} &= 0. \end{aligned}}$$

Figure 17 shows the corresponding intrinsic distance maps from a common narrow reference point $(\mu_\star, \sigma_\star) = (0, 0.22)$: Fisher–Rao contours drift toward larger σ, while Wasserstein–Otto contours remain Euclidean circles.

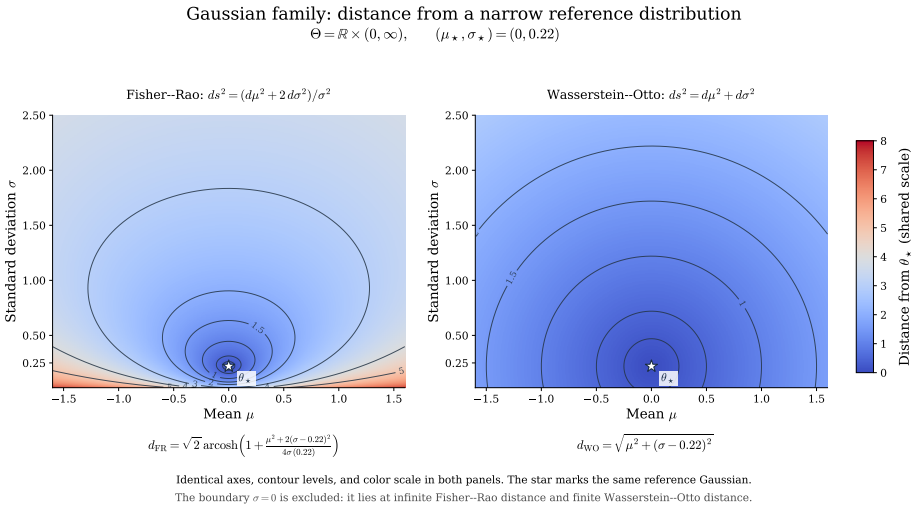


FIGURE 17. Distances from $(\mu, \sigma) = (0, 0.22)$ in the Gaussian parameter half-plane, with a common color scale. Fisher–Rao (left), $ds^2 = (d\mu^2 + 2d\sigma^2)/\sigma^2$, gives hyperbolic distance contours extending toward larger scales. Wasserstein–Otto (right), $ds^2 = d\mu^2 + d\sigma^2$, gives circles restricted to $\sigma > 0$. The boundary $\sigma = 0$ lies at infinite Fisher–Rao distance and finite Wasserstein–Otto distance.

For the centered, unit-variance Laplace reference density,

$$p(x) = \frac{1}{\sqrt{2}} e^{-\sqrt{2}|x|}, \qquad p'(x) = -\sqrt{2}\ \mathrm{sign}(x) p(x) \quad \text{for } x \neq 0,$$

we have $A_p = 2$, $B_p = 0$, and $C_p = \int_{\mathbb{R}} (1 - \sqrt{2}|x|)^2 p(x)\,dx = 1$. Thus

$$\boxed{\begin{aligned} ds^2_{\mathrm{FR}} &= \frac{2\,d\mu^2 + d\sigma^2}{\sigma^2}, & K_{\mathrm{FR}} &= -1, \\ ds^2_{\mathrm{WO}} &= d\mu^2 + d\sigma^2, & K_{\mathrm{WO}} &= 0. \end{aligned}}$$

In both examples, σ is the standard deviation. Their Wasserstein–Otto metrics coincide, while their Fisher–Rao metrics and curvatures differ. This comparison isolates the distinction: Wasserstein–Otto uses only the first two reference moments, fixed here by standardization, whereas Fisher–Rao retains information about the shape of the density through p' and assigns a scale-dependent cost to parameter displacements.

6. Multivariate Gaussian family

We now consider Gaussian densities on $\mathbb{R}^n$. The mean is a vector $\mu \in \mathbb{R}^n$, and the covariance is a symmetric positive-definite matrix $\Sigma \in S_n^{++}(\mathbb{R})$. The parameter space is therefore

$$\Theta = \mathbb{R}^n \times S_n^{++}(\mathbb{R}),$$

and the family is given by

$$p_{\mu,\Sigma}(x) = \frac{1}{(2\pi)^{n/2}\det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right), \qquad x \in \mathbb{R}^n.$$

Each density is strictly positive and has finite second moment, so $(\mu, \Sigma) \mapsto p_{\mu,\Sigma}$ takes values in $\mathcal{P}_{2,+}(\mathbb{R}^n)$. The mean determines its position, while the covariance determines its dispersion and orientation. For $n = 1$, $\Sigma = \sigma^2$ recovers the Gaussian location–dispersion family of Section 5.3.

6.1. **Tangent directions and density differential**

Let $S_n(\mathbb{R})$ denote the vector space of real symmetric $n \times n$ matrices. Since $S_n^{++}(\mathbb{R})$ is an open subset of $S_n(\mathbb{R})$,

$$T_\mu \mathbb{R}^n = \mathbb{R}^n, \qquad T_\Sigma S_n^{++}(\mathbb{R}) = S_n(\mathbb{R}).$$

Consequently, the tangent space of the product parameter manifold is

$$\boxed{T_{(\mu,\Sigma)}\Theta = \mathbb{R}^n \times S_n(\mathbb{R}).}$$

A tangent direction is thus a pair $(d\mu, d\Sigma)$, with $d\mu \in \mathbb{R}^n$ and $d\Sigma$ symmetric. The matrix $d\Sigma$ need not be positive definite: it describes a variation of the covariance, whereas Σ itself must remain positive definite. For every such direction, $\Sigma + t\, d\Sigma$ remains positive definite for sufficiently small $|t|$.

We recall the differential briefly. For a detailed computation, see Pierre Guilbert's *Pullback Fisher–Rao Geometries of Parametric Probability Families: An Expository Atlas*, Section 4.2, "Multivariate Gaussian", paragraph "Differential" [1]. At fixed x, it is convenient to differentiate

$$\log p_{\mu,\Sigma}(x) = -\frac{n}{2}\log(2\pi) - \frac{1}{2}\log\det\Sigma - \frac{1}{2}(x-\mu)^\top\Sigma^{-1}(x-\mu).$$

The matrix identities

$$d(\log\det)_\Sigma(d\Sigma) = \operatorname{tr}(\Sigma^{-1}d\Sigma), \qquad d(\Sigma \mapsto \Sigma^{-1})_\Sigma(d\Sigma) = -\Sigma^{-1}d\Sigma\,\Sigma^{-1}$$

give the two parameter contributions

$$\begin{aligned} d_\mu \log p_{\mu,\Sigma}(d\mu)(x) &= (x-\mu)^\top\Sigma^{-1}d\mu, \\ d_\Sigma \log p_{\mu,\Sigma}(d\Sigma)(x) &= -\frac{1}{2}\operatorname{tr}(\Sigma^{-1}d\Sigma) \\ &\quad + \frac{1}{2}(x-\mu)^\top\Sigma^{-1}d\Sigma\,\Sigma^{-1}(x-\mu). \end{aligned}$$

Adding these contributions and using $dp = p\, d\log p$, the density differential on the product manifold is

$$\boxed{\begin{aligned} &dp_{\mu,\Sigma}(d\mu, d\Sigma)(x) \\ &\quad = p_{\mu,\Sigma}(x)\Big[(x-\mu)^\top\Sigma^{-1}d\mu - \frac{1}{2}\operatorname{tr}(\Sigma^{-1}d\Sigma) \\ &\qquad\qquad + \frac{1}{2}(x-\mu)^\top\Sigma^{-1}d\Sigma\,\Sigma^{-1}(x-\mu)\Big]. \end{aligned}}$$

This is the density variation induced by $(d\mu, d\Sigma)$ at fixed observation x. Its integral is zero: the mean term integrates to zero, and the expectation of

the quadratic term is $\operatorname{tr}(\Sigma^{-1}d\Sigma)$, which cancels the trace term. It therefore preserves the total probability, as required of a tangent variation.

6.2. **Wasserstein–Otto metric**

For a tangent direction $(d\mu, d\Sigma) \in T_{(\mu,\Sigma)}\Theta = \mathbb{R}^n \times S_n(\mathbb{R})$, the Wasserstein–Otto construction requires a potential whose gradient transports the density with the prescribed variation. We therefore seek ϕ satisfying

$$-\operatorname{div}(p_{\mu,\Sigma}\nabla\phi) = dp_{\mu,\Sigma}(d\mu, d\Sigma).$$

Once this equation is solved, the squared pullback norm is

$$g^{\mathrm{WO}}_{\mu,\Sigma}((d\mu, d\Sigma), (d\mu, d\Sigma)) = \int_{\mathbb{R}^n} p_{\mu,\Sigma}(x)\|\nabla\phi(x)\|^2\, dx.$$

For Gaussian densities, a quadratic potential suffices. This reduces the weighted Poisson equation to a linear equation for a symmetric matrix; see Malagò, Montrucchio, and Pistone [10, Sections 1.2 and 4] for the Gaussian potential representation and its Lyapunov formulation.

6.2.1. **The transport potential**

Let $A \in S_n(\mathbb{R})$ be a symmetric matrix to be determined, and consider

$$\phi(x) = d\mu^\top(x-\mu) + \frac{1}{2}(x-\mu)^\top A(x-\mu).$$

All derivatives in the continuity equation are with respect to the observation variable x, with the parameters and tangent direction held fixed. Since A is symmetric,

$$\nabla_x\phi(x) = d\mu + A(x-\mu), \qquad \operatorname{div}_x(\nabla_x\phi) = \operatorname{tr}(A).$$

The first term translates the distribution; the second deforms its covariance. Using $\nabla_x p_{\mu,\Sigma}(x) = -p_{\mu,\Sigma}(x)\Sigma^{-1}(x-\mu)$, the product rule gives

$$\begin{aligned}-\operatorname{div}_x(p_{\mu,\Sigma}\nabla_x\phi)(x) = p_{\mu,\Sigma}(x)\Big[&(x-\mu)^\top\Sigma^{-1}d\mu \\ &+ (x-\mu)^\top\Sigma^{-1}A(x-\mu) - \operatorname{tr}(A)\Big].\end{aligned}$$

The mean term already agrees with the density differential in Section 6.1. To match the covariance terms, choose A so that

$$\boxed{\Sigma A + A\Sigma = d\Sigma.}$$

Indeed, multiplying this identity on both sides by Σ^{-1} gives

$$A\Sigma^{-1} + \Sigma^{-1}A = \Sigma^{-1}d\Sigma\,\Sigma^{-1}.$$

A scalar quadratic form equals the quadratic form of the symmetric part of its matrix. Hence

$$\begin{aligned}(x-\mu)^\top\Sigma^{-1}A(x-\mu) &= \frac{1}{2}(x-\mu)^\top(\Sigma^{-1}A + A\Sigma^{-1})(x-\mu) \\ &= \frac{1}{2}(x-\mu)^\top\Sigma^{-1}d\Sigma\,\Sigma^{-1}(x-\mu).\end{aligned}$$

Taking the trace after multiplying $\Sigma A + A\Sigma = d\Sigma$ by Σ^{-1} also gives

$$2\operatorname{tr}(A) = \operatorname{tr}(\Sigma^{-1}d\Sigma).$$

Both covariance terms therefore coincide with those of $dp_{\mu,\Sigma}(d\mu, d\Sigma)$. No commutation between Σ and $d\Sigma$ is assumed.

The equation $\Sigma A + A\Sigma = d\Sigma$ is a Lyapunov equation. It has a unique symmetric solution for every symmetric $d\Sigma$. To see this, observe that the linear map $A \mapsto \Sigma A + A\Sigma$ sends $S_n(\mathbb{R})$ into itself and that, for $A \neq 0$,

$$\operatorname{tr}\big(A(\Sigma A + A\Sigma)\big) = 2\operatorname{tr}(A\Sigma A) > 0,$$

because Σ is positive definite. Its kernel is therefore zero, and it is invertible on the finite-dimensional space $S_n(\mathbb{R})$. With the notation introduced in Section 2,

$$A = \mathcal{L}_\Sigma(d\Sigma), \qquad \Sigma\mathcal{L}_\Sigma(d\Sigma) + \mathcal{L}_\Sigma(d\Sigma)\Sigma = d\Sigma.$$

The resulting affine velocity is square-integrable under the Gaussian density. Since it is a gradient and satisfies the continuity equation, it is the minimal-energy velocity selected by the Wasserstein–Otto metric, as established by the weighted Helmholtz decomposition above. Figure 18 shows how this operator turns a prescribed covariance variation into that linear gradient field, and how a finite step of the resulting flow matches $\Sigma + \varepsilon d\Sigma$ only to first order.

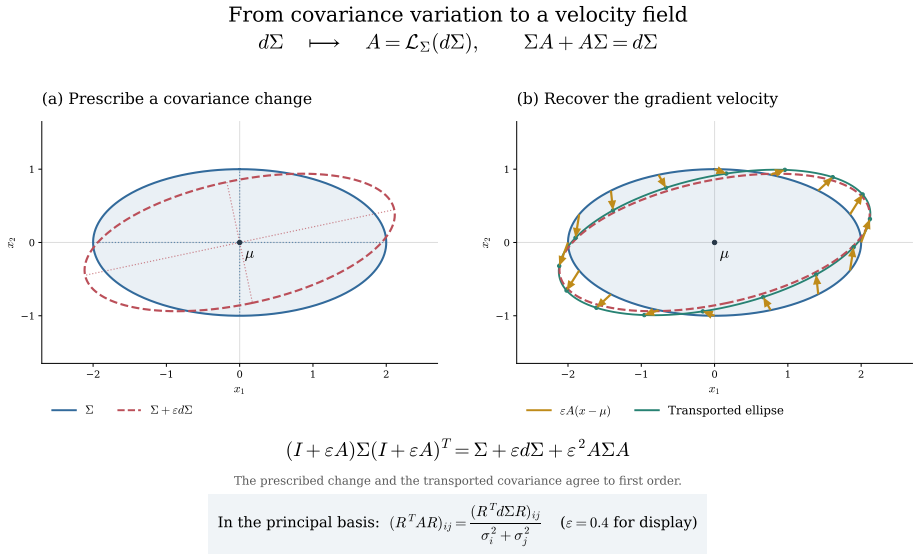


FIGURE 18. The Lyapunov operator converts $d\Sigma$ into the gradient velocity $v(x) = A(x-\mu)$, where $A = \mathcal{L}_\Sigma(d\Sigma)$ solves $\Sigma A + A\Sigma = d\Sigma$. Left: density-level ellipses for Σ and $\Sigma + \varepsilon d\Sigma$. Right: arrows εv transport the initial ellipse to covariance $(I+\varepsilon A)\Sigma(I+\varepsilon A)^\top = \Sigma + \varepsilon d\Sigma + \varepsilon^2 A\Sigma A$. Agreement with the prescribed change is first-order. Here $\mu = 0$, $\Sigma = \operatorname{diag}(4, 1)$, $d\Sigma = \left(\begin{smallmatrix} 1.2 & 2 \\ 2 & -0.3 \end{smallmatrix}\right)$, and $\varepsilon = 0.4$. A symmetric gradient deformation can change both scales and orientation.

6.2.2. **The pullback metric**

For two parameter directions $(d\mu_1, d\Sigma_1)$ and $(d\mu_2, d\Sigma_2)$, write

$$A_1 = \mathcal{L}_\Sigma(d\Sigma_1), \qquad A_2 = \mathcal{L}_\Sigma(d\Sigma_2).$$

Substituting their velocity fields into the ambient inner product gives

$$\begin{aligned} &g^{\mathrm{WO}}_{\mu,\Sigma}\big((d\mu_1, d\Sigma_1), (d\mu_2, d\Sigma_2)\big) \\ &\quad = \int_{\mathbb{R}^n} p_{\mu,\Sigma}(x)\big(d\mu_1 + A_1(x-\mu)\big)^\top \big(d\mu_2 + A_2(x-\mu)\big)\, dx \\ &\quad = d\mu_1^\top d\mu_2 + \operatorname{tr}(A_1 \Sigma A_2). \end{aligned}$$

Here the mixed terms vanish because $\int_{\mathbb{R}^n}(x-\mu)p_{\mu,\Sigma}(x)\,dx = 0$. The covariance term follows from

$$\int_{\mathbb{R}^n}(x-\mu)(x-\mu)^\top p_{\mu,\Sigma}(x)\,dx = \Sigma$$

and the symmetry of A_1, A_2, Σ. To express it directly through a covariance direction, use the Lyapunov equation for A_2:

$$\begin{aligned} \frac{1}{2}\operatorname{tr}(A_1 d\Sigma_2) &= \frac{1}{2}\operatorname{tr}\big(A_1(\Sigma A_2 + A_2\Sigma)\big) \\ &= \operatorname{tr}(A_1 \Sigma A_2). \end{aligned}$$

The last equality uses invariance of the trace under transposition and cyclic permutation. We obtain the Gaussian Wasserstein–Otto metric

$$\boxed{\begin{aligned} &g^{\mathrm{WO}}_{\mu,\Sigma}\big((d\mu_1, d\Sigma_1), (d\mu_2, d\Sigma_2)\big) \\ &\quad = d\mu_1^\top d\mu_2 + \frac{1}{2}\operatorname{tr}\big(\mathcal{L}_\Sigma(d\Sigma_1)\, d\Sigma_2\big). \end{aligned}}$$

Equivalently, the line element is

$$\boxed{ds^2_{\mathrm{WO}} = \|d\mu\|^2 + \frac{1}{2}\operatorname{tr}\big(\mathcal{L}_\Sigma(d\Sigma)\, d\Sigma\big).}$$

The covariance contribution is positive for every nonzero $d\Sigma$, since

$$\frac{1}{2}\operatorname{tr}\big(\mathcal{L}_\Sigma(d\Sigma)\, d\Sigma\big) = \operatorname{tr}\big(\mathcal{L}_\Sigma(d\Sigma)\, \Sigma\, \mathcal{L}_\Sigma(d\Sigma)\big) > 0.$$

Thus the formula defines a positive-definite metric on $\mathbb{R}^n \times S_n^{++}(\mathbb{R})$. Mean and covariance directions are orthogonal. The mean block is Euclidean and independent of Σ; the covariance block depends on Σ through the Lyapunov equation. In particular, the full metric is not constant in the covariance entries. The covariance geometry is also studied from the matrix viewpoint in [15].

Reduction to the one-dimensional Gaussian family. For $n = 1$, write $\Sigma = \sigma^2$ and $d\Sigma = 2\sigma\, d\sigma$. The Lyapunov equation becomes

$$2\sigma^2 A = 2\sigma\, d\sigma, \qquad A = \frac{d\sigma}{\sigma}.$$

Consequently,

$$\frac{1}{2}\mathcal{L}_{\sigma^2}(d\Sigma)\, d\Sigma = \frac{1}{2}\frac{d\sigma}{\sigma}(2\sigma\, d\sigma) = d\sigma^2,$$

and the metric reduces to $ds^2_{\mathrm{WO}} = d\mu^2 + d\sigma^2$, in agreement with Section 5.3. This also distinguishes the covariance coordinate Σ from the standard-deviation coordinate σ used on the line.

6.3. **Fisher–Rao metric**

We now evaluate the Fisher–Rao metric on the same parameter and tangent spaces. The computation uses the density differential from Section 6.1 and the second and fourth moments of a Gaussian distribution. For the detailed calculation, see P. Guilbert, *Pullback Fisher–Rao Geometries of Parametric Probability Families: An Expository Atlas*, Section 4.2, "Multivariate Gaussian", under "Metric" [1]. For the classical Riemannian treatment of the multivariate normal model, see Skovgaard [16].

For $(d\mu_1, d\Sigma_1), (d\mu_2, d\Sigma_2) \in \mathbb{R}^n \times S_n(\mathbb{R})$, the pullback is

$$\begin{aligned} &g^{\mathrm{FR}}_{\mu,\Sigma}((d\mu_1, d\Sigma_1), (d\mu_2, d\Sigma_2)) \\ &\quad = \int_{\mathbb{R}^n} \frac{dp_{\mu,\Sigma}(d\mu_1, d\Sigma_1)(x)\, dp_{\mu,\Sigma}(d\mu_2, d\Sigma_2)(x)}{p_{\mu,\Sigma}(x)}\, dx. \end{aligned}$$

The logarithmic differential consists of a term linear in $x - \mu$, associated with $d\mu$, and a centered quadratic term associated with $d\Sigma$. We evaluate their products separately.

Mean directions. The covariance identity is

$$\int_{\mathbb{R}^n} (x-\mu)(x-\mu)^\top p_{\mu,\Sigma}(x)\, dx = \Sigma.$$

It gives

$$\begin{aligned} &\int_{\mathbb{R}^n} ((x-\mu)^\top \Sigma^{-1} d\mu_1)((x-\mu)^\top \Sigma^{-1} d\mu_2) p_{\mu,\Sigma}(x)\, dx \\ &\quad = d\mu_1^\top \Sigma^{-1} \Sigma \Sigma^{-1} d\mu_2 = d\mu_1^\top \Sigma^{-1} d\mu_2. \end{aligned}$$

Mixed directions. The product of a mean contribution and a covariance contribution is odd in $x - \mu$. Its integral against the centered Gaussian density is zero. Thus mean and covariance directions are orthogonal for Fisher–Rao, as for Wasserstein–Otto.

Covariance directions. For $X \sim \mathcal{N}(\mu, \Sigma)$ and symmetric matrices B, C, the Gaussian fourth-moment identity reads [17]

$$\mathrm{Cov}\Big((X-\mu)^\top B(X-\mu), (X-\mu)^\top C(X-\mu)\Big) = 2\,\mathrm{tr}(B\Sigma C\Sigma).$$

In the density differential, the two matrices are

$$B = \Sigma^{-1} d\Sigma_1 \Sigma^{-1}, \qquad C = \Sigma^{-1} d\Sigma_2 \Sigma^{-1}.$$

For $i = 1, 2$, the corresponding quadratic form has expectation

$$\mathrm{tr}(\Sigma^{-1} d\Sigma_i).$$

This is exactly the trace term subtracted in the differential. The covariance block is therefore one quarter of the covariance of these two quadratic forms.

Applying the identity above gives

$$\begin{aligned} &g^{\mathrm{FR}}_{\mu,\Sigma}((0,d\Sigma_1),(0,d\Sigma_2)) \\ &\quad= \frac{1}{2}\operatorname{tr}\Big((\Sigma^{-1}d\Sigma_1\Sigma^{-1})\Sigma(\Sigma^{-1}d\Sigma_2\Sigma^{-1})\Sigma\Big) \\ &\quad= \frac{1}{2}\operatorname{tr}(\Sigma^{-1}d\Sigma_1\Sigma^{-1}d\Sigma_2). \end{aligned}$$

Combining the three blocks yields

$$\boxed{\begin{aligned} &g^{\mathrm{FR}}_{\mu,\Sigma}((d\mu_1,d\Sigma_1),(d\mu_2,d\Sigma_2)) \\ &\quad= d\mu_1^{\top}\Sigma^{-1}d\mu_2 + \frac{1}{2}\operatorname{tr}(\Sigma^{-1}d\Sigma_1\Sigma^{-1}d\Sigma_2). \end{aligned}}$$

Equivalently,

$$\boxed{ds^2_{\mathrm{FR}} = d\mu^{\top}\Sigma^{-1}d\mu + \frac{1}{2}\operatorname{tr}(\Sigma^{-1}d\Sigma\,\Sigma^{-1}d\Sigma).}$$

The covariance contribution can also be written as

$$\frac{1}{2}\operatorname{tr}\Big((\Sigma^{-1/2}d\Sigma\,\Sigma^{-1/2})^2\Big).$$

It is nonnegative and vanishes only when $d\Sigma = 0$. Together with positive definiteness of Σ^{-1}, this establishes positive definiteness of the pullback.

For mean variations, Fisher–Rao weights directions by Σ^{-1}: smaller variance means greater cost. The transport metric instead gives the Euclidean squared norm $\|d\mu\|^2$.

For covariance variations, Fisher–Rao uses Σ^{-1} on both sides of $d\Sigma$, while the transport metric uses the inverse Lyapunov operator. The mixed block is zero for both metrics. However, the mean block of Fisher–Rao depends on Σ, so this orthogonality does not make it a product of independent mean and covariance metrics.

For $n = 1$, substituting $\Sigma = \sigma^2$ and $d\Sigma = 2\sigma\,d\sigma$ gives

$$ds^2_{\mathrm{FR}} = \frac{d\mu^2}{\sigma^2} + \frac{1}{2}\frac{(2\sigma\,d\sigma)^2}{\sigma^4} = \frac{d\mu^2 + 2\,d\sigma^2}{\sigma^2},$$

in agreement with the Gaussian example in Section 5.3.

6.4. Comparison: mean, principal scales, rotations, and geodesics

The two metrics act on the same tangent space $T_{(\mu,\Sigma)}\Theta = \mathbb{R}^n \times S_n(\mathbb{R})$, but assign different costs to its mean and covariance components. We first compare their action on one tangent direction, then describe the resulting paths in parameter space.

6.4.1. Mean and covariance directions

For a pure mean variation, the squared norms are

$$\boxed{\begin{aligned} &g^{\mathrm{FR}}_{\mu,\Sigma}((d\mu,0),(d\mu,0)) = d\mu^{\top}\Sigma^{-1}d\mu, \\ &g^{\mathrm{WO}}_{\mu,\Sigma}((d\mu,0),(d\mu,0)) = \|d\mu\|^2. \end{aligned}}$$

Fisher–Rao measures the displacement relative to the current dispersion: shifting a narrow distribution is more costly than shifting a broad one by the same amount. The transport cost of a translation is its squared Euclidean length, independently of the covariance.

For a pure covariance variation, the comparison is

$$\boxed{\begin{aligned} g^{\mathrm{FR}}_{\mu,\Sigma}\big((0,d\Sigma),(0,d\Sigma)\big) &= \frac{1}{2}\operatorname{tr}(\Sigma^{-1}d\Sigma\,\Sigma^{-1}d\Sigma), \\ g^{\mathrm{WO}}_{\mu,\Sigma}\big((0,d\Sigma),(0,d\Sigma)\big) &= \frac{1}{2}\operatorname{tr}\big(\mathcal{L}_\Sigma(d\Sigma)\,d\Sigma\big). \end{aligned}}$$

The first expression compares the covariance change with the covariance itself. The second measures the kinetic energy of the affine gradient field producing that change. To make their respective weights explicit, we decompose the covariance into its principal scales and orientation.

6.4.2. **Rotation–scale coordinates for the covariance**

Write the spectral decomposition in terms of standard deviations along the principal axes:

$$\Sigma = R\,\operatorname{diag}(\sigma_1^2,\dots,\sigma_n^2)R^\top, \qquad \sigma_i > 0, \qquad R \in SO(n).$$

Here R is a rotation matrix whose columns are an oriented orthonormal basis of eigenvectors, and $SO(n)$ denotes the group of orthogonal matrices of determinant one. Locally, when the eigenvalues are distinct, we fix their order and choose the eigenvectors smoothly. The covariance is then described by the n principal scales and the rotation R. Figure 19 shows this for a three-dimensional Gaussian: the principal axes of the density ellipsoid are the columns of R, and the semi-axis lengths are the σ_i.

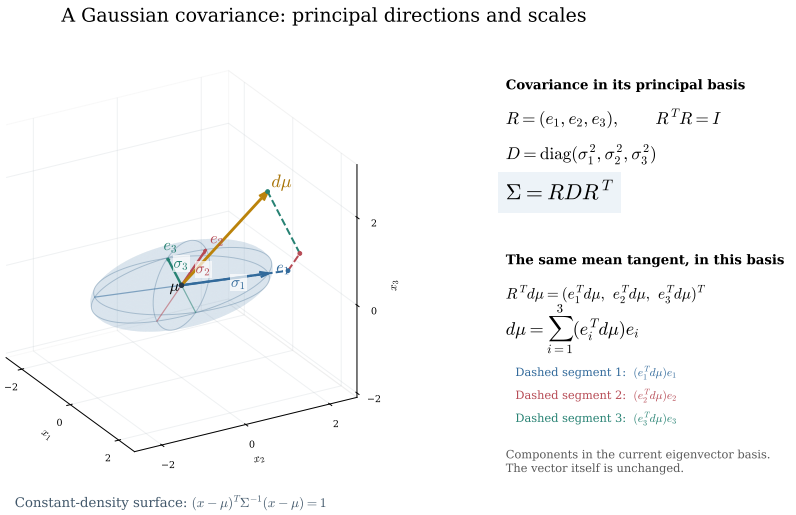


FIGURE 19. Spectral decomposition $\Sigma = RDR^\top$ of a three-dimensional Gaussian covariance, with $R = (e_1, e_2, e_3)$ and $D = \mathrm{diag}(\sigma_1^2, \sigma_2^2, \sigma_3^2)$. The density-level ellipsoid $(x - \mu)^\top \Sigma^{-1}(x - \mu) = 1$ has semiaxes $\sigma_i e_i$. The mean tangent $d\mu$ is resolved in this principal basis as $R^\top d\mu$. Here $(\sigma_1, \sigma_2, \sigma_3) = (2.2, 1.25, 0.7)$ and $R^\top d\mu = (2.7, 0.6, 1.6)^\top$.

A rotation direction $dR \in T_R SO(n)$ satisfies $R^\top dR + (dR)^\top R = 0$. Define its infinitesimal rotation matrix by

$$\Omega = R^\top dR, \qquad \Omega^\top = -\Omega, \qquad dR = R\Omega.$$

Differentiating the spectral decomposition gives

$$\begin{aligned} R^\top d\Sigma R = {} & \mathrm{diag}(2\sigma_1 d\sigma_1, \ldots, 2\sigma_n d\sigma_n) \\ & + \Omega\,\mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2) - \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)\Omega. \end{aligned}$$

In particular,

$$(R^\top d\Sigma R)_{ii} = 2\sigma_i d\sigma_i, \qquad (R^\top d\Sigma R)_{ij} = (\sigma_j^2 - \sigma_i^2)\Omega_{ij} \quad (i \neq j).$$

The diagonal entries record scale changes; the off-diagonal entries record changes of orientation. These are entries in the current principal-axis basis, not new independent covariance parameters.

In this basis, multiplication by Σ^{-1} divides the ith component by σ_i^2. The Lyapunov equation, in turn, gives

$$\left(R^\top \mathcal{L}_\Sigma(d\Sigma)R\right)_{ij} = \frac{(R^\top d\Sigma R)_{ij}}{\sigma_i^2 + \sigma_j^2}.$$

Thus the two covariance norms can be evaluated entry by entry:

$$\frac{1}{2}\operatorname{tr}(\Sigma^{-1}d\Sigma\,\Sigma^{-1}d\Sigma) = \frac{1}{2}\sum_{i,j=1}^{n}\frac{(R^\top d\Sigma R)_{ij}^2}{\sigma_i^2\sigma_j^2},$$
$$\frac{1}{2}\operatorname{tr}(\mathcal{L}_\Sigma(d\Sigma)\,d\Sigma) = \frac{1}{2}\sum_{i,j=1}^{n}\frac{(R^\top d\Sigma R)_{ij}^2}{\sigma_i^2+\sigma_j^2}.$$

Substitution of the diagonal and off-diagonal entries yields the full line elements in rotation–scale coordinates:

$$\boxed{ds_{\mathrm{FR}}^2 = \sum_{i=1}^{n}\frac{(R^\top d\mu)_i^2}{\sigma_i^2} + 2\sum_{i=1}^{n}\frac{d\sigma_i^2}{\sigma_i^2} + \sum_{i<j}\frac{(\sigma_i^2-\sigma_j^2)^2}{\sigma_i^2\sigma_j^2}\Omega_{ij}^2,}$$

$$\boxed{ds_{\mathrm{WO}}^2 = \|d\mu\|^2 + \sum_{i=1}^{n}d\sigma_i^2 + \sum_{i<j}\frac{(\sigma_i^2-\sigma_j^2)^2}{\sigma_i^2+\sigma_j^2}\Omega_{ij}^2.}$$

Here $d\sigma_i^2$ means $(d\sigma_i)^2$, the squared scale direction. Also, $(R^\top d\mu)_i$ is the component of the mean direction in the current principal-axis basis. We have reparameterized only Σ: $R^\top d\mu$ is not being identified with the differential of $R^\top\mu$, which would also contain a term involving dR.
Changes of principal scale. Fix the orientation R and the mean μ. The two squared norms reduce to

$$2\sum_i\left(\frac{d\sigma_i}{\sigma_i}\right)^2 \quad \text{for Fisher–Rao,} \qquad \sum_i(d\sigma_i)^2 \quad \text{for Wasserstein–Otto.}$$

Equal relative scale changes have equal Fisher cost, while equal absolute scale changes have equal transport cost. The scale part of Fisher–Rao is Euclidean in $\log\sigma_i$, up to the factor two; the transport scale part is Euclidean in σ_i. Figure 20 shows the corresponding distance maps in the ordered half-plane $\sigma_1 > \sigma_2 > 0$: Fisher–Rao contours become circles in logarithmic scale coordinates, while Wasserstein–Otto contours are ordinary circles.

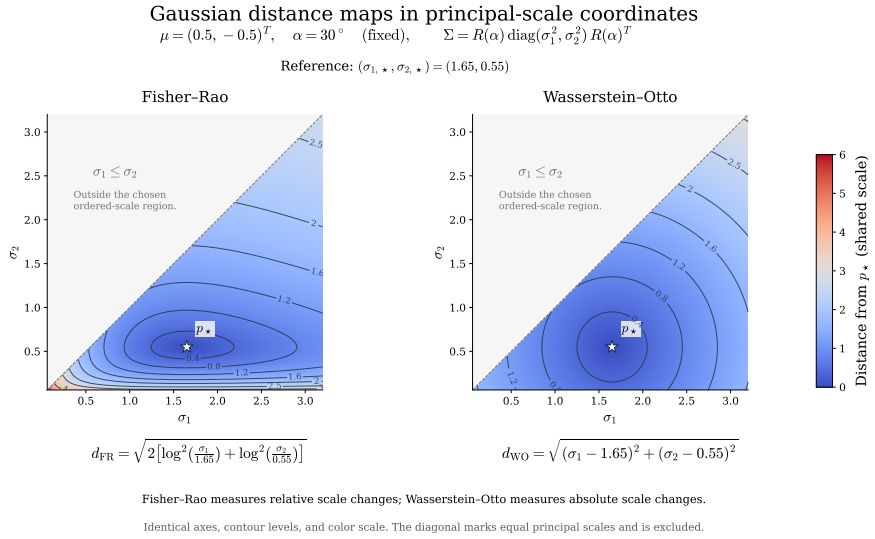


FIGURE 20. Gaussian distance maps at fixed $\mu = (0.5, -0.5)^\top$ and orientation $\alpha = 30°$, with reference principal standard deviations $(1.65, 0.55)$. Left: $d_{\mathrm{FR}} = \sqrt{2[\log^2(\sigma_1/1.65) + \log^2(\sigma_2/0.55)]}$. Right: $d_{\mathrm{WO}} = \sqrt{(\sigma_1 - 1.65)^2 + (\sigma_2 - 0.55)^2}$. Colors and contours show distances on $\sigma_1 > \sigma_2 > 0$. The exact Gaussian geodesics remain in this slice, interpolating scales geometrically for FR and arithmetically for WO, so these are also full Gaussian distances. At equal scales the spectral coordinates become singular, while both metrics remain regular.

Changes of orientation. For a rotation mixing principal axes i and j, the two coefficients are

$$\frac{(\sigma_i^2 - \sigma_j^2)^2}{\sigma_i^2 \sigma_j^2} \quad \text{and} \quad \frac{(\sigma_i^2 - \sigma_j^2)^2}{\sigma_i^2 + \sigma_j^2}.$$

The Fisher coefficient depends only on the ratio of the two scales. Multiplying all scales by the same positive factor leaves it unchanged. The transport coefficient is multiplied by the square of that factor: rotating a larger distribution moves its mass farther. In two dimensions, if R is parameterized by an angle α, then $\Omega_{12}^2 = (d\alpha)^2$, so these are directly the coefficients of the angular squared displacement. Figure 21 plots the resulting Fisher–Rao and Wasserstein–Otto distances as the orientation varies at fixed scales: the Fisher distance depends only on the scale ratio, while the transport distance grows with the absolute scales.

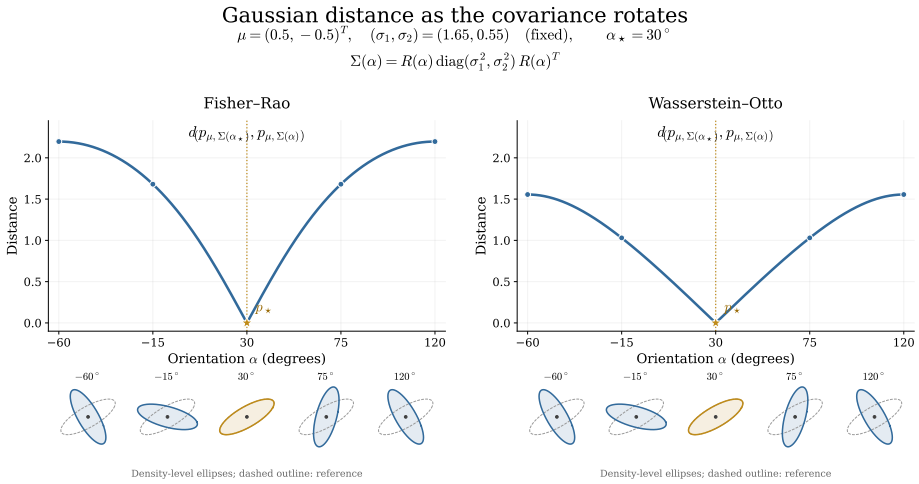


FIGURE 21. Gaussian endpoint distances as orientation varies at fixed $\mu = (0.5, -0.5)^\top$ and principal standard deviations $(\sigma_1, \sigma_2) = (1.65, 0.55)$, relative to $\alpha_\star = 30°$. With $\delta = \alpha - \alpha_\star$, the exact distances are $d_{\mathrm{FR}} = 2\,\mathrm{asinh}(|\sigma_1^2 - \sigma_2^2|\,|\sin\delta|/(2\sigma_1\sigma_2))$ and $d_{\mathrm{WO}} = [2(\sigma_1^2 + \sigma_2^2) - 2\sqrt{4\sigma_1^2\sigma_2^2 + (\sigma_1^2 - \sigma_2^2)^2\cos^2\delta}]^{1/2}$. Connecting geodesics may vary the scales; these are not lengths constrained to rotation. Both distances have period 180°, with maxima at $|\delta| = 90°$: $2\log(\sigma_1/\sigma_2) \approx 2.197$ and $\sqrt{2}(\sigma_1 - \sigma_2) \approx 1.556$. Thumbnails show the density levels $(x-\mu)^\top\Sigma(\alpha)^{-1}(x-\mu) = 1$.

Both coefficients vanish when $\sigma_i = \sigma_j$. A rotation within the corresponding eigenspace leaves Σ unchanged, so it induces no density variation. This is a redundancy of the rotation–scale representation, not a degeneracy of either metric on $S_n^{++}(\mathbb{R})$. At repeated eigenvalues, these variables are not a regular coordinate chart; the original matrix formulas remain valid.

Scale and rotation directions are orthogonal in the displayed formulas. They are not independent geometries: the rotation coefficients depend on the principal scales. Consequently, linear interpolation of scales and rotation angles does not in general produce a geodesic of either full metric.

6.4.3. **Geodesics in the Gaussian parameter space**

All parameter paths below describe Gaussian densities $p_{\mu(t),\Sigma(t)}$. Their length is computed from the corresponding pullback metric on Θ. A Fisher–Rao geodesic here is intrinsic to the Gaussian family; it is not asserted to be a geodesic in the full space of densities equipped with the ambient Fisher–Rao metric.

Wasserstein–Otto: independent mean and covariance paths. The mean block is Euclidean, and the covariance block depends only on Σ, so this metric

is a Riemannian product. For endpoints (μ_0, Σ_0) and (μ_1, Σ_1), define the symmetric positive-definite transport matrix

$$T = \Sigma_0^{-1/2}(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2})^{1/2}\Sigma_0^{-1/2}, \qquad T\Sigma_0 T = \Sigma_1.$$

Matrix square roots here are the positive-definite square roots. The optimal affine transport sends x to $\mu_1 + T(x - \mu_0)$. Its displacement interpolation remains Gaussian and gives the constant-speed geodesic [10, Sections 3 and 4]:

$$\boxed{\begin{aligned} &\mu(t) = (1-t)\mu_0 + t\mu_1, \\ &\Sigma(t) = ((1-t)I + tT)\Sigma_0((1-t)I + tT), \qquad 0 \le t \le 1. \end{aligned}}$$

Since $(1-t)I + tT$ is positive definite, the covariance remains positive definite. This path realizes both the intrinsic Gaussian distance and the ambient Wasserstein distance, with

$$\begin{aligned} &W_2^2(p_{\mu_0,\Sigma_0}, p_{\mu_1,\Sigma_1}) \\ &\quad = \|\mu_1 - \mu_0\|^2 + \operatorname{tr}\Big(\Sigma_0 + \Sigma_1 - 2(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2})^{1/2}\Big). \end{aligned}$$

The covariance path is generally not linear in the entries of Σ. For the Gaussian distance formula above, see Dowson and Landau [18]. If the endpoint covariances commute, they admit a common principal-axis basis. In that basis the scales interpolate linearly:

$$\sigma_i(t) = (1-t)\sigma_{i,0} + t\sigma_{i,1}.$$

When the endpoint covariances do not commute, the matrix formula determines both the changing scales and orientation; neither is generally an affine function of time.

Fisher–Rao: coupling between mean and covariance. For the full Gaussian family, the geodesic equations are [11, Section 2.2]

$$\boxed{\begin{aligned} &\ddot{\mu}(t) = \dot{\Sigma}(t)\Sigma(t)^{-1}\dot{\mu}(t), \\ &\ddot{\Sigma}(t) = \dot{\Sigma}(t)\Sigma(t)^{-1}\dot{\Sigma}(t) - \dot{\mu}(t)\dot{\mu}(t)^{\top}. \end{aligned}}$$

Dots denote differentiation with respect to the geodesic parameter t. These equations follow from the energy associated with the Fisher–Rao line element and display the coupling absent from the transport metric. A changing covariance affects the mean velocity, and a moving mean affects the covariance evolution. Even with identical endpoint covariances, a nonconstant mean path with covariance fixed throughout is not a geodesic: the second equation would require $\dot{\mu}(t)\dot{\mu}(t)^{\top} = 0$.

A minimizing path can instead broaden the distribution in directions of travel, reducing the mean cost before restoring the endpoint covariance. This already occurs in the one-dimensional Gaussian example. In higher dimensions, the mean path can also curve in response to anisotropy. One must solve the coupled equations with the endpoint conditions, rather than independently interpolate the mean, scales, and angles.

Fixed mean: an explicit covariance comparison. When $\mu_0 = \mu_1$, a constant mean minimizes the mean contribution for any covariance path. The Fisher–Rao minimizing covariance path is [11, Section 2.3]

$$\boxed{\mu(t) = \mu_0, \qquad \Sigma(t) = \Sigma_0^{1/2}(\Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2})^t\Sigma_0^{1/2}.}$$

Matrix powers are defined by applying the scalar power to the positive eigenvalues. This path solves the covariance geodesic equation with $\dot{\mu} = 0$ and is the geodesic of the affine-invariant covariance metric appearing in Fisher–Rao. Its length is

$$d_{g^{\mathrm{FR}}}((\mu_0, \Sigma_0), (\mu_0, \Sigma_1)) = \left[\frac{1}{2}\operatorname{tr}\Big(\big[\log(\Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2})\big]^2\Big)\right]^{1/2}.$$

For commuting endpoint covariances, the common eigenvectors remain fixed and the principal scales satisfy

$$\boxed{\begin{aligned}\sigma_i^{\mathrm{FR}}(t) &= \sigma_{i,0}^{1-t}\sigma_{i,1}^{t},\\ \sigma_i^{\mathrm{WO}}(t) &= (1-t)\sigma_{i,0} + t\sigma_{i,1}.\end{aligned}}$$

The Fisher path is linear in logarithmic scales; the transport path is linear in the scales themselves. The Fisher formula here requires a common mean. It must not be combined with an arbitrary moving mean to claim a geodesic of the full Gaussian family. Likewise, a path obtained by varying only the orientation while holding all scales fixed is a constrained path, not generally the minimizing path between its endpoints for either metric. Figure 22 compares the full Fisher–Rao and Wasserstein–Otto geodesics between two bivariate Gaussians whose means, scales, and orientations all differ: the Fisher–Rao mean path curves through its coupling with covariance, while the Wasserstein–Otto mean path stays straight.

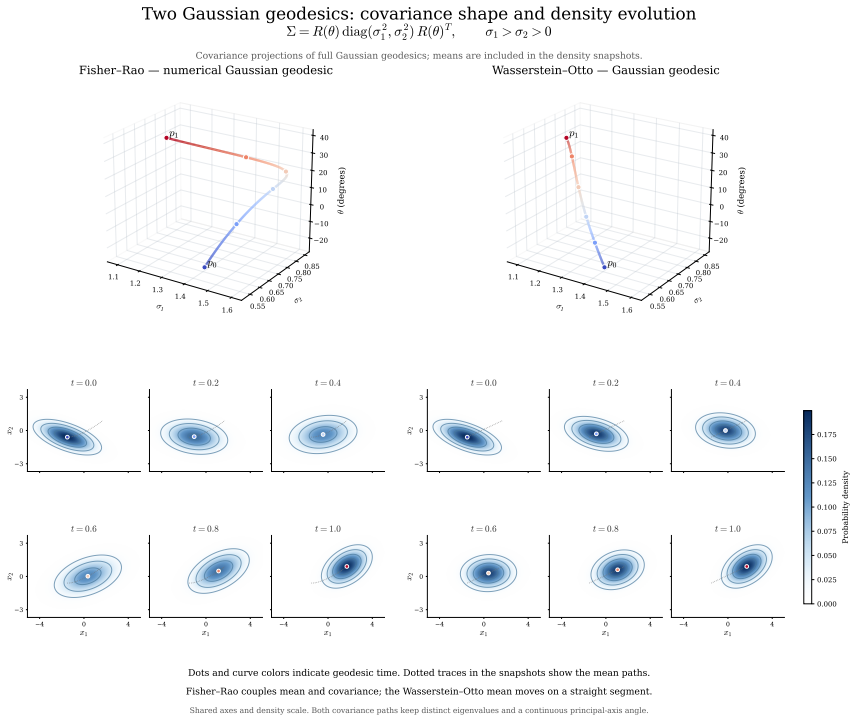


FIGURE 22. Fisher–Rao and Wasserstein–Otto geodesics between bivariate Gaussians with means $(-1.5, -0.6)^\top$, $(1.7, 0.9)^\top$, principal standard deviations $(1.45, 0.55)$, $(1.15, 0.70)$, and orientations $-20°$, $35°$. Top: covariance paths in $(\sigma_1, \sigma_2, \theta)$ coordinates, remaining away from spectral-coordinate singularities. Bottom: density snapshots at equal constant-speed times $t = 0, 0.2, \ldots, 1$, on a common density scale. The Fisher–Rao path is a numerical solution of the coupled mean–covariance geodesic equations [11, Section 2.2]; the Wasserstein–Otto path is exact Gaussian displacement interpolation [10, Sections 3–4]. Coupling with covariance curves the Fisher–Rao mean path, while the Wasserstein–Otto mean follows a straight segment.

7. Gibbs families

A Gibbs family is a family of densities of the form

$$p_\theta(x) = \frac{1}{Z_\theta} e^{-U_\theta(x)}, \qquad Z_\theta = \int_E e^{-U_\theta(x)}\, dx,$$

where U_θ is the potential, or energy, and Z_θ is the normalization constant. We take $E = \mathbb{R}^n$ and let Θ be an open subset of $\mathbb{R}^k$ on which $0 < Z_\theta < \infty$. We assume that the densities have finite second moments and that the potentials have the spatial and parameter regularity needed below. In particular, differentiation under the integral sign is justified, the parameter derivatives of U_θ are square-integrable under p_θ, and integration by parts has no boundary contribution at infinity. We write $\mathbb{E}_\theta[f] = \int_E f(x) p_\theta(x)\, dx$.

The family is called *linear* when the potential is linear in the parameters:

$$U_\theta(x) = \phi(x)^T\theta, \qquad \phi\colon E \longrightarrow \mathbb{R}^k.$$

The components of ϕ may be nonlinear functions of x: linearity here concerns θ. We first allow a general nonlinear dependence $\theta \mapsto U_\theta$; the linear case will then follow by replacing $\nabla_\theta U_\theta$ with ϕ. Figure 23 shows four two-parameter examples, two linear and two nonlinear, in one and two spatial dimensions.

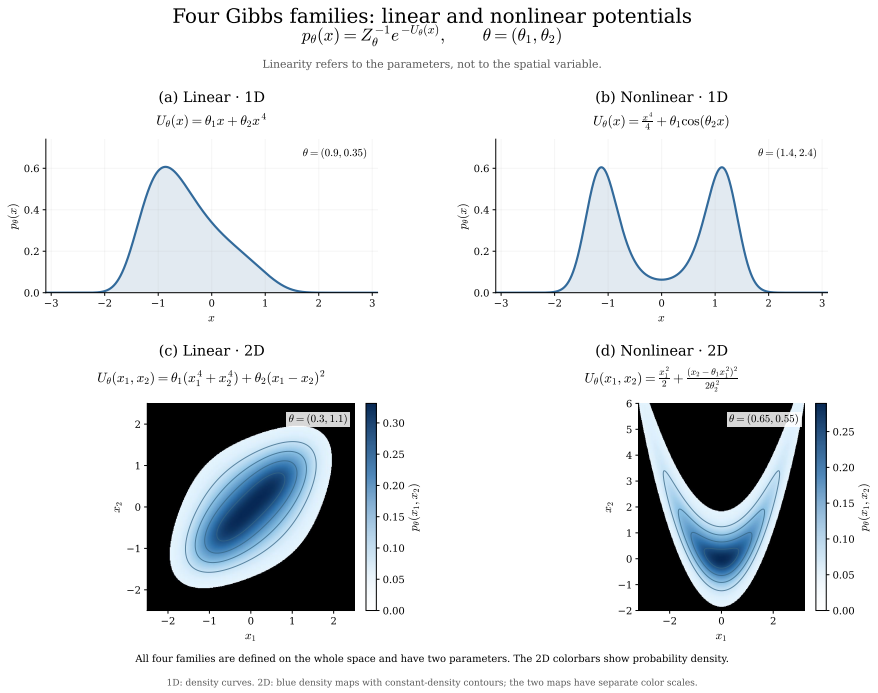


FIGURE 23. Four Gibbs families $p_\theta = Z_\theta^{-1} e^{-U_\theta}$, linear or nonlinear in θ. (a) Linear 1D: $U_\theta(x) = \theta_1 x + \theta_2 x^4$, $\Theta = \mathbb{R} \times (0, \infty)$, $\theta = (0.9, 0.35)$. (b) Nonlinear 1D: $U_\theta(x) = x^4/4 + \theta_1 \cos(\theta_2 x)$, $\Theta = (0, \infty)^2$, $\theta = (1.4, 2.4)$. (c) Linear 2D: $U_\theta(x_1, x_2) = \theta_1(x_1^4 + x_2^4) + \theta_2(x_1 - x_2)^2$, $\Theta = (0, \infty)^2$, $\theta = (0.3, 1.1)$. (d) Nonlinear 2D: $U_\theta(x_1, x_2) = x_1^2/2 + (x_2 - \theta_1 x_1^2)^2/(2\theta_2^2)$, $\Theta = \mathbb{R} \times (0, \infty)$, $\theta = (0.65, 0.55)$. In (d), the first marginal is standard Gaussian and the second conditional law has mean $\theta_1 x_1^2$ and standard deviation θ_2, giving $Z_\theta = 2\pi\theta_2$. All four laws have finite moments of every order on their full domains. The first three normalizations use numerical quadrature; 2D contours mark 8%, 25%, 50%, and 80% of peak density.

7.1. Differentiation with respect to the parameters

Since $\Theta \subset \mathbb{R}^k$ is open, its tangent space is

$$T_\theta\Theta \simeq \mathbb{R}^k.$$

We use the parameter gradient $\nabla_\theta = (\partial_{\theta_1}, \dots, \partial_{\theta_k})^T$; the spatial variable x is held fixed. Applying the product rule directly to $p_\theta(x) = Z_\theta^{-1} e^{-U_\theta(x)}$

gives

$$\begin{aligned}\nabla_\theta p_\theta(x) &= -\frac{\nabla_\theta Z_\theta}{Z_\theta^2} e^{-U_\theta(x)} - \frac{1}{Z_\theta} e^{-U_\theta(x)} \nabla_\theta U_\theta(x) \\ &= p_\theta(x) \left(-\frac{\nabla_\theta Z_\theta}{Z_\theta} - \nabla_\theta U_\theta(x) \right).\end{aligned}$$

Differentiating the normalization under the integral sign,

$$\begin{aligned}\nabla_\theta Z_\theta &= -\int_E \nabla_\theta U_\theta(x) e^{-U_\theta(x)}\, dx \\ &= -Z_\theta\, \mathbb{E}_\theta[\nabla_\theta U_\theta].\end{aligned}$$

Substitution therefore yields

$$\boxed{\nabla_\theta p_\theta(x) = -p_\theta(x)(\nabla_\theta U_\theta(x) - \mathbb{E}_\theta[\nabla_\theta U_\theta]).}$$

For a tangent direction $d\theta \in T_\theta\Theta$, the differential is obtained by pairing this gradient with $d\theta$:

$$\begin{aligned}d_\theta p_\theta(d\theta) &= (\nabla_\theta p_\theta)^T d\theta \\ &= -p_\theta \big(\nabla_\theta U_\theta - \mathbb{E}_\theta[\nabla_\theta U_\theta]\big)^T d\theta, \qquad d_\theta U_\theta(d\theta) = (\nabla_\theta U_\theta)^T d\theta.\end{aligned}$$

Its integral vanishes, as required for a tangent density variation. Normalization therefore removes the mean energy variation: only its fluctuations change the density. In particular, adding a parameter-dependent constant to $U_\theta(x)$ changes Z_θ but leaves p_θ unchanged.

7.2. Fisher–Rao metric

We briefly recall the computation. For more detail, see P. Guilbert, *Pullback Fisher–Rao Geometries of Parametric Probability Families: An Expository Atlas*, Sections 7.1–7.3 for nonlinear Gibbs families and Sections 6.1–6.3 for linear Gibbs families [1]. Substituting the density differential into the Fisher–Rao metric gives

$$\begin{aligned}g_\theta^{\mathrm{FR}}(d\theta_1, d\theta_2) &= \int_E \frac{d_\theta p_\theta(d\theta_1) d_\theta p_\theta(d\theta_2)}{p_\theta}\, dx \\ &= \mathrm{Cov}_\theta(d_\theta U_\theta(d\theta_1), d_\theta U_\theta(d\theta_2))\,.\end{aligned}$$

Thus, if $G^{\mathrm{FR}}(\theta)$ denotes the matrix of this pullback in the coordinates $\theta_1, \dots, \theta_k$,

$$\boxed{\begin{aligned}G_{ij}^{\mathrm{FR}}(\theta) &= \mathrm{Cov}_\theta\Big(\partial_{\theta_i} U_\theta, \partial_{\theta_j} U_\theta\Big)\,, \\ g_\theta^{\mathrm{FR}}(d\theta, d\theta) &= \mathrm{Var}_\theta[d_\theta U_\theta(d\theta)].\end{aligned}}$$

A parameter direction is costly when its energy variation differs strongly between states carrying probability under p_θ.

For a nonlinear potential, another differentiation of $\log Z_\theta$ yields

$$G_{ij}^{\mathrm{FR}}(\theta) = \partial_{\theta_i}\partial_{\theta_j} \log Z_\theta + \mathbb{E}_\theta[\partial_{\theta_i}\partial_{\theta_j} U_\theta].$$

The correction term need not vanish. In the linear case, however, $\partial_{\theta_i} U_\theta = \phi_i$ and the second parameter derivatives of U_θ are zero, so

$$\boxed{G^{\mathrm{FR}}(\theta) = \mathrm{Cov}_\theta(\phi(X)) = \mathrm{Hess}_\theta \log Z_\theta, \qquad U_\theta(x) = \phi(x)^T \theta.}$$

This is a Hessian metric in the natural parameters [14, Section 3.9]. It is generally variable, even though the potential is linear in those parameters.

7.3. Wasserstein–Otto metric

7.3.1. From the density variation to the transport potential

Fix θ and a parameter direction $d\theta$. The corresponding density variation is $u = d_\theta p_\theta(d\theta)$. We seek its minimal-energy velocity as a spatial gradient, as in [12, Section 3.1],

$$v(x) = \nabla_x \psi_{\theta,d\theta}(x), \qquad -\operatorname{div}_x(p_\theta \nabla_x \psi_{\theta,d\theta}) = u.$$

The potential $\psi_{\theta,d\theta}$ depends linearly on the tangent direction once its additive constant is fixed. We choose $\mathbb{E}_\theta[\psi_{\theta,d\theta}] = 0$. To ensure a well-defined finite-energy solution for every parameter direction, we assume a weighted Poincaré inequality at each parameter:

$$\mathbb{E}_\theta[(f - \mathbb{E}_\theta f)^2] \le \frac{1}{\lambda_\theta}\mathbb{E}_\theta[\|\nabla_x f\|^2], \qquad \lambda_\theta > 0.$$

Together with the square-integrability assumptions, this gives existence and uniqueness of the centered weak solution in the weighted Sobolev space. This is an additional assumption on the Gibbs family, not a consequence of normalization alone.

Substitution of the density differential gives

$$-\operatorname{div}_x(p_\theta \nabla_x \psi_{\theta,d\theta}) = -p_\theta(d_\theta U_\theta(d\theta) - \mathbb{E}_\theta[d_\theta U_\theta(d\theta)]).$$

The normalization Z_θ is constant with respect to x, hence

$$\nabla_x p_\theta = -p_\theta \nabla_x U_\theta.$$

Expanding the divergence and dividing by $p_\theta > 0$ therefore yields

$$\boxed{\Delta_x \psi_{\theta,d\theta} - \nabla_x U_\theta \cdot \nabla_x \psi_{\theta,d\theta} = d_\theta U_\theta(d\theta) - \mathbb{E}_\theta[d_\theta U_\theta(d\theta)].}$$

On the left, all derivatives act on the spatial variable x. On the right, $d_\theta U_\theta(d\theta)$ differentiates the potential with respect to its parameters. The two operations have different roles: the parameter variation prescribes a density change, and the spatial Poisson equation reconstructs the velocity that produces it.

It is useful to write the positive weighted operator as

$$L_\theta(f) := -\frac{1}{p_\theta}\operatorname{div}_x(p_\theta \nabla_x f) = -\Delta_x f + \nabla_x U_\theta \cdot \nabla_x f.$$

Integration by parts gives

$$\mathbb{E}_\theta[f\, L_\theta(h)] = \mathbb{E}_\theta[\nabla_x f \cdot \nabla_x h].$$

Thus L_θ is symmetric and nonnegative in $L^2(p_\theta dx)$, with constants in its kernel. Its inverse below is always taken on the centered subspace, where the Poincaré inequality makes it well defined. With this sign convention,

$$\psi_{\theta,d\theta} = -L_\theta^{-1}(d_\theta U_\theta(d\theta) - \mathbb{E}_\theta[d_\theta U_\theta(d\theta)])\,.$$

7.3.2. **The pullback metric and the linear case**

The Wasserstein–Otto metric is the energy of these velocities:

$$\boxed{g_\theta^{\mathrm{WO}}(d\theta_1, d\theta_2) = \mathbb{E}_\theta[\nabla_x \psi_{\theta,d\theta_1} \cdot \nabla_x \psi_{\theta,d\theta_2}]\,.}$$

To compute its matrix, let $\psi_{\theta,i}$ be the centered solution for variation in the ith parameter:

$$L_\theta(\psi_{\theta,i}) = -(\partial_{\theta_i} U_\theta - \mathbb{E}_\theta[\partial_{\theta_i} U_\theta]).$$

Then $\psi_{\theta,d\theta} = \sum_i d\theta_i \psi_{\theta,i}$ and

$$\begin{aligned} G_{ij}^{\mathrm{WO}}(\theta) &= \mathbb{E}_\theta[\nabla_x \psi_{\theta,i} \cdot \nabla_x \psi_{\theta,j}] \\ &= \mathbb{E}_\theta[\psi_{\theta,i} L_\theta(\psi_{\theta,j})] \\ &= -\mathbb{E}_\theta\Big[\psi_{\theta,i}(\partial_{\theta_j} U_\theta - \mathbb{E}_\theta[\partial_{\theta_j} U_\theta])\Big]\,. \end{aligned}$$

Equivalently, moving the inverse operator onto the centered energy variation gives

$$\boxed{\begin{aligned} G_{ij}^{\mathrm{WO}}(\theta) = \mathbb{E}_\theta\Big[&(\partial_{\theta_i} U_\theta - \mathbb{E}_\theta[\partial_{\theta_i} U_\theta]) \\ &L_\theta^{-1}(\partial_{\theta_j} U_\theta - \mathbb{E}_\theta[\partial_{\theta_j} U_\theta])\Big]. \end{aligned}}$$

The two minus signs from the density variations have cancelled. The operator acts on functions of x at the current parameter θ; it is not an inverse of the Fisher matrix on Θ.

For $U_\theta = \phi^T \theta$, this becomes

$$\boxed{G_{ij}^{\mathrm{WO}}(\theta) = \mathbb{E}_\theta\Big[(\phi_i - \mathbb{E}_\theta \phi_i) L_\theta^{-1}(\phi_j - \mathbb{E}_\theta \phi_j)\Big]\,.}$$

The source functions are now the centered sufficient statistics, but L_θ still depends on the parameter through $U_\theta = \sum_i \theta_i \phi_i$. In particular, linearity of the energy does not remove the Poisson solve. Numerically, one solves k centered Poisson problems at each parameter and takes their gradient inner products to obtain the metric matrix.

7.4. **Comparison and geodesics**

7.4.1. **Fluctuation cost and transport cost**

Both pullbacks act on the same vector $d\theta \in T_\theta\Theta$ and on the same density differential. To display their difference, denote this single centered energy variation by

$$h_\theta(d\theta) := d_\theta U_\theta(d\theta) - \mathbb{E}_\theta[d_\theta U_\theta(d\theta)].$$

The two squared norms are

$$\boxed{\begin{aligned} g_\theta^{\mathrm{FR}}(d\theta, d\theta) &= \mathbb{E}_\theta[h_\theta(d\theta)^2], \\ g_\theta^{\mathrm{WO}}(d\theta, d\theta) &= \mathbb{E}_\theta\Big[h_\theta(d\theta)\, L_\theta^{-1}(h_\theta(d\theta))\Big]\,. \end{aligned}}$$

Fisher–Rao measures the size of the relative density change:

$$\frac{d_\theta p_\theta(d\theta)}{p_\theta} = -h_\theta(d\theta).$$

Wasserstein–Otto also accounts for the spatial arrangement of this change: it measures the least kinetic energy needed to redistribute mass in E. Two energy variations with equal variance can therefore have different transport costs.

For example, if a centered spatial function h satisfies $L_\theta(h) = \lambda h$ with $\lambda > 0$, then a direction with $h_\theta(d\theta) = h$ has

$$g_\theta^{\mathrm{WO}}(d\theta, d\theta) = \frac{1}{\lambda}\, g_\theta^{\mathrm{FR}}(d\theta, d\theta).$$

More generally, whenever the centered operator admits a complete orthonormal eigenbasis, the same expansion gives a sum of squared mode coefficients for Fisher–Rao, and the same sum weighted by inverse eigenvalues for Wasserstein–Otto. Small eigenvalues correspond to spatial modes that are expensive to realize by transport. This explains why the two matrices on Θ need not have the same principal directions or relative eigenvalues.

The Poincaré inequality provides the local comparison

$$g_\theta^{\mathrm{WO}}(d\theta, d\theta) \le \lambda_\theta^{-1} g_\theta^{\mathrm{FR}}(d\theta, d\theta).$$

Indeed, writing $f = L_\theta^{-1}\big(h_\theta(d\theta)\big)$, integration by parts, Cauchy–Schwarz and Poincaré give

$$\begin{aligned}\mathbb{E}_\theta[\|\nabla_x f\|^2] &= \mathbb{E}_\theta[h_\theta(d\theta) f] \\ &\le \big(\mathbb{E}_\theta[h_\theta(d\theta)^2]\big)^{1/2} \big(\lambda_\theta^{-1}\mathbb{E}_\theta[\|\nabla_x f\|^2]\big)^{1/2}.\end{aligned}$$

The comparison factor depends on the current law and spatial scale; there is no universal ordering of the two squared norms with factor one.

Their null directions nevertheless coincide. Either norm vanishes exactly when $d_\theta U_\theta(d\theta)$ is constant p_θ-almost everywhere, equivalently when $d_\theta p_\theta(d\theta) = 0$. Under the stated existence assumptions, both pullbacks are therefore positive definite precisely where $\theta \mapsto p_\theta$ is an immersion. In the linear case, this means that no nonzero linear combination of the ϕ_i is constant almost everywhere. When the matrices are positive definite, their unit ellipsoids $d\theta^T G(\theta) d\theta = 1$ describe which parameter variations are locally costly for each metric.

7.4.2. **Intrinsic geodesics of the Gibbs family**

A curve is now a path $t \mapsto \theta(t)$ in Θ, representing the densities $p_{\theta(t)}$. Its length and energy for either metric are

$$\begin{aligned}L_g[\theta] &= \int_0^1 \sqrt{\dot\theta(t)^T G(\theta(t))\dot\theta(t)}\, dt, \\ \mathcal{E}_g[\theta] &= \frac{1}{2}\int_0^1 \dot\theta(t)^T G(\theta(t))\dot\theta(t)\, dt.\end{aligned}$$

On a regular parameter region where the metric is sufficiently smooth, constant-speed geodesics are stationary paths of this energy. The coordinate equations are

$$\frac{d}{dt}\left(\sum_{j=1}^k G_{ij}(\theta(t))\dot\theta_j(t)\right) = \frac{1}{2}\sum_{j,\ell=1}^k \partial_{\theta_i} G_{j\ell}(\theta(t))\dot\theta_j(t)\dot\theta_\ell(t), \qquad i = 1, \dots, k.$$

The intrinsic distance is the infimum of L_g over paths in Θ with the prescribed endpoints. A minimizing constant-speed path, when it exists, satisfies these equations; a stationary solution alone need not be globally minimizing.

For Fisher–Rao, the energy integrates the variance of the instantaneous energy variation $d_\theta U_{\theta(t)}(\dot\theta(t))$. For Wasserstein–Otto, the same variation prescribes a Poisson problem at each time, and the energy integrates the resulting squared velocity. Thus the path selection differs even when both metrics have the same null directions. Neither metric generally makes straight parameter segments geodesic.

In particular, the linear Gibbs case gives $G^{\mathrm{FR}} = \operatorname{Hess}\log Z$, but a Hessian metric need not be constant or Riemannian-flat. Straight lines in natural parameters are the exponential interpolation paths of the family; they are not, in general, its Levi–Civita geodesics [14, Sections 3.8–3.9]. The Hessian structure also does not transfer automatically to G^{WO}, which includes the parameter-dependent inverse spatial operator.

Finally, these are intrinsic geodesics of the chosen Gibbs family. The ambient Fisher–Rao geodesic obtained in square-root density coordinates, or the ambient Wasserstein displacement interpolation, need not remain in that family. Their distances are lower bounds for the corresponding intrinsic distances, with equality when an ambient minimizing path is realized by a path in the family. In the general nonlinear case, neither normalization nor the Gibbs form alone provides a closed formula for these parameter-space geodesics or their curvature.

8. Push-forward families

8.1. Transporting a reference density

Let q be a fixed positive probability density on $E = \mathbb{R}^n$, and let $\Theta \subset \mathbb{R}^k$ be open. A push-forward family is obtained by transporting this reference law through parameter-dependent maps

$$\phi_\theta \colon E \longrightarrow E, \qquad p_\theta = (\phi_\theta)_\sharp q.$$

Thus, if X has density q, then $\phi_\theta(X)$ has density p_θ. We assume that each ϕ_θ is a diffeomorphism, with sufficient regularity in both its spatial variable and its parameters. The change-of-variables formula reads

$$\boxed{p_\theta(y) = \frac{q(\phi_\theta^{-1}(y))}{|\det D_x\phi_\theta(\phi_\theta^{-1}(y))|}.}$$

Here $D_x\phi_\theta$ denotes the spatial Jacobian matrix. Equivalently,

$$p_\theta(\phi_\theta(x))\,|\det D_x\phi_\theta(x)| = q(x).$$

The variable x labels a reference point carrying the density q; $y = \phi_\theta(x)$ is its current position, where the transported density is evaluated. These two variables must be distinguished when we differentiate with respect to θ.

We assume that $\int_E \|\phi_\theta(x)\|^2 q(x)\,dx < \infty$, so $p_\theta \in \mathcal{P}_{2,+}(E)$, and that the derivatives and weighted integrals below exist. Integration by parts is understood with decay conditions eliminating boundary terms. The finite-energy Poisson solutions used for Wasserstein–Otto are also assumed to exist; as in Section 7.3, a weighted Poincaré inequality is a sufficient condition when the sources divided by p_θ are square-integrable.

Affine maps already recover several familiar examples. A standard Gaussian reference and $\phi_{\mu,B}(x) = \mu + Bx$, with B invertible, give a Gaussian with mean μ and covariance BB^T. Nonlinear maps can bend the reference density: in dimension two,

$$\phi_\theta(x_1, x_2) = (x_1,\, \theta_1 x_1^2 + \theta_2 x_2), \qquad \theta_1 \in \mathbb{R}, \quad \theta_2 > 0,$$

is a diffeomorphism with determinant θ_2. Applied to a standard Gaussian reference, it gives the curved Gibbs density illustrated in Figure 23. A single density family can therefore admit both an energy description and a transport description.

8.2. **Differentiation with respect to the parameters**

We distinguish two differentiation operations. The notation D_x denotes the differential with respect to the spatial variable x, with the parameter fixed; d_θ denotes the differential with respect to the parameter, with the spatial point fixed. In particular, we set

$$J_\theta(x) := D_x \phi_\theta(x), \qquad (J_\theta(x))_{ab} = \partial_{x_b} (\phi_\theta)_a(x).$$

The matrix $J_\theta(x)$ is the spatial Jacobian, but it still depends on θ. Since $T_\theta\Theta \simeq \mathbb{R}^k$, a parameter direction $d\theta$ gives the vector

$$d_\theta \phi_\theta(d\theta)(x) = \sum_{i=1}^{k} \partial_{\theta_i} \phi_\theta(x)\, d\theta_i.$$

For a scalar function of the current position y, its spatial differential is represented by the gradient: $D_y p_\theta(y)(a) = \nabla_y p_\theta(y) \cdot a$ for $a \in \mathbb{R}^n$. The density variation we seek is $d_\theta p_\theta(d\theta)(y)$, where y is held fixed.

Start from the density formula. At a fixed reference point x, the change-of-variables identity is

$$p_\theta(\phi_\theta(x)) = \frac{q(x)}{|\det J_\theta(x)|}.$$

We differentiate each side along $\theta(s) = \theta + s\,d\theta$ at $s = 0$, holding x and hence $q(x)$ fixed.

Differentiate the right-hand side. Only the denominator depends on θ. The derivative of its reciprocal gives

$$\frac{d}{ds}\Big|_{s=0} \frac{q(x)}{|\det J_{\theta(s)}(x)|} = -\frac{q(x)}{|\det J_\theta(x)|^2}\, d_\theta(|\det J_\theta|)(d\theta)(x).$$

The determinant derivative is

$$\begin{aligned} &d_\theta(|\det J_\theta|)(d\theta)(x) \\ &\qquad = |\det J_\theta(x)|\, \operatorname{tr}\Big(J_\theta(x)^{-1} d_\theta J_\theta(d\theta)(x)\Big). \end{aligned}$$

This also holds when the determinant is negative: its sign is locally constant because the Jacobian is invertible. Substitution gives

$$\begin{aligned}\frac{d}{ds}\Big|_{s=0} \frac{q(x)}{|\det J_{\theta(s)}(x)|} &= -\frac{q(x)}{|\det J_\theta(x)|}\ \mathrm{tr}\Big(J_\theta(x)^{-1} d_\theta J_\theta(d\theta)(x)\Big) \\ &= -p_\theta(\phi_\theta(x))\ \mathrm{tr}\Big(J_\theta(x)^{-1} d_\theta J_\theta(d\theta)(x)\Big)\,.\end{aligned}$$

The last line substitutes $p_\theta(\phi_\theta(x)) = q(x)/|\det J_\theta(x)|$.
Differentiate the left-hand side. Here there are two dependencies on θ: the density p_θ changes, and its evaluation point $\phi_\theta(x)$ changes. The chain rule separates these two contributions:

$$\begin{aligned}\frac{d}{ds}\Big|_{s=0} p_{\theta(s)}(\phi_{\theta(s)}(x)) &= d_\theta p_\theta(d\theta)(\phi_\theta(x)) \\ &\quad + \nabla_y p_\theta(\phi_\theta(x)) \cdot d_\theta\phi_\theta(d\theta)(x).\end{aligned}$$

The first term differentiates $p_\theta(y)$ at fixed y, then evaluates at $y = \phi_\theta(x)$. The second differentiates its evaluation point, with the density function held fixed. Equating the derivatives of the two sides and isolating the first term gives

$$\begin{aligned}d_\theta p_\theta(d\theta)(\phi_\theta(x)) = &-\nabla_y p_\theta(\phi_\theta(x)) \cdot d_\theta\phi_\theta(d\theta)(x) \\ &- p_\theta(\phi_\theta(x))\ \mathrm{tr}\Big(J_\theta(x)^{-1} d_\theta J_\theta(d\theta)(x)\Big)\,.\end{aligned}$$

Express the parameter derivative of the Jacobian. Since $J_\theta = D_x\phi_\theta$, the assumed C^2 regularity allows us to exchange the spatial and parameter derivatives:

$$d_\theta(D_x\phi_\theta) = D_x(d_\theta\phi_\theta).$$

Thus, for the fixed direction $d\theta$,

$$d_\theta J_\theta(d\theta)(x) = D_x\big(d_\theta\phi_\theta(d\theta)\big)(x).$$

This is the identity used in the next subsection to express the trace as the divergence of the induced velocity. The density differential above contains both motion through the density profile and change of the volume element; the velocity formulation will combine them.

8.3. **The induced spatial velocity**

The trajectory of a reference point under the parameter path $\theta + s\,d\theta$ has velocity $d_\theta\phi_\theta(d\theta)(x)$ at $s = 0$. Reading this velocity at the current position defines

$$\boxed{v_{\theta,d\theta}(y) := d_\theta\phi_\theta(d\theta)(\phi_\theta^{-1}(y)).}$$

Equivalently, $v_{\theta,d\theta}(\phi_\theta(x)) = d_\theta\phi_\theta(d\theta)(x)$. The tangent vector $d\theta$ belongs to $T_\theta\Theta$, whereas $v_{\theta,d\theta}$ is a vector field on E. Differentiating the last identity with respect to x gives

$$D_y v_{\theta,d\theta}(\phi_\theta(x))\, D_x\phi_\theta(x) = D_x\big(d_\theta\phi_\theta(d\theta)\big)(x).$$

By cyclic invariance of the trace,

$$\mathrm{div}_y\, v_{\theta,d\theta}(\phi_\theta(x)) = \mathrm{tr}\Big((D_x\phi_\theta(x))^{-1} D_x\big(d_\theta\phi_\theta(d\theta)\big)(x)\Big)\,.$$

Substituting into the density differential and using the bijectivity of ϕ_θ gives, at every current position y,

$$\begin{aligned} u(y) = d_\theta p_\theta(d\theta)(y) &= -\nabla_y p_\theta(y) \cdot v_{\theta,d\theta}(y) - p_\theta(y)\,\mathrm{div}_y\, v_{\theta,d\theta}(y) \\ &= -\,\mathrm{div}_y(p_\theta v_{\theta,d\theta})(y). \end{aligned}$$

Hence the induced velocity satisfies the continuity equation

$$\boxed{d_\theta p_\theta(d\theta) + \mathrm{div}_y(p_\theta v_{\theta,d\theta}) = 0.}$$

Its sign reflects conservation of mass. The integral of u vanishes, but the weighted divergence need not vanish pointwise. Along a parameter path $\theta(t)$, the same formula reads

$$\partial_t p_{\theta(t)} + \mathrm{div}_y(p_{\theta(t)} v_{\theta(t),\dot\theta(t)}) = 0.$$

Figure 24 shows this construction for a sinusoidal shear and vertical scaling: the parameter direction $d\theta$ becomes a spatial velocity field $v_{\theta,d\theta}$ on the current coordinates y, and a finite step of that field matches the nearby grid exactly when the map is affine in θ.

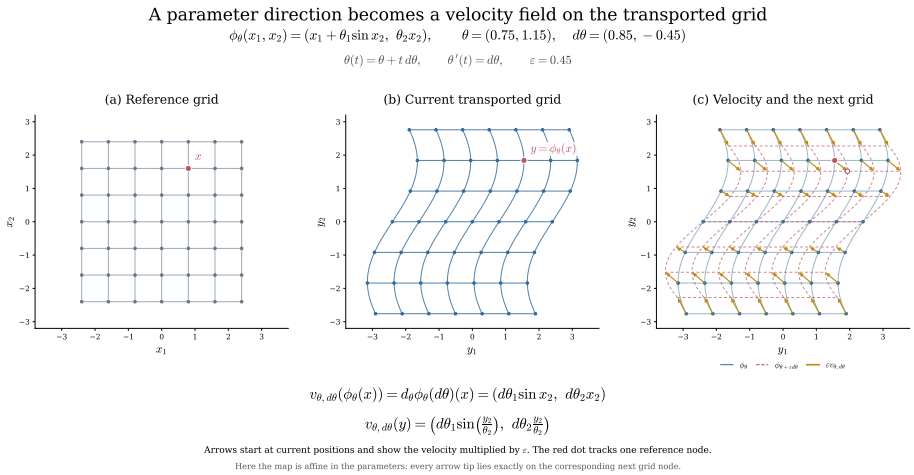


FIGURE 24. Map-induced velocity for $\phi_\theta(x_1, x_2) = (x_1 + \theta_1 \sin x_2, \theta_2 x_2)$, $\Theta = \mathbb{R} \times (0, \infty)$. Panels show the reference grid, its image at $\theta = (0.75, 1.15)$, and its change along $d\theta = (0.85, -0.45)$ with $\varepsilon = 0.45$. At the current position y, $v_{\theta,d\theta}(y) = (d\theta_1 \sin(y_2/\theta_2), d\theta_2 y_2/\theta_2)$. Arrows represent εv and reach the next grid exactly because the map is affine in θ; generally this agreement is first-order. The field realizes $d_\theta p_\theta(d\theta) = -\,\mathrm{div}(p_\theta v)$ and may require gradient projection to evaluate the Wasserstein–Otto metric.

8.4. **Wasserstein–Otto metric**

The map-induced field realizes the correct density variation, but it need not be a gradient and need not minimize kinetic energy. Applying the weighted Helmholtz decomposition of Section 4.2.3, write

$$v_{\theta,d\theta} = \nabla_y \psi_{\theta,d\theta} + w_{\theta,d\theta}, \qquad \mathrm{div}_y(p_\theta w_{\theta,d\theta}) = 0.$$

The two terms are orthogonal in $L^2(p_\theta dy; \mathbb{R}^n)$. The centered potential, fixed by $\int_E \psi_{\theta,d\theta} p_\theta \, dy = 0$, solves

$$\boxed{-\operatorname{div}_y(p_\theta \nabla_y \psi_{\theta,d\theta}) = -\operatorname{div}_y(p_\theta v_{\theta,d\theta}) = d_\theta p_\theta(d\theta).}$$

Its gradient is the minimum-energy velocity for this density variation. The pullback metric is therefore

$$\boxed{g_\theta^{\mathrm{WO}}(d\theta_1, d\theta_2) = \int_E p_\theta \, \nabla_y \psi_{\theta,d\theta_1} \cdot \nabla_y \psi_{\theta,d\theta_2} \, dy.}$$

For coordinate computations, set

$$v_{\theta,i}(y) = \partial_{\theta_i} \phi_\theta(\phi_\theta^{-1}(y)), \qquad i = 1, \dots, k,$$

and let $\psi_{\theta,i}$ solve the same Poisson equation with $v_{\theta,i}$ on its right-hand side. Linearity in $d\theta$ gives

$$G_{ij}^{\mathrm{WO}}(\theta) = \int_E p_\theta \, \nabla_y \psi_{\theta,i} \cdot \nabla_y \psi_{\theta,j} \, dy.$$

The distinction between the prescribed map and the density is visible in the energy identity

$$\begin{aligned} \int_E q(x) \|d_\theta \phi_\theta(d\theta)(x)\|^2 \, dx &= \int_E p_\theta(y) \|v_{\theta,d\theta}(y)\|^2 \, dy \\ &= g_\theta^{\mathrm{WO}}(d\theta, d\theta) + \int_E p_\theta(y) \|w_{\theta,d\theta}(y)\|^2 \, dy. \end{aligned}$$

Thus the squared $L^2(q)$ norm of the map variation is an upper bound for the density's Wasserstein–Otto squared norm. They coincide exactly when the map-induced velocity has no weighted divergence-free component. In particular, when every coordinate velocity is an admissible gradient,

$$G_{ij}^{\mathrm{WO}}(\theta) = \int_E q(x) \, \partial_{\theta_i} \phi_\theta(x) \cdot \partial_{\theta_j} \phi_\theta(x) \, dx.$$

This expression applies only when the coordinate velocities are admissible gradients.

8.5. **Fisher–Rao metric**

Here the density differential can be inserted directly. For the detailed calculation and interpretation, see P. Guilbert, *Pullback Fisher–Rao Geometries of Parametric Probability Families: An Expository Atlas*, Sections 8.1–8.3 [1]. The formula is

$$\boxed{g_\theta^{\mathrm{FR}}(d\theta_1, d\theta_2) = \int_E \frac{\operatorname{div}_y(p_\theta v_{\theta,d\theta_1}) \operatorname{div}_y(p_\theta v_{\theta,d\theta_2})}{p_\theta} \, dy.}$$

In coordinates, this gives

$$G_{ij}^{\mathrm{FR}}(\theta) = \int_E \frac{\operatorname{div}_y(p_\theta v_{\theta,i}) \operatorname{div}_y(p_\theta v_{\theta,j})}{p_\theta} \, dy.$$

The corresponding squared norm can also be read as

$$g_\theta^{\mathrm{FR}}(d\theta, d\theta) = \int_E p_\theta \left(\operatorname{div}_y v_{\theta,d\theta} + \frac{\nabla_y p_\theta}{p_\theta} \cdot v_{\theta,d\theta} \right)^2 \, dy.$$

The expression inside parentheses has zero mean under p_θ, since its integral against p_θ is the integral of a divergence. Fisher–Rao therefore measures the variance of this relative compression and advection rate. It does not measure $\operatorname{div} v$ alone: even a spatially constant translation field can change a nonuniform density. The relation between Fisher–Rao geometry and metrics on diffeomorphism groups is developed, for compact sample manifolds, in [13, Section 3.3].

8.6. **Comparison of the metrics**

What is being compared? Fix a parameter θ and a direction $d\theta$. The map gives the velocity $v_{\theta,d\theta}$, and the continuity equation gives the density variation

$$u = -\operatorname{div}_y(p_\theta v_{\theta,d\theta}).$$

The two squared costs answer different questions:

$$g_\theta^{\mathrm{FR}}(d\theta, d\theta) = \int_E p_\theta \left(\frac{u}{p_\theta}\right)^2 dy,$$
$$g_\theta^{\mathrm{WO}}(d\theta, d\theta) = \min_{-\operatorname{div}_y(p_\theta a)=u} \int_E p_\theta \|a\|^2 \, dy.$$

Fisher–Rao asks how large the relative density change u/p_θ is at each spatial point. Wasserstein–Otto asks how much kinetic energy is needed to produce that change by moving mass. Its minimization is over spatial velocity fields a, not over parameter directions. The following situations separate the effects of motion, compression, spatial scale, and the choice of a velocity realizing u.

1. Motion that leaves the density unchanged. Take the standard isotropic Gaussian on $\mathbb{R}^2$ and let ϕ_α be a rotation through angle α. The density remains the same for every α. The induced field is nevertheless nonzero:

$$v_{\alpha,d\alpha}(y) = d\alpha\,(-y_2, y_1).$$

Its ordinary divergence is zero, and it is tangent to the circular density contours. Therefore $\nabla_y p \cdot v_{\alpha,d\alpha} = 0$ as well, and

$$u = 0, \qquad g_\alpha^{\mathrm{FR}}(d\alpha, d\alpha) = 0, \qquad g_\alpha^{\mathrm{WO}}(d\alpha, d\alpha) = 0.$$

Figure 8 shows this density-preserving motion. Both metric costs vanish: $u = 0$, and zero velocity realizes it. The prescribed rotation nevertheless has kinetic energy $\int p\|v_{\alpha,d\alpha}\|^2 = 2(d\alpha)^2$.

This is the precise meaning of the invariance with respect to motions that preserve density: adding any w with $\operatorname{div}_y(p_\theta w) = 0$ to a velocity leaves u, and hence both metric costs, unchanged. It can change the energy of that particular velocity. Both pullbacks consequently have the same kernel,

$$\{d\theta : \operatorname{div}_y(p_\theta v_{\theta,d\theta}) = 0\}.$$

Under the stated regularity assumptions, they are positive definite exactly when the parameter-to-density map is an immersion. Distinct transport maps alone need not give distinct densities.

2. Translation: motion without volume compression. For $\phi_\mu(x) = x + \mu$, the density is $p_\mu(y) = q(y - \mu)$ and the velocity is the constant field $v_{\mu,d\mu} = d\mu$. Thus

$$\operatorname{div}_y v_{\mu,d\mu} = 0, \qquad u(y) = -\nabla_y p_\mu(y) \cdot d\mu.$$

Unlike the rotation above, this motion crosses the density contours. The density at a fixed location changes even though small volumes are not compressed. The velocity is already a gradient, giving

$$g_\mu^{\mathrm{WO}}(d\mu, d\mu) = \|d\mu\|^2,$$
$$g_\mu^{\mathrm{FR}}(d\mu, d\mu) = \int_E \frac{(\nabla_x q(x) \cdot d\mu)^2}{q(x)}\, dx.$$

For a Gaussian reference with covariance Σ, these become

$$g_\mu^{\mathrm{WO}}(d\mu, d\mu) = \|d\mu\|^2, \qquad g_\mu^{\mathrm{FR}}(d\mu, d\mu) = d\mu^T \Sigma^{-1} d\mu.$$

Wasserstein–Otto charges for the displacement itself. Fisher–Rao charges more for the same displacement when the density is narrow in that direction: its profile changes more sharply at fixed spatial points. Figure 25 compares a narrow and a broad Gaussian under the same translational $d\mu$: the Wasserstein–Otto integrals coincide, while the Fisher–Rao cost is larger for the narrow density.

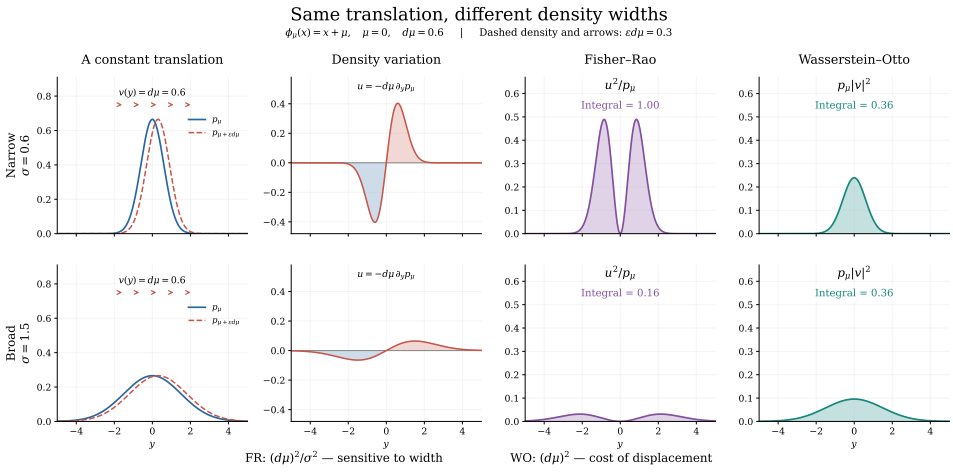


FIGURE 25. Translation of Gaussian densities at $\mu = 0$, with $d\mu = 0.6$ and scales $\sigma = 0.6$ (top), 1.5 (bottom). Columns show the translated density, variation u, Fisher–Rao integrand u^2/p, and Wasserstein–Otto integrand $p|v|^2$. The constant velocity $v = d\mu$ is already a gradient. Its squared tangent costs are $g^{\mathrm{FR}} = (d\mu)^2/\sigma^2$ and $g^{\mathrm{WO}} = (d\mu)^2 = 0.36$: broadening lowers the Fisher cost while leaving the transport cost unchanged. The displayed displacement uses $\varepsilon = 0.5$; costs use the unscaled tangent.

3. Dilation: absolute and relative scale changes. Take $q = \mathcal{N}(0, I_n)$ and $\phi_\sigma(x) = \sigma x$, with $\sigma > 0$. Then $p_\sigma = \mathcal{N}(0, \sigma^2 I_n)$ and

$$v_{\sigma,d\sigma}(y) = \frac{d\sigma}{\sigma} y, \qquad \frac{u(y)}{p_\sigma(y)} = \frac{d\sigma}{\sigma}\left(\frac{\|y\|^2}{\sigma^2} - n\right).$$

The field is a gradient. Using $\mathbb{E}[\|Y\|^2] = n\sigma^2$ and $\mathrm{Var}(\|Y\|^2/\sigma^2) = 2n$, we obtain

$$\boxed{g_\sigma^{\mathrm{FR}}(d\sigma, d\sigma) = 2n\left(\frac{d\sigma}{\sigma}\right)^2, \qquad g_\sigma^{\mathrm{WO}}(d\sigma, d\sigma) = n(d\sigma)^2.}$$

For the same absolute increment $d\sigma$, Fisher–Rao charges more at small σ, while the WO cost is independent of σ. For the same relative increment $d\sigma/\sigma$, the Fisher cost is constant, while the WO cost increases with σ^2: a proportional expansion of a larger distribution moves mass farther.

4. Shear: why the induced velocity must be projected. In dimension two, let $q = \mathcal{N}(0, I_2)$ and $\phi_\theta(x_1, x_2) = (x_1 + \theta x_2, x_2)$. At $\theta = 0$, $p_0 = q$ and

$$v_{0,d\theta}(y) = d\theta\,(y_2, 0), \qquad u(y) = d\theta\, y_1 y_2 q(y).$$

Again the ordinary divergence is zero, but the weighted divergence is not. The shear deforms the circular density contours. Its velocity splits explicitly as

$$v_{0,d\theta}(y) = \frac{d\theta}{2}(y_2, y_1) + \frac{d\theta}{2}(y_2, -y_1).$$

The first term is the gradient of $d\theta\, y_1 y_2/2$. The second is a rotational field with zero q-weighted divergence. Only the first is needed to produce u. Since the standard Gaussian coordinates are independent with unit variance,

$$\begin{aligned} g_0^{\mathrm{FR}}(d\theta, d\theta) &= (d\theta)^2 \mathbb{E}_q[Y_1^2 Y_2^2] = (d\theta)^2, \\ g_0^{\mathrm{WO}}(d\theta, d\theta) &= \frac{(d\theta)^2}{4}\mathbb{E}_q[Y_1^2 + Y_2^2] = \frac{1}{2}(d\theta)^2, \\ \int_E q\|v_{0,d\theta}\|^2\, dy &= (d\theta)^2. \end{aligned}$$

The raw velocity overestimates the WO squared norm by a factor of two; its equality with the Fisher cost is specific to this example. Figure 26 shows the Helmholtz split of this shear: the same density variation u is produced by v and by its gradient part alone, while the discarded rotation halves the kinetic energy.

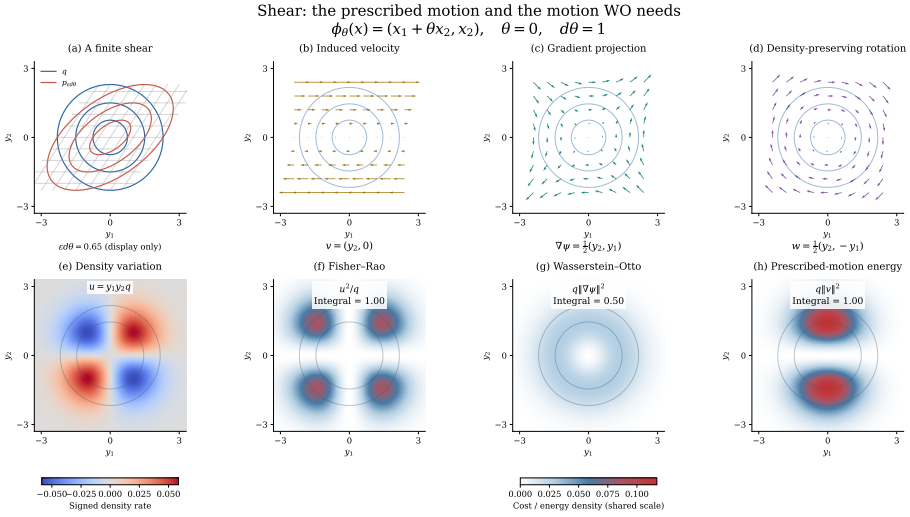


FIGURE 26. Shear of $q = \mathcal{N}(0, I_2)$ by $\phi_\theta(x) = (x_1 + \theta x_2, x_2)$, evaluated at $\theta = 0$, $d\theta = 1$. The prescribed velocity decomposes as $v = (y_2, 0) = \frac{1}{2}(y_2, y_1) + \frac{1}{2}(y_2, -y_1)$ into a gradient and a weighted divergence-free rotation. Only the gradient contributes to the Wasserstein–Otto squared norm, $g^{\mathrm{WO}} = 1/2$. The density variation $u = y_1 y_2 q$ gives $g^{\mathrm{FR}} = 1$, while raw kinetic energy $\int q\|v\|^2 = 1$ overestimates the WO cost by a factor of two. The deformation panel uses $\varepsilon = 0.65$ and arrows use a display factor 0.28; costs are unscaled.

5. Rapid spatial variation at a fixed velocity amplitude. On the line, take a standard Gaussian reference and, for a fixed $k > 0$, the diffeomorphisms

$$\phi_\theta(x) = x + \theta \sin(kx), \qquad |\theta| k < 1.$$

At $\theta = 0$, the velocity is $v_{0,d\theta}(y) = d\theta \sin(ky)$. It is already a spatial gradient. Writing $v = v_{0,d\theta}$ in the following identities, the continuity equation gives

$$\frac{u(y)}{q(y)} = y\, v(y) - v'(y), \qquad g_0^{\mathrm{WO}}(d\theta, d\theta) = \int_{\mathbb{R}} q(y) v(y)^2 \, dy.$$

For the standard Gaussian, integration by parts also gives

$$g_0^{\mathrm{FR}}(d\theta, d\theta) = \int_{\mathbb{R}} q(y)(v'(y) - y v(y))^2 \, dy = \int_{\mathbb{R}} q(y)\left(v'(y)^2 + v(y)^2\right) dy.$$

Indeed, the cross term is $-\int yq\,(v^2)' = \int (1-y^2) q\, v^2$, which cancels the $y^2 v^2$ contribution. For the sinusoidal field this yields

$$g_0^{\mathrm{WO}}(d\theta, d\theta) = \frac{(d\theta)^2}{2}(1 - e^{-2k^2}),$$

$$g_0^{\mathrm{FR}}(d\theta, d\theta) = g_0^{\mathrm{WO}}(d\theta, d\theta) + \frac{(d\theta)^2 k^2}{2}(1 + e^{-2k^2}).$$

As k increases at fixed $d\theta$, the speeds remain bounded, and the WO squared cost tends to $(d\theta)^2/2$. The Fisher squared cost grows like $(d\theta)^2 k^2/2$: neighboring points move in opposite directions over shorter spatial intervals, producing larger alternating density changes. This comparison holds at the

reference density and at fixed velocity amplitude; it does not compare directions with a fixed density-variation norm. Figure 27 compares $k = 1$ and $k = 4$ at the same $d\theta$: the WO integrand stays controlled by the bounded speeds, while the FR integrand picks up the stronger alternating compressions.

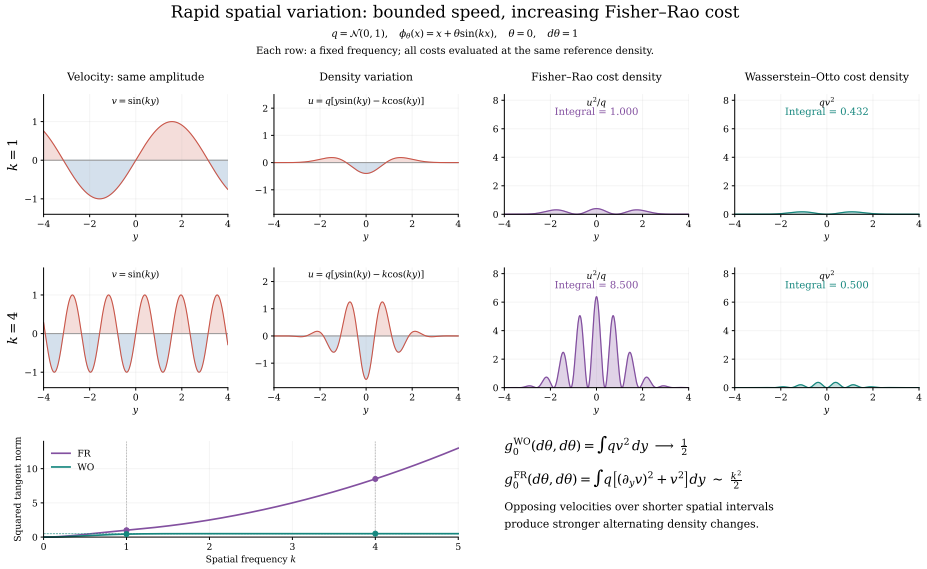


FIGURE 27. Rapid spatial variations in $\phi_\theta(x) = x + \theta\sin(kx)$, $|\theta|k < 1$, applied to $q = \mathcal{N}(0,1)$ at $\theta = 0$, $d\theta = 1$. Rows $k = 1, 4$ show $v(y) = \sin(ky)$, $u(y) = q(y)[y\sin(ky) - k\cos(ky)]$, and the cost densities u^2/q and qv^2. The lower plot compares their whole-space integrals as frequency increases. At fixed velocity amplitude, $g^{\mathrm{WO}} = (1 - e^{-2k^2})/2$ remains bounded and tends to $1/2$, whereas $g^{\mathrm{FR}} = g^{\mathrm{WO}} + k^2(1 + e^{-2k^2})/2 \sim k^2/2$. These are squared tangent norms.

8.7. Geodesics

Translation with fixed shape. For the translation family $p_\mu(y) = q(y - \mu)$ considered above, both matrices are constant in the translation parameter μ, so straight mean paths are intrinsic geodesics of this fixed-shape family, with different lengths. Allowing the covariance to vary gives a larger family and can change the Fisher–Rao minimizing path, as in Section 6.4.

Dilation of an isotropic Gaussian. For $p_\sigma = \mathcal{N}(0, \sigma^2 I_n)$, the two metrics give $2n(d\sigma/\sigma)^2$ and $n(d\sigma)^2$, respectively. In this one-parameter family, the constant-speed geodesics are:

$$\sigma_{\mathrm{FR}}(t) = \sigma_0^{1-t}\sigma_1^t, \qquad \sigma_{\mathrm{WO}}(t) = (1-t)\sigma_0 + t\sigma_1.$$

The FR geodesic allocates equal time to equal logarithmic scale changes; the WO geodesic allocates equal time to equal absolute scale changes. The respective distances are $\sqrt{2n}\,|\log(\sigma_1/\sigma_0)|$ and $\sqrt{n}\,|\sigma_1 - \sigma_0|$. The first two columns of Figure 28 show this comparison for $n = 2$, $\sigma_0 = 1$, and $\sigma_1 = 2$:

the two paths traverse the same densities with different time parameterizations.

Geodesics in the general parameter space. For either pullback matrix $G(\theta)$, the energy of a parameter path is

$$\mathcal{E}_g[\theta] = \frac{1}{2}\int_0^1 \dot\theta(t)^T G(\theta(t))\dot\theta(t)\,dt.$$

Intrinsic geodesics are stationary paths of this energy, with the coordinate equations given in Section 7.4; minimizing paths determine the intrinsic distance. Along such a path, Fisher–Rao penalizes the relative density rate, while Wasserstein–Otto penalizes the gradient component of $v_{\theta(t),\dot\theta(t)}$. The resulting geodesics generally differ, and neither the parameter path nor the path of transport maps is necessarily straight.

For Wasserstein–Otto, there is a useful special case. Given endpoint maps ϕ_{θ_0} and ϕ_{θ_1}, the common reference law defines a coupling by transporting the same reference point to both endpoints. It gives the bound

$$W_2^2(p_{\theta_0}, p_{\theta_1}) \le \int_E q(x)\|\phi_{\theta_1}(x) - \phi_{\theta_0}(x)\|^2\,dx.$$

If this coupling is optimal, its displacement interpolation is the ambient Wasserstein geodesic [19]. To realize it within the chosen parameterization, a sufficient condition is the existence of a path $\theta(t) \in \Theta$ such that

$$\phi_{\theta(t)}(x) = (1-t)\phi_{\theta_0}(x) + t\phi_{\theta_1}(x), \qquad 0 \le t \le 1.$$

Under both conditions, this path also realizes the intrinsic distance. Optimality of the coupling and membership in the family are separate requirements. Membership likewise determines whether the ambient Fisher–Rao geodesic in square-root coordinates realizes an intrinsic geodesic.

Straight map interpolation with a nonoptimal coupling. The third column of Figure 28 illustrates why optimality of the endpoint coupling matters. For $q = \mathcal{N}(0, I_2)$, the maps $x \mapsto 2x$ and $x \mapsto 2R_{\pi/2}x$ both produce $\mathcal{N}(0, 4I_2)$, where $R_{\pi/2}$ is a quarter-turn. Interpolating the identity with the second map gives

$$x \longmapsto \big((1-t)I_2 + 2tR_{\pi/2}\big)x, \qquad \Sigma(t) = \big((1-t)^2 + 4t^2\big)I_2.$$

These maps remain invertible and their densities stay in the isotropic Gaussian family. Nevertheless, the scale initially decreases from 1 to $\sqrt{0.8}$ at $t = 1/5$, before increasing to 2. This detour minimizes neither metric's length. The endpoint coupling has cost $\mathbb{E}_q\|2R_{\pi/2}X - X\|^2 = 10$, whereas the optimal radial coupling has cost $\mathbb{E}_q\|2X - X\|^2 = W_2^2 = 2$. The tracked particles reveal the rotational motion that the circular density levels alone cannot show.

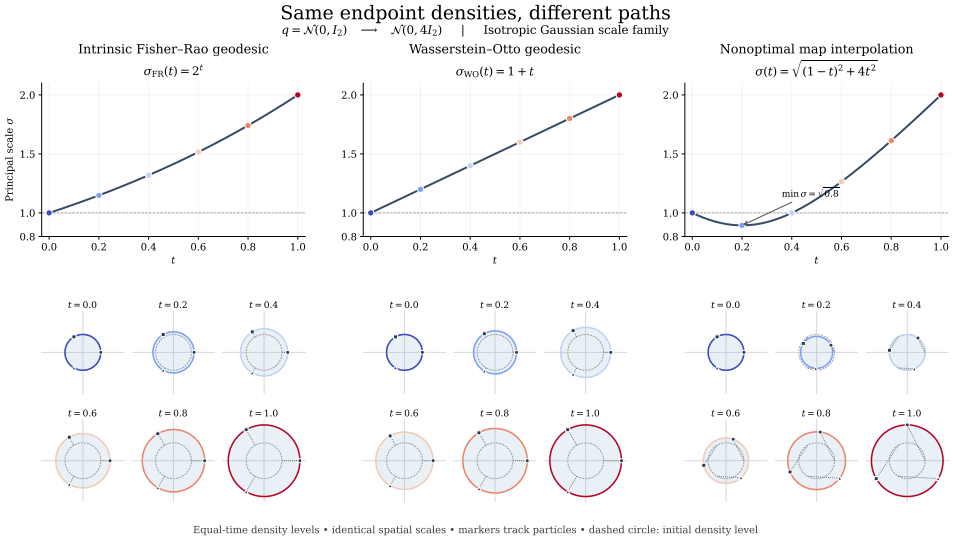


FIGURE 28. Dilation paths from $\mathcal{N}(0, I_2)$ to $\mathcal{N}(0, 4I_2)$. Top: intrinsic Fisher–Rao, Wasserstein–Otto, and nonoptimal straight map interpolation. Bottom: density levels $y^\top \Sigma(t)^{-1} y = 1$ at six equal times, with common spatial scales; markers track three particles and dotted lines show their past motion. FR and WO traverse the same densities at different constant metric speeds. The third path stays in the family but contracts before expanding. Colors indicate time; dashed circles mark the initial density level.

9. Conclusion and perspectives

For a density variation u, the two geometries studied in this atlas measure different aspects of the same change:

$$g_p^{\mathrm{FR}}(u,u) = \int_E \frac{u^2}{p^2}\, p\, dx, \qquad g_p^{\mathrm{WO}}(u,u) = \int_E p\|v\|^2\, dx, \qquad u = -\operatorname{div}(pv),$$

where, in the WO expression, v is the minimum-energy velocity realizing u, hence the appropriate gradient field. Fisher–Rao measures relative density change; Wasserstein–Otto measures the motion of probability mass through the sample space. A fixed translation is therefore more costly for a narrow Gaussian under FR, but has the same WO cost whatever its width. Rapid spatial oscillations of a fixed-amplitude velocity can increase the FR cost without making the WO cost unbounded. Density-preserving motions cost nothing in either geometry, even when a chosen transport map moves particles. Pulling these metrics back to a family retains these distinctions, while constraining the admissible variations and geodesics to that family.

These differences suggest concrete questions for nonlinear optimization and machine learning. Given finitely many samples from an unknown density, one may fit p_θ by minimizing an empirical loss $\mathcal{L}(\theta)$. Where the pullback

matrix $G(\theta)$ is positive definite, its natural gradient flow is [20]

$$\dot{\theta} = -G(\theta)^{-1}\nabla_\theta \mathcal{L}(\theta).$$

The metric determines which density changes count as small optimization steps: FR uses relative change, whereas WO uses transport effort [12]. For the same loss, these choices can change the trajectory and conditioning, but not the stationary points where G is invertible. With few samples, it is natural to ask how this choice interacts with model restrictions, regularization and early stopping. Neither metric alone resolves the underdetermination of density estimation, and a cheaper tangent direction does not by itself imply better generalization.

A further perspective is to choose a geometry through its symmetries. Both pullbacks are intrinsic under changes of parameter coordinates; this property does not distinguish FR from WO. The distinction concerns transformations of the *sample space*: FR is invariant under smooth invertible changes of observations, whereas WO preserves costs under isometries of its fixed ground metric. Diffeomorphism invariance leads to uniqueness results for FR under suitable hypotheses [3]; isometry invariance alone does not characterize WO. This suggests studying geometries adapted to other transformation groups or additional structure, such as anisotropic transport costs. A concrete example is covariance-modulated optimal transport [21], whose covariance-dependent Mahalanobis cost yields an affine-invariant transport geometry on distributions with finite, positive-definite covariance. Which symmetries should be preserved, which density changes should be expensive, and which computations remain feasible are then linked questions in the design of a learning method.

Statement on AI-Assisted Translation

This manuscript was originally written in French. Artificial intelligence tools were used to assist in translating the manuscript into English. The translation workflow was configured with author-defined instructions intended to preserve, as closely as possible, the tone, terminology, lexical choices, and style of mathematical exposition of the original French text.

The author has carefully reviewed and, where necessary, revised the complete English version. The author therefore takes full responsibility for the wording, accuracy, and scientific content of the English manuscript, notwithstanding the use of AI-assisted translation.

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